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IMPROVING PEDESTRIAN AND BICYCLIST SAFETY AT INTERSECTIONS FINAL REPORT

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LIST OF ACRONYMS

AASHTO	American Association of State Highway and Transportation Officials
AADT	Annual Average Daily Traffic
ADT	Average Daily Traffic
Caltrans	California Department of Transportation
CMF	Crash Modification Factor
DCMF	Development of Crash Modification Factors
DOT	Department of Transportation
FARS	Fatality Analysis Reporting System
FDW	Flashing Don't Walk
FHWA	Federal Highway Administration
FYA	Flashing Yellow Arrow
GIS	Geographic Information System
HAWK	High-Intensity Activated Cross Walk Beacon (also known as Pedestrian Hybrid Beacon)
HSIP	Highway Safety Improvement Program
HSIS	Highway Safety Information System
HSM	Highway Safety Manual
ITDP	Institute for Transportation and Development Policy
KAB	Killed, disabling (A), or evident injury (B) severity crash
LED	Light-emitting Diode
LPI	Leading Pedestrian Interval
MassDOT	Massachusetts Department of Transportation
MPH	Miles per Hour
MPO	Metropolitan Planning Organization
MUTCD	Manual of Uniform Traffic Control Devices
MV	Motor Vehicle
NACTO	National Association of City Transportation Officials
NC	North Carolina
NCDOT	North Carolina Department of Transportation
NCHRP	National Cooperative Highway Research Program

PBCAT	Pedestrian and Bicycle Crash Analysis Tool
PCS	Pedestrian countdown signals
PEDSAFE	Pedestrian Safety Guide and Countermeasure Selection System
PennDOT	Pennsylvania Department of Transportation
PET	Post Encroachment Time
PHB	Pedestrian Hybrid Beacon
RNS	Research Needs Statement
RRFB	Rectangular Rapid Flash Beacon
RSA	Road Safety Audit
SLC	Salt Lake City
TRB	Transportation Research Board
TWLTL	Two-way Left Turn Lane
UDOT	Utah Department of Transportation
UNC HSRC	University of North Carolina Highway Safety Research Center
UTA	Utah Transit Authority
YBP	Yellow Border on Pedestrian Signal

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ABSTRACT

The intent of this project was to build upon the existing foundation of best practices and research to develop user-friendly guidance for transportation practitioners about effective design and operational treatments that can improve pedestrian and bicycle safety at a variety of intersection types. The resulting guidance helps practitioners understand and treat a variety of intersection risk factors influencing pedestrian and bicyclist safety. In this project, the team identified areas where best practices or the effectiveness of treatments is unclear and performed original evaluations to address one key knowledge gap: our understanding of the potential safety benefits of protected intersections.

To better understand the safety impacts of protected intersections, the project team conducted a limited research effort of two protected intersections in the United States (U.S.), each compared to a nearby untreated intersection with similar characteristics. The study used observations of road user behavior to investigate whether protected intersection design could potentially reduce crash risk or injury severity for crashes involving motorists turning right or left across the path of parallel pedestrians and bicyclists, which are among the most frequent intersection crash types. The findings show that, for the intersections studied, the protected intersection was associated with some potential safety benefits (more bicyclists making two-stage lefts and lower motorist right turn speed) and some indications that they may be less safe than the comparison intersections (protected intersection had lower bicyclists and pedestrian signal compliance and a lower percentage of left turning vehicles yielding to bicyclists). The results also show the need for further research on protected intersections to understand what elements are most effective and in what conditions they are best suited.

SUMMARY

In recent years, people walking and bicycling have made up almost 20 percent of all traffic fatalities on roadways in the U.S., despite making fewer than 4 percent of all commute trips (NHTSA 2018; U.S. Census Bureau 2018). The rate of motor-vehicle crashes has reached historic lows in recent years, but this incidence of pedestrian and bicycle crashes has remained steady and continues to rise in some places. Intersections present a particular safety challenge for many nonmotorized roadway users. Nearly 20 percent of pedestrian fatalities and 30 percent of bicyclist fatalities occur at intersections, and an even larger proportion of non-fatal injury collisions occur at these locations. Many common pedestrian and bicycle crash types involve conflicts between pedestrians, bicyclists, or drivers attempting to turn at or traverse through an intersection. While some crashes can be attributed to inappropriate behaviors (such as failure to yield to other road users or failure to adhere to the traffic signal), many could also be attributed to a lack of facilities or appropriate design or signal treatments to safely accommodate all road users at these critical junctures.

As pedestrian and bicycle crashes and injuries continue to persist, state and local transportation practitioners need guidance on what could be done. Intersection design guidance to proactively prevent pedestrian and bicycle crashes and reduce the severity of injuries that occur is limited in the U.S., in part due to a traditional focus on vehicle traffic flow efficiency. The problem of intersection safety for nonmotorized users is particularly complex due to the wide range of intersection configurations (e.g., number and types of lanes, signals) and mix of road users—and therefore potential crash types—that exist in a full range of land uses. Thus, guidance is sorely needed that can help practitioners discern the best treatment according to the particular safety issue and context.

The intent of this project was to build upon the existing foundation of best practices and research to develop user-friendly guidance for transportation practitioners about effective design and operational treatments that can improve pedestrian and bicycle safety at intersections. In this project, the team identified areas where best practices or the effectiveness of treatments is unclear and performed original evaluations to address one key knowledge gap: our understanding of the potential safety benefits of protected intersections. Using the research documented in this report, the team developed guidance on how agencies can analyze the safety and effectiveness of treatments and evaluate trade-offs in the process of project decision-making and monitoring. The guidance is documented in the 15-63 Guidebook (hereafter referred to as “the Guide”): *Guidance to Improve Pedestrian and Bicyclist Safety at Intersections*. The research and guidance produced in this project will help enable transportation agencies to improve their processes for managing safety risks for all road users at intersections.

Research tasks needed to develop the Guide are documented in the five chapters of this report:

- Chapter 1 documents current literature on pedestrian and bicycle intersection safety countermeasures
- Chapter 2 summarizes interview practitioners to understand guidance needs
- Chapter 3 details common pedestrian and bicycle crash types at intersections
- Chapter 4 identifies and prioritizes gaps in our knowledge of pedestrian and bicycle countermeasure effectiveness
- Chapter 5 presents the original research conducted to gain understanding to inform one such research gap: protected intersections

Major research findings from each research task are summarized below.

Literature Review

The literature review of pedestrian and bicyclist safety countermeasure effectiveness is documented in **Chapter 1**. The team explored existing syntheses of literature as well as more recently completed studies, based on a scan of a wide range of sources. We found that while some treatments already have at least one study evaluating safety effectiveness, most treatments lack sufficient investigation on their safety benefits. Even those treatments for which crash modification factors (CMFs) have been estimated often need additional study concerning under what conditions they are most effective. Our team identified countermeasures where little existing research, or only some completed studies, revealed a gap in understanding regarding the safety effects that might be expected from proper application of those countermeasures.

Interviews with Practitioners

As documented in **Chapter 2**, the team interviewed practitioners from state, local, and regional agencies to better understand the needs of transportation professionals with respect to guidance on intersection safety countermeasures. In those interviews, the project team discovered that agencies around the country desire additional research and guidance on pedestrian and bicycle safety at intersections and have specific interest in the following treatments: Pedestrian Hybrid Beacons (PHBs/HAWK), rectangular rapid flashing beacons (RRFB), leading pedestrian intervals (LPI), protected intersections, protected left turn phasing, crossing treatments at freeway ramps, marked crosswalks, right-turn-on-red, and intersection curb radius as well as guidance on pedestrian refuge islands. Interviewees desired guidance on how to address right and left turn crashes, mid-block crashes, and speed and distraction related crashes. Cities expressed specific interest in traffic calming measures and more systemic city-wide safety improvements. Counties expressed interest in suburban conditions including understanding the impacts of channelized right-turn lanes and raised crosswalks. States are interested in Highway Safety Improvement Program (HSIP) funding and crash-based safety analysis to identify locations to treat.

Crash Types Assessment

As documented in **Chapter 3**, the team assessed common crash types involving pedestrians and bicyclists at intersection locations using state, city, and national-level data. Based on the analysis of crash types in three crash databases coded using the Pedestrian and Bicycle Crash Analysis Tool (PBCAT), the team identified common crash types which we recommend as high priority for this project.

A majority of bicycle and pedestrian fatalities involved through movements with one or the other party failing to yield or comply with traffic controls. Left turn crash types were more common than right-turning types for both pedestrian and bicyclist fatalities. The analysis also highlighted that intersection and intersection-related pedestrian and bicyclist fatalities mostly occurred at four-way intersections in urban areas and involved motorists that were initially traveling on two-way undivided roadways.

When the pedestrian failed to yield, the most frequent pedestrian crash type by far, the fatally-injured pedestrians were most often struck while attempting to cross a road divided by a median of some type (including painted medians). While the highest percent (46 percent) of intersection or intersection-related pedestrian fatalities were at signalized intersections, the highest percent (48 percent) of bicyclist intersection or intersection-related fatalities were at unsignalized intersections. In crashes where neither party faced traffic control, fatally-injured bicyclists were also commonly struck at intersections with median-divided roads.

Common conditions for bicycle intersection crashes were daylight, whereas pedestrian fatalities were more common in dark but lighted conditions. Common speed limits were 30 to 35 miles per hour (mph) for both bicycle and pedestrian intersection and intersection-related crashes.

Gaps in Knowledge

Considering the inputs from the literature review, practitioner interviews and assessment of crash types, the team developed a series of tables to demonstrate overlap between gaps in the literature, priorities gleaned from practitioner interviews, and the ability of countermeasures to address common crash types. These tables, found in **Chapter 4**, present countermeasures that address pedestrian and bicyclist safety at signalized locations, unsignalized locations, and both types of locations.

Using the tables, the team identified a list of five high-priority treatments for future study based on whether the treatment has not been previously studied, addresses high priority crash types, and was of interest to interviewees:

- Curb extensions and curb radius reduction
- Four-way stop
- Pedestrian Hybrid Beacon for bicycles
- Protected intersections
- Speed management

The project team and panel acknowledge early in the process that only one of these could be studied with the existing research budget. The NCHRP panel selected protected intersections due to the lack of study of this intersection type in the U.S. in current literature, interest from practitioners, and its potential to address some prominent crash types. At the time of the project initiation only 10 protected intersections had been installed in the U.S., making it an important intersection type to study and evaluate as its use grows. Furthermore, there are specific components of the protected intersection that have the potential for improving safety. The panel saw value in understanding the safety performance of the protected intersection design as a whole, as well as some of the benefits of the individual components of the design.

Protected Intersection Analysis

To better understand the safety impacts of protected intersections, the project team conducted a limited research effort of two protected intersections in the U.S., each compared to a nearby untreated intersection with similar characteristics. The study, documented in **Chapter 5**, used observations of road user behavior to investigate whether protected intersection design reduces crash risk or injury severity for crashes involving motorists turning right or left across the path of parallel pedestrians and bicyclists, which are among the most frequent intersection crash types.

Due to the low numbers of interactions at both treated and comparison sites, only a very dramatic change in behavior between comparison and the treated intersection could be measured in this study. No such dramatic difference was consistently observed, and the results are mostly equivocal.

However, some observations can be made from this analysis:

- In general, pedestrian and bicyclist signal compliance was better at comparison sites than at treatment sites in both jurisdictions. This may be because those walking and bicycling felt more comfortable at the treated intersection, and thus felt less need to comply with the signal.

- The percent of left turning motorists yielding to bicyclists was lower at the protected intersections studied than at the comparison.
- The protected intersections studied saw a higher percent of bicyclists making two-stage left turns than at their respective comparison intersections.
- Pedestrians waited in the pedestrian crossing island of the protected intersection roughly half the time in Salt Lake City, but much less often at the Berkeley protected intersection.
- Motor vehicle right-turning speeds were significantly lower at the Salt Lake City protected intersection studied than at the comparison intersections.

These findings show that, for the intersections studied, the protected intersection was associated with some potential safety benefits (more bicyclists making two-stage lefts and lower motorist right turn speed) and some indications that they may be less safe than the comparison intersections (protected intersection had lower bicyclists and pedestrian signal compliance and lower percent of left turning vehicles yielding to bicyclists).

The results of the research do not reject the hypothesis that the protected intersections studied improve safety for pedestrians and bicyclists, neither do they prove it. When considering the statistically significant results from this research report along with findings from San Francisco and international studies of protected intersections (San Francisco Municipal Transportation Agency 2018, Kidholm et al. 2017), we conclude that protected intersections have potential to reduce motorist right turns speeds and as a result have promise as a bicyclist safety countermeasure in the United States.

The findings provide some evidence of potential safety benefits from protected intersections but also some evidence of the potential for reduced safety from lower pedestrian and bicyclist compliance with signals. The results also show the need for further research on protected intersections to understand what elements are most effective and in what conditions they are best suited.

More research is needed to understand and better quantify the benefits of protected intersections and other pedestrian and bicycle intersection safety treatments.

CHAPTER 1: Review of Literature on Pedestrian and Bicyclist Intersection Treatments

In this Chapter, we summarize existing literature on pedestrian and bicycle intersection treatments currently used in the U.S., distinguishing the treatments and their usefulness by intersection type: signalized, unsignalized, or both. For each treatment, we first describe the treatment and summarize the state of the practice and relevant literature. The treatments studied in this memo are listed in Table 1. A detailed summary of literature review findings is available in Appendix A.

The purpose of this review is not to be an exhaustive list of all research on a treatment, but instead to summarize the state of the research in order to identify research gaps. In this review we summarize the past reviews, highlight key findings from past reviews, and focus on research published in the past two years that would not have been included in past reviews, such as NCHRP Synthesis 498. This section concludes with a table of the treatments and a summary of the crash types treated, safety benefits, and research gaps remaining for each.

Safety benefits described include crash reduction (all modes), yielding, conflicts, speed reduction, and other observed benefits. Studies that result in crash modification factors (CMFs) are reported first since these give some of the best quantification of safety benefits. However, not all CMFs are created equal, and it should be noted that CMFs are rated by the CMF Clearinghouse (www.cmfclearinghouse.org) using a five-star system with five stars being the most reliable. Due to small sample sizes, relatively few crashes, crash reporting issues, infrequent use of treatments, lack of exposure data especially before treatment installation and other data problems, there are few pedestrian and no bicycle treatment CMFs with over 3 stars. Despite these limitations, CMFs are reported, indicating the best of our knowledge on the safety impact of a given treatment. Studies that do not include CMFs are discussed following CMF studies.

The review draws heavily from previous literature reviews listed below:

- NCHRP Synthesis 498: Application of Pedestrian Crossing Treatments for Streets and Highways (Thomas et al. 2016)
- Pedestrian and Bicycle Safety Guide and Countermeasure Selection Systems (PEDSAFE and BIKESAFE) (Zegeer et al. 2013; Sundstrom and Nabors 2014)
- FHWA Separated Bike Lane Planning and Design Guide 2015 (Federal Highway Administration 2015)
- MassDOT Separated Bike Lane Planning and Design Guide 2015 (MassDOT 2015)
- NCHRP Report 672, Roundabouts: An Informational Guide, 2nd Edition (Rodegerdts et al. 2010)
- NCHRP Report 674: Crossing Solutions at Roundabouts and Channelized Turn Lanes for Pedestrians with Vision Disabilities (Schroeder et al. 2011)
- NCHRP Report 659: Guide for the Geometric Design of Driveways (Gattis et al. 2010)
- Development of Crash Modification Factors for Uncontrolled Pedestrian Crossing Treatments. Pre-publication draft of NCHRP Research Report 841 (Zegeer et al. 2016)
- City of Portland Bicycle Plan for 2030 (Portland Bureau of Transportation 2010)
- NACTO Urban Bikeway Design Guide Second Edition 2014 (NACTO 2014)
- Designing Walkable Urban Thoroughfares: A Context Sensitive Approach (Institute of Transportation Engineers 2010)
- Smart Transportation Guidebook (PennDOT and New Jersey DOT 2008)

Table 1. List of Treatments Studied

Intersection Type	Treatment Type	Modes Treated
Signalized	Barnes Dance	Pedestrian
	Bicycle Signals	Bicycle
	Bike Box*	Bicycle
	Flashing Yellow Arrow	Pedestrian
	Leading Bicycle Interval (LBI)	Bicycle
	Leading Pedestrian Interval (LPI)*	Pedestrian
	No Turn on Red Signs	Both
	Passive Detection – Bicycle	Bicycle
	Passive Detection – Pedestrian	Pedestrian
	Pedestrian Countdown Signals*	Pedestrian
	Protected Intersections	Bicycle
	Protected Phases	Both
	Signal Timing (and effects of Delay)	Both
	Two Stage Bicycle Turn Queue Boxes	Bicycle
Unsignalized	Driveway Treatments	Pedestrian
	Four-Way Stop	Pedestrian
	In-Pavement Warning Lights	Pedestrian
	In-Road Signs	Both
	Interchange Treatments	Both
	Median Islands	Both
	Pedestrian Hybrid Beacon	Both
	Rectangular Rapid Flash Beacon	Both
	Roundabouts	Both
Both	Access Management	Both
	Advance Stop/Yield Lines	Both
	Colored Pavement Markings Through Intersections	Bicycle
	Crosswalk Marking	Pedestrian
	Curb Extension	Both
	Curb Radius Reduction	Pedestrian
	Gateway Treatment	Pedestrian
	Lighting	Both
	Mixing Zone Treatments	Bicycle
	Number of Through Lanes and Turn Lanes	Both
	Parking Restrictions	Pedestrian
	Pedestrian Crosswalk Yellow Warning Sign	Pedestrian
	Raised Crossing	Both
	Speed Management	Both

*Treatment may have a positive safety impact on the other mode as well.

The research team created a list of the prominent pedestrian and bicycle treatments used in the U.S. We recognize that the list is not exhaustive, but have tried to include all major treatments and all treatments that would be of interest for future research. Some recently developed treatments are not included because we chose to focus on treatments that are farther along in their development and use. While many pedestrian and bicycle intersection treatments have been investigated internationally, this review focuses on treatments that have been implemented in the U.S. in order to maximize the applicability to

U.S. practitioners. In some cases, research conducted internationally is included in the literature review on a given treatment.

Design principles are important to pedestrian and bicyclist safety at intersections. For example, roadway designs that prioritize motor vehicle mobility over pedestrian and bicyclist safety create unsafe intersections for people walking and bicycling (limited access highway ramps, diverging diamond interchanges, etc.). Similarly, geometric design and corridor signal timing that result in or allow high speed motor vehicle traffic can exacerbate the problems we seek to address in this project. Alternative design principles, such as limiting the tangent length of horizontal curves to less than 400 feet (less than 250 feet was used in one town in the Netherlands (Foletta 2014)), building in curves and narrowing roadways in urban areas, and designing corridor signal timing to slow motorists to a safe speed in urban areas, can address pedestrian and bicyclist intersection safety at the roadway design level. These principles slow motor vehicle speeds, which has been shown to reduce total crashes and is likely to also improve safety of those who walk and bicycle (Elvik and Vaa 2004).

Guidance on such fundamental principles are included in the Guide. However, this literature review focuses on treatments that can be implemented at a given intersection, instead of those that are best applied on the corridor level.

Also along the lines of design principles, it may appear that some of the treatments are very basic, such as provision of signalized intersections, sidewalks, and pedestrian signals. These are included because, unfortunately, many communities do not consistently provide them where they are needed to enable pedestrian and bicycle travel. For example, a study in Copenhagen found that after signalizing 54 intersections, the intersections saw a 30 percent reduction in crashes involving bicyclists and 16 percent reduction in crashes involving pedestrians (Jensen 2010). These basic facilities can be treatments in themselves, although due to the need to coordinate with warrants or transportation policies, we do not focus on these explicitly in our review; rather, guidance on these basics of pedestrian and bicycle intersection design are provided in the Guide.

In addition to the work presented in this literature review, there are many ongoing research projects that relate to bicycle and pedestrian safety. Below is a list of such projects that the team is aware of and feels are most relevant to pedestrian and bicycle intersection safety:

- FHWA Development of Crash Modification Factors (DCMF) Program Task B5 is evaluating LPIs and protected left turn phasing for pedestrian safety. Because data collection and some analysis for this project are being conducted by project team members, there is a good chance for collaboration with this project.
- FHWA DCMF Program Task A5 is evaluating pedestrian countdown signals in three cities: Toronto, Philadelphia, and Charlotte.
- NCHRP 07-25 Guide for Pedestrian and Bicycle Safety at Alternative Intersections and Interchanges
- NCHRP 17-84 Pedestrian and Bicycle Safety Performance Functions for the Highway Safety Manual
- Addressing Bicycle-Vehicle Conflicts with Alternate Signal Control Strategies (Portland State University)
- Contextual Guidance at Intersections for Protected Bicycle Lanes (Portland State University)
- Effectiveness of Bicycle Signals for Improving Safety and Multimodal Mobility at Urban Intersections (Western Michigan University)

CHAPTER 2: Summary of Interviews with Practitioners

This section is a summary of the interviews conducted with several cities, states, and counties on treatments for pedestrian and bicycle crossings at intersections. The research team interviewed eight city agencies, four state Departments of Transportation (DOTs), and three counties, as listed in Table 2.

Table 2. Interviewees

City/Region/State	Scale	Agency	Person(s) Interviewed, Position
Alaska	State	Alaska DOT	Scott Thomas, Regional traffic and safety Engineer
Austin, TX	Large City	Austin Department of Public Works	Nathan Wilkes, Senior Active Transportation Designer
Boston, MA	Large City	Boston Transportation Department	Charlotte Fleetwood, Planner
California	State	Caltrans	Tammy Quigley, Transportation Planner
Coral Gables, FL	Small/Rural City	Department of Public Works	Jessica Keller, Assistant Director of Public Works
Fairfax County, VA	County	Fairfax County DOT	Adam Lind, Bicycle Program Coordinator
Florence, SC	Small/Rural City	City of Florence	Clint Moore, Project Manager
Florida	State	Florida DOT	Mary O'Brien, State Bicycle/Pedestrian Coordinator
Hennepin County, MN	County	Hennepin County DOT	Robert Byers, Engineer
Minnesota	State	Minnesota DOT	Melissa Barnes, Pedestrian and Bicycle Safety Engineer Office of Traffic Safety
Overland Park, KS	Medium City	City of Overland	Brian Shields, Engineer
Philadelphia, PA	Very Large City	Office of Transportation and Infrastructure Systems	Kasim Ali, Traffic Engineer Kelly Yemen, Director of Complete Streets Jeanette Brugger, Bicycle and Pedestrian Coordinator
Seattle, WA	Large City	Seattle DOT	Dongho Chang, Engineer
Shasta County, CA (City of Redding)	County	City of Redding	Chuck Aukland Melissa Estrada, Transportation Planner
Tucson, AZ	Medium City	City of Tucson	Ann Chanecka, Bicycle and Pedestrian Coordinator Dion Schwartz, Traffic Engineer

The information gathered from the interviews provides insight into how agencies and jurisdictions identify safety challenges at intersections for pedestrians and bicyclists and how they evaluate treatments

and identify countermeasures to implement. The interviews also focused on understanding where agencies lacked guidance on countermeasures. The following sections summarize overall themes identified from the interviews.

2.1 General Themes

2.1.1 IDENTIFYING ISSUES AT INTERSECTIONS

Many agencies described their process for identifying safety issues at intersections as reactive based on public input or crash data that they analyze on an annual basis. The current practice for many agencies is largely based on public complaints of feeling unsafe or frequent “near misses.” Public complaints identify issues at specific locations and, in some places, can trigger a crash analysis along with field observations to evaluate possible countermeasures.

Additionally, many agencies also conduct annual crash analyses to rank and prioritize intersections. These agencies evaluate crashes over a three- to five-year period or require a certain threshold of crashes to be met before certain countermeasures are considered. In some cases, a fatality automatically triggers a review, but does not necessarily lead to treatments or improvements.

The adoption of Vision Zero programs and policies is growing in the U.S., although still in the early stages of development for many agencies. Where adopted, Vision Zero has become a mechanism for agencies to approach safety at intersections more systematically. Agencies that have adopted Vision Zero policies have started to dedicate staff, either partially or entirely, to developing a proactive plan to evaluate safety issues and more systematically apply countermeasures. Some agencies noted that there has been a more “corridor-wide” approach to identifying safety challenges and countermeasures, in contrast to examining the intersections in an isolated manner.

2.1.2 COUNTERMEASURE CONSIDERATIONS

While many agencies identified a desire to “push the envelope” and try newer treatments, it is often easiest to implement treatments that are “tried and true” per the Manual of Uniform Traffic Control Devices (MUTCD) standards or American Association of State Highway and Transportation Officials (AASHTO) guidelines (FHWA 2012, AASHTO 2018). These documents serve as the universal resources for design guidance considered safe and acceptable. Agencies also have their own toolboxes that serve as a supplement to the MUTCD and AASHTO guides (FHWA 2012, AASHTO 2004, AASHTO 2018, AASHTO 2019 Forthcoming). The MUTCD warrant analyses are used when evaluating intersections and the impacts of adding a signal or stop control.

Many agencies cited FHWA’s Safety Effects of Marked Versus Unmarked Crosswalks at Uncontrolled Locations (“the Zegeer study”) for guidance on crosswalks at uncontrolled intersections. This study provides recommendations for some but not all pedestrian crossing situations. The authors advise that installing crosswalks at uncontrolled intersections on roadways with more than 12,000 average daily traffic (ADT) supplemented with additional treatments such as narrowing roads and providing medians or crossing islands, traffic calming treatments, traffic signals, and pedestrian signals when warranted. Note that recent research provides additional guidance for uncontrolled pedestrian crossings (Blackburn 2018).

Though recent research has documented safety benefits of advance stop-yield lines and signs, median islands, pedestrian hybrid beacons, and rectangular rapid flashing beacons at these types of locations, many agencies still struggle to provide safe crossing treatments across major arterials. Identifying which treatments should be used under what speed, traffic, number of lanes, and other conditions is a challenge for some agencies.

Additionally, many agencies spoke of the need for “lighter, quicker, more flexible” countermeasures that can be combined with existing projects such as roadway repaving projects. There was a clear desire for additional guidance about how to use temporary materials to create barriers, channelizations, etc. in lieu of constructing expensive concrete countermeasures – at least for the first phase of a phased approach.

Finally, many agencies use or are held to thresholds for determining whether a safety concern is “real” or “perceived.” For instance, for speeding to be considered an issue, the 85th percentile speed must exceed the speed limit by 10 mph. Coarse thresholds like these limit the ability of agencies to respond to perceived safety and address walkability and bikeability; even when incidents have occurred, these thresholds can be limiting within the context of constrained resources.

2.1.3 ADDITIONAL RESEARCH FOR COUNTERMEASURES

A number of countermeasures were identified as promising, but in need of further research. The countermeasures identified by most of the interviewees are summarized in Table 3. The table includes each countermeasure and the additional research and/or guidance that interviewees recommended.

Table 3. Countermeasures Identified for Additional Research

Countermeasure	Recommended Additional Research/Guidance
Pedestrian Hybrid Beacon (HAWK)	● Effectiveness under different conditions ● Clarity about the conditions under which they are appropriate for use
Rectangular Rapid Flash Beacon (RRFB)	● Verification of how effective they are in reducing crashes and in which contexts they are most effective ● Additional research on when they are effective alone or in concert with other treatments ● Guidance about the balance between use and overuse
Leading Pedestrian Interval (LPI)	● Appropriate time for the LPI and its impact on pedestrian safety
Protected Intersections	● Protected intersections in general and how they compare to bicycle lanes and two-stage turn boxes ● Protected intersection design and how the design impacts pedestrians. ● Guidance about how pedestrians and bicyclists interact and what design elements are needed for both at protected intersections
Protected-Permitted Left Turns/Flashing Yellow Arrow (FYA)	● Appropriate use of permitted and protected left turns and the impact on pedestrian safety ● Lagging left turns and the impact on pedestrian safety
Crossing Treatments at Freeway Ramps	● Guidance on when and how to implement crossing treatments at freeway ramps; some agencies will not put a crossing facility at the end of freeway ramps and instead divert pedestrians to cross three legs of an intersection
Marked Crosswalks	● Additional research on crosswalks and specifically how their presence on different types of roadways affect driver and pedestrian compliance and behavior ● Further understanding of the difference in the safety impacts between standard crosswalks and enhanced crosswalks
Right Turn on Red	● Additional research on the effectiveness of “No Right Turn on Red” in reducing crashes
Intersection Curb Radius	● Before and after effects on traffic operations (particularly for heavy vehicles) and pedestrian safety for curb extensions at intersections

Additional countermeasures that jurisdictions identified for further guidance and research are outlined below. These countermeasures were not as widely identified as the ones in Table 3, but were still prevalent in many of the interview discussions.

- Many agencies identified pedestrian refuges as a typical treatment for pedestrian crossings, but would like better guidance on when they should be used and the impact they have on safety.
- Several agencies also requested additional research on raised crossings, their safety impacts, and when they are appropriate.
- Several agencies highlighted the “blank out” No Right Turn on Red light-emitting diode (LED) sign, a new treatment California and other agencies have been applying for handling right turn on reds. Many wanted to better understand how they were working and whether they would start to be implemented more nationally.
- California has also experimented with a “Yellow Border on Pedestrian Signal” (YBP) as shown in Figure 1. The yellow border turns on when the pedestrian crossing button is pushed and turns off when the pedestrian countdown begins. This is intended to alert drivers of a potential pedestrian crossing and inform pedestrians that they have been detected. The experiment has proved to be a positive enhancement so far, but there is additional research needed to determine how it can be applied nationally. (Stinger 2014)

2.1.4 CRASH TYPES RESEARCH

Agencies were asked to identify whether there are specific pedestrian or bicycle crash types they would like to see studied further. While there was general consensus that the profession would benefit from a more thorough understanding of crash types in general, key crash types identified included:

- Right-turn and left-turn related crashes in general and the countermeasures that reduce and impact those crashes.
- Left-turn crashes at signalized intersections and the impacts protected, permitted, split and lagging phasing treatments have on crashes.
- Crashes at mid-block crosswalks and crosswalks at unsignalized intersections, particularly along high-speed corridors.
- Crashes related to inattention, both with the fault of pedestrian and drivers. In particular, how does education impact these crashes?
- Speed-related crashes on arterials and collectors.

2.1.5 ADDITIONAL GUIDANCE

Agencies were asked to identify types of guidance that would be helpful to their practice. In many cases, agencies stressed the importance of the guidance being (1) nationally accepted by the engineering community and (2) supported by strong research. Types of guidance identified by agencies included:

- Guidance on short- and long-term cost-benefit trade-offs for countermeasures.
- Specific guidance related to signal timing and best practices in signal timing when considering pedestrian and bicycle safety. There was also an expressed interest in this type of guidance including research on the trade-offs of using certain types of signal timing phases to improve pedestrian and bicycle crossing conditions.
- Relatedly, guidance on when to phase-separate signals for bicyclists; current guidance from the MUTCD Interim Approval (IA-16) is restrictive and impractical in many cases.

- Guidance on talking points with stakeholders regarding the specific impacts and justification for using certain countermeasures – especially ones that are newer or experimental. This could come in the form of a compiled online resource or “statistics library.”
- Guidance on best barriers for protected bicycle lanes and safety impacts of each.
- Guidance on the best design application of separated and protected bicycle lanes at intersections, particularly at constrained intersections or where parking is in high demand.



Figure 1 Yellow LED Border on Pedestrian Signal (Source: Rob Stinger) (Stinger 2014)

2.1.6 SAFETY DATA AVAILABILITY

Agencies indicated during the interview what type of data they had available and whether they would be able to share the data with the team. A summary of the data available by agency is provided in Table 4.

Table 4. Summary of Safety Data Available by Agencies Interviewed

Agency	Data Available (Format)	Contact Person
Alaska DOT	State Crash Data	Jon Knowles, Engineer
Austin Department of Public Works	PBCAT data, Aggregate crash data, Ranking of crash data based on severity	Nathan Wilkes, Active Transportation Designer
Boston Transportation Department	N/A	Charlotte Fleetwood, Planner
Caltrans	Crash data is available at: http://iswitrs.chp.ca.gov/Reports/jsp/userLogin.jsp	Tammy Quigley, Transportation Planner
Coral Gables Department of Public Works	Annual Crash Data	Jessica Keller, Assistant Director of Public Works
Fairfax County DOT	Police Data for the past 5 years	Adam Lind, Bicycle Program Coordinator
City of Florence, SC	Crash Data (GIS)	Clint Moore, Project Manager
Florida DOT	Crash Data	Mary O'Brien, State Bicycle/Pedestrian Coordinator
Hennepin County DOT	Limited Data Examples – Need to check with the state about whole data set	Robert Byers, Engineer
Minnesota DOT	Crash data, 8 Permanent traffic counters and few temporary counters	Melissa Barnes, Engineer
City of Overland Park	2000 Crash Data and 2008 Crash Videos	Brian Shields, Engineer
Philadelphia Office of Transportation and Infrastructure Systems	Crash data is on OpendataPHL, Count data is on DVRPC – for bikes, peds, cars, Ideal places to look at Data: Roosevelt Boulevard, North Broad Street, Market (by City Hall) – they are great examples of pedestrian safety needing to be balanced with high traffic volume through put, Vision Zero efforts should be coming out with new data by September	Kasim Ali, Traffic Engineer Kelly Yemen, Director of Complete Streets Jeanette Brugger, Bicycle and Pedestrian Coordinator
Seattle DOT	Annual Crash Data	Dongho Chang, Engineer
City of Redding	Database of crash data and crash rates	Chuck Aukland Melissa Estrada, Transportation Planner
City of Tucson	Has lots of data on the HAWK – but have not been able to digest it enough to show latent demand, Tucson MPO Data, Have 5+ years of crash data	Ann Chanecka, Bicycle and Pedestrian Coordinator

2.2 City-Specific Themes

While there were general themes noted across all agencies, several city, state, and county specific themes were noted over the interviews. The cities and jurisdictions interviewed appear to have either adopted or are migrating toward Vision Zero as a program/policy that monitors safety performance. This comes in the form of looking at crash data to identify “hot spots” but also looking at public input to evaluate priorities and countermeasures more holistically. The Vision Zero adoption also tackles coupling engineering treatments with improved enforcement and education.

Several cities also indicated they have advisory groups where safety concerns can be discussed with the community. This approach may include community “traffic calming and safety committees” or city-wide safety improvement committees.

Cities also identified a notable trend in moving towards analyzing corridors over intersections. Many agencies feel that most streets require safety concerns to be handled systematically across a corridor versus at an isolated intersection.

2.3 County-Specific Themes

County agencies identified several concerns that were specific to suburban-based contexts. These included further research and guidance on treatments such as:

- Intersection crossings on “ring roads” and suburban arterials (four lane, divided and undivided roads).
- Appropriate treatments for free-right turn or channelized right turns. What are the impacts to removing a free or channelized right turn? Can speed tables or raised crosswalks be placed at channelized right turns to deter speeding?
- Compliance at crosswalks specific to suburban land uses and roadways.

Additionally, counties that were interviewed pointed to their repaving programs as a way to address safety issues in design. As repaving work is done, they look at crash data to identify additional treatments they can do along with repaving.

2.4 State-Specific Themes

In the state agency context, federal programs like the HSIP drive much of the approach to identifying challenges and potential countermeasures. This is particularly true when funding sources to address these issues are dependent on federal funds. Many states have also utilized crash data on an annual basis to find “hot spot” crash locations along corridors and at intersections. This type of crash analysis usually leads to a routine of identifying priority locations for improvements and subsequently handling a location-specific study for further analysis on possible improvements.

The National Association of City Transportation Officials (NACTO) has identified that states and cities often have different goals and objectives related to use of HSIP funds. There are cases where national standards like the MUTCD conflict with local policies and this results in challenges for agencies trying to deliver projects.

2.5 Summary of Interview Findings

In the interviews, the project team discovered that agencies around the country desire additional research and guidance on pedestrian and bicycle safety at intersections and have specific interest in the following treatments: HAWK (high-intensity activated crosswalk beacon), RRFB, LPI, protected intersections, protected left turn phasing, crossing treatments at freeway ramps, marked crosswalks, right turn on red, and intersection curb radius as well as guidance on pedestrian refuge islands. Interviewees desired guidance on how to address right and left turn crashes, mid-block crashes, and speed and distraction related crashes. Cities expressed specific interest in traffic calming measures and more systemic city-wide safety improvements. Counties expressed interest in suburban conditions including understanding the impacts of channelized right-turn lanes and raised crosswalks. Agencies, especially states, are interested in HSIP funding and crash-based safety analysis to identify locations to treat.

CHAPTER 3: Crash Types Assessment

This chapter describes the research gathered and reviewed related to risk factors and analysis methods. This research informed the development of content provided in Step 3 of the Guide.

To understand common types of crashes involving pedestrians and bicyclists at intersections, we explored existing crash datasets coded with the PBCAT tool. Three existing datasets were used in this preliminary exploration:

- At the national level - the Fatality Analysis and reporting System (FARS) for 2014 and 2015.
- At the state level - the North Carolina (NC) crash database for 2008 through 2014.
- At the city level - the crash database for Boulder, Colorado for 2008 through 2014.

Using the PBCAT tool provided a consistent crash coding framework for all three databases.

Table 6 summarizes crash location characteristics by database, using the PBCAT-derived definitions of whether a crash occurred at an intersection or related to an intersection. For fatalities especially, it is important to note that the crash type coded often relies only on the report of the survivor, usually the driver, not the pedestrian or bicyclist, and this may bias the crash type selection.

Table 5. Number and Percent of Fatalities (FARS) or Crashes occurring at Intersection or Intersection-related Locations (NC and Boulder)

General Location Factors	Data Source			
	Mode	FARS [1] <i>N = 10,723 Pedestrian fatalities [1] N = 2,398 Bicyclist fatalities [2]</i>	NC Total (2008-14) KAB only [3] <i>N = 8,906 Pedestrian crashes N = 3,116 Bicyclist crashes</i>	Boulder (2008-14) <i>N = 381 Pedestrian crashes N = 1,266 Bicyclist crashes</i>
Number and percent of people struck or crashes at intersection or intersection related	Pedestrian	2,954 28%	2,257 25%	228 60%
	Bicycle	902 38%	1485 50%	828 65%

[1] Pedestrians includes persons on personal conveyances such as skateboards and wheelchairs. Two years (2014-2015) of FARS data were used for pedestrian fatalities.

[2] Bicyclists includes persons riding a two-wheeled bicycle (not motorized scooters) or other type of pedal-cycle. Three years (2014-2016) of FARS data were used for bicyclist fatalities.

[3] KAB = Killed, disabling (A), or evident injury (B) severity crash. KAB crashes involve a fatal, disabling, or evident type injury (to one or more pedestrians or bicyclists).

Table 6. Summary of Pedestrian and Bicycle Intersection Crash Characteristics by Database

<i>Intersection and Intersection-Related Top Frequency Characteristics</i>				
Location Characteristics	Mode	<u>FARS</u> (2,954 Pedestrian, 2 years; 902 Bicyclist fatalities, 3 years)	<u>NC KAB [1]</u> (2,257 Pedestrian, 1,485 Bicyclist crashes)	<u>Boulder [2]</u> (228 Bicyclist crashes)
Percent in Urban areas	Pedestrian	89%	85%	All (n/a)
	Bicycle	80%	81%	All (n/a)
Top type of trafficway involved (for motorist's roadway) [3]	Pedestrian	51% 2-way, undivided	66% 2-way, undivided	Not available
	Bicycle	58% 2-way, undivided (not including 2-way, center turn lane)	74% 2-way, undivided (including 2-way center turn lane)	Not available
Top types of Intersection	Pedestrian	62% 4-way	45% no special feature [4] (4-way was 2nd with 29%)	Not available
	Bicycle	59% 4-way	33% 4-way	
Top Traffic Control types (for motorist)	Pedestrian	46% no control for striking motor vehicle 46% traffic signal (with/without ped signals);	39% traffic signal (with/without ped signals); 35% no control at intersection	61% traffic signal
	Bicycle	64% no control for striking motor vehicle	17% no control at intersection	51% traffic signal
Light Conditions	Pedestrian	47% dark, lighted	52% daylight	Not available
	Bicycle	56% daylight	75% daylight	Not available
Speed Limits (for motorist's roadway) [1]	Pedestrian	37%, 30 –35 mph	50%, 30–35 mph	Not available
	Bicycle	26%, 30 –35 mph; 17% ea, 45 and 55 mph roads	48%, 30 – 35 mph	Not available
Top Crash type	Pedestrian	<i>34% Pedestrian Failed to Yield</i>	<i>28% Pedestrian Failed to Yield</i>	<i>32% Pedestrian Failed to Yield</i>
	Bicycle	<i>15% Bicyclist Ride Through – Signalized Intersection</i>	<i>16% Motorist Drive Out – Sign-Controlled Intersection</i>	<i>18% Motorist Drive Out – Sign-Controlled Intersection</i>

[1] NC KAB = Killed, disabling (A), or evident injury (B) severity crash.

[2] Boulder = bicyclist crashes of all severity.

[3] The investigating officer deemed that there were "no special features" (non-intersection) at the crash location, but when the crash was PBCAT crash-typed, the coders determined from review of the crash report, diagram, and narrative that the crash did occur at or related to an intersection. Two-thirds of the "no special feature" crashes were deemed to be "related to an intersection" in NC crash typing – that is, the crash occurred outside the bounds of the intersection + crosswalk area, but within 50 feet of an intersection

[4] The investigating officer deemed that there were "no special features" (non-intersection) at the crash location, but when the crash was PBCAT crash-typed, the coders determined from review of the crash report, diagram, and narrative that the crash did occur at or related to an intersection. Two-thirds of the "no special feature" crashes were deemed to be "related to an intersection" in NC crash typing – that is, the crash occurred outside the bounds of the intersection + crosswalk area, but within 50 feet of an intersection.

3.1 Types of Crashes Involving Pedestrians

Table 7 below shows a summary of the top 10 types of crashes involving pedestrians nationally (FARS data), in North Carolina (statewide data) and in the city of Boulder, CO.

While there are other frameworks for examining events that led up to crashes, the availability of PBCAT crash types at a national, state, and local level provided a helpful way to examine the types of conflicts (which include behavioral elements such as failure to yield or driving or riding out at controlled locations) that led up to crashes. The distributions of crash types vary by severity, location, and design factors as well as by local cultures, enforcement, and populations involved. However, the similarity of top crash types across all three databases suggests that there are common high-frequency crash types at intersections that are associated with fatalities, with injury and fatal crashes (NC), and with all severity pedestrian crashes in a city (Boulder).

Table 7. Top 10 Types of Crashes Involving Fatally Injured Pedestrians and Comparisons*

U.S. Pedestrian Fatalities (FARS 2014-15)			Ped. Fatal, Disabling, and Evident Injury Crashes (NC 2008-14)			Ped. all Severity Crashes NC 2008-14			Ped. all Severity Crashes NC 2008-14			Ped. all Severity Crashes Boulder, CO 2008-14		
All Area Types			All Area Types			Urban (inside Municipalities)			Rural (outside Municipalities)			Urban		
Crash Type	Freq	%	Crash Type	Freq	%	Crash Type	Freq	%	Crash Type	Freq	%	Crash Type	Freq	%
Pedestrian Failed to Yield	1018	34.5	Pedestrian Failed to Yield	620	27.5	Pedestrian Failed to Yield	822	19.7	Pedestrian Failed to Yield	136	22.9	Motorist Left Turn - Parallel Paths	72	31.6
Motorist Failed to Yield	427	14.5	Motorist Left Turn - Parallel Paths	363	16.1	Motorist Left Turn - Parallel Paths	791	19.0	Motorist Left Turn - Parallel Paths	63	10.6	Dash	36	15.8
At Intersection - Other / Unknown	344	11.6	Dash	237	10.5	Motorist Failed to Yield	389	9.3	Walking Along Rdway With Traffic - From Behind	43	7.3	Motorist Failed to Yield	26	11.4
Motorist Left Turn - Parallel Paths	309	10.5	Motorist Failed to Yield	123	5.4	Dash	321	7.7	Dash	36	6.1	Motorist Right Turn - Parallel Paths	26	11.4
Dash	235	8.0	Walking Along Rdway With Traffic - From Behind	74	3.3	Motorist Right Turn - Parallel Paths	177	4.2	Motorist Failed to Yield	28	4.7	Motorist Right Turn - Perpend. Paths	14	6.1
Motorist Right Turn - Parallel Paths	72	2.4	Intersection - Other / Unknown	56	2.5	Motorist Right Turn - Perpend. Paths	174	4.2	Motor Vehicle Loss of Control	25	4.2	Dart-Out	6	2.6
Walking/Running Along Roadway With Traffic - From Behind	54	1.8	Motor Vehicle Loss of Control	54	2.4	At Intersection - Other / Unknown	131	3.1	Walking in Roadway	24	4.0	At Intersection - Other / Unknown	6	2.6
Dart-Out	46	1.6	Motorist Right Turn - Parallel Paths	54	2.4	Walking Along Rdway With Traffic - From Behind	130	3.1	Standing in Roadway	20	3.4	Motorist Right Turn on Red - Perpendicular Paths	10	4.4
Lying in Roadway	44	1.5	Walking in Roadway	51	2.3	Motorist Right Turn on Red - Perpend. Paths	116	2.8	Working in Roadway	20	3.4	Multiple Threat	3	1.3
Vehicle-Vehicle / Object	39	1.3	Play Vehicle-Related	49	2.2	Motor Vehicle Loss of Control	79	1.9	Walking Along Rdway Against Traffic - From Front	19	3.2	Pedestrian Failed to Yield	3	1.3
Subtotal Top 10	2588	87.6	Subtotal Top 10	1681	74.5	Subtotal Top 10	3130	75.1	Subtotal Top 10	414	69.8	Subtotal Top 10	202	88.6
Total	2954		Total	2257		Total	4166		Total	593		Total	228	
*The FARS data count people who died at the scene or within 30 days of being struck. The other data sets count crashes with the different combinations of pedestrian injury severity or location types indicated. Crash types were ranked in terms of frequency. Shading indicates crash types that are in the FARS top 10 and at least one other column.														

Crash types among the top 10 types for pedestrian fatalities nationwide are shaded in the FARS column if they are also found in the top 10 most frequent crash types (shaded cells) in North Carolina (among those killed or receiving disabling or evident injuries), and in Boulder among crashes of all severity (bicycle crashes only).

It is clear by examining the distribution of fatalities nationally, as well as various subsets by injury severity or rural/urban designation in North Carolina and in Boulder, CO, that seven of the top 10 crash types for pedestrian fatalities nationally are important in all of these datasets. See Table 8 for these distributions of top 10 ranked crash types.

Among the top 10, five pedestrian crash types for pedestrian fatalities nationwide merit further study and examination of available countermeasures in phase 2. These include:

Pedestrian Failed to Yield (34 percent) is the top-ranked crash type in all distributions with 34 percent of fatalities being coded as this type (Figure 2). Note that this crash type does not necessarily imply fault, but to the best of coders' judgment upon review of crash diagrams, narrative, and data elements, the pedestrian apparently did not have right-of-way at the time of the crash. Possible contributing factors of this crash type may include that a pedestrian has difficulty selecting a gap in traffic on a multi-lane road with high traffic volumes and high speeds. Although half of these occurred at signalized locations; nationally, only one-third of those were known to have pedestrian signal heads of some type. Furthermore, nearly half (49 percent) of these occurred at intersections lacking either a traffic signal or other types of traffic control for the motorist. Nighttime conditions make it even more difficult for pedestrians to perceive the speed of oncoming traffic and select a safe gap when there are no signals or traffic controls. Fifty-six percent (compared to 48 percent for all types) of these crashes occurred at night on streets indicated to have lighting. However, even with lighting, there may be insufficient or improper lighting for pedestrians crossing to be seen by motorists. Fully 79 percent of this type of fatalities occurred at night. Examination of FARS data for this type showed that the largest number (38 percent) occurred on two-way, undivided facilities, but 45 percent occurred on a road divided by some type of median (Figure 3). Another 14 percent occurred when the motor vehicle was on a road with a continuous two-way center turn lane. Forty-one percent of these fatalities occurred on roads with speed limits of 35 or 40 mph; another 25 percent occurred on roads with speed limits of 45 mph — both of these percentages are higher than for all types of pedestrian crashes.

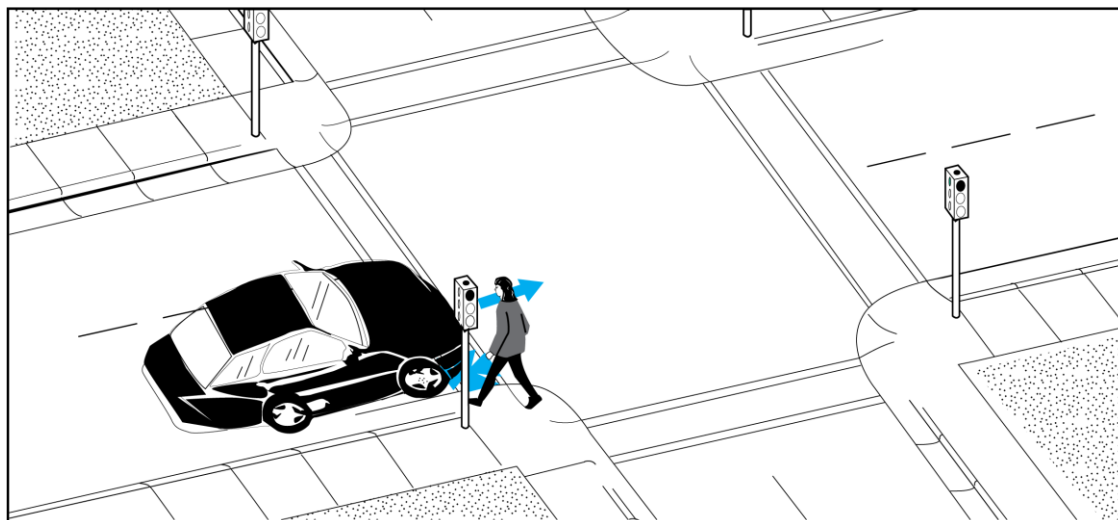


Figure 2. *Pedestrian Failed to Yield* (Source: PBCAT)

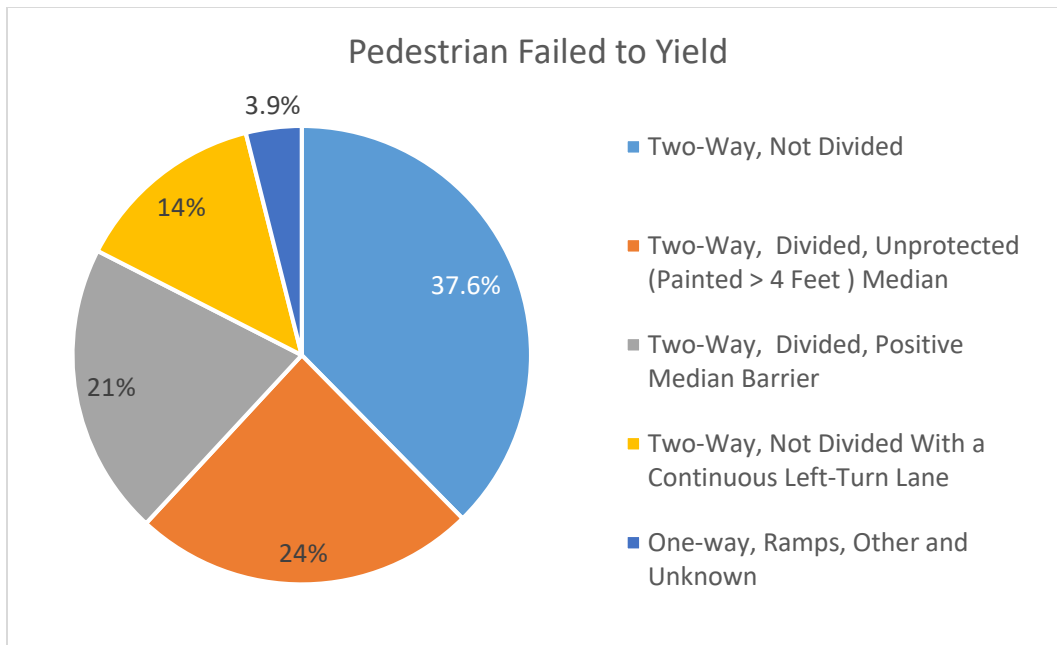


Figure 3. Pedestrian Failed to Yield (Motorist Traveling Straight) Fatalities and Motorist Traffic-Way Type

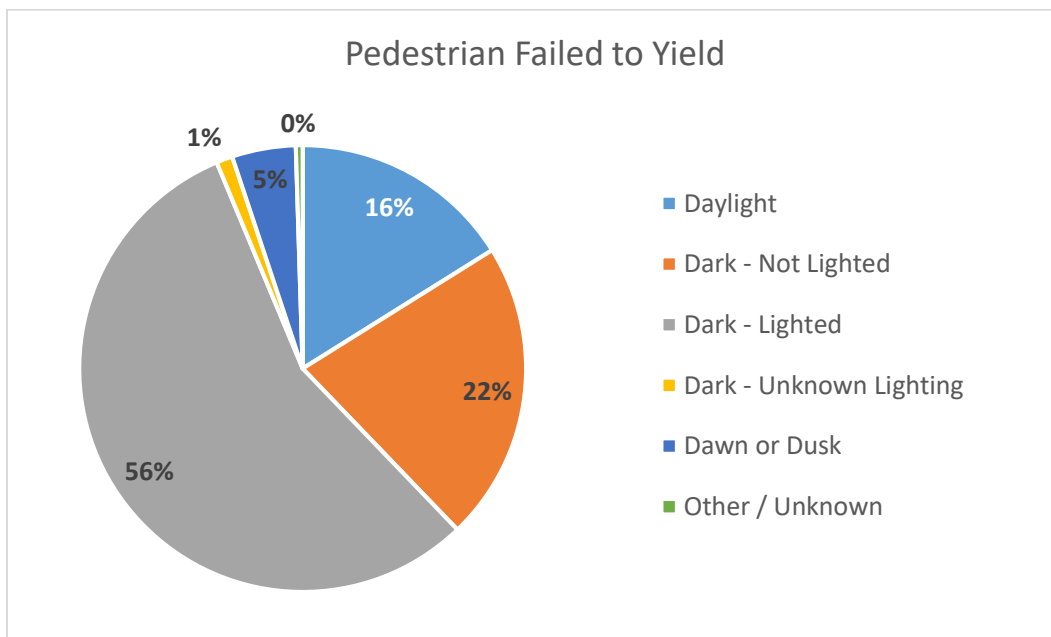


Figure 4. Pedestrian Failed to Yield Fatalities and Light Conditions at Time of Crash

Motorist Failed to Yield (15 percent) does not imply fault but implies that the motor vehicle apparently did not have right-of-way before striking a crossing pedestrian (Figure 5). Motorists may be unwilling to yield at marked crosswalks, due to higher motor vehicle speeds. Visibility or sight-distance as well as motorist distraction may contribute. Motor vehicles may also run stop signs or signals, or drive out after stopping, failing to notice a pedestrian approaching from either the right or left. Examination of the FARS data suggested that the motor vehicle was most often on a two-way, undivided facility (57 percent), most

with no traffic controls at the location, but these crashes also occurred often on two-way roads divided by some type of median (25 percent) and on roads with a continuous two-way left turn lane (15 percent). (Another 12 percent of crashes were indicated to occur at intersections, but it was unknown who failed to yield or there were other circumstances.)

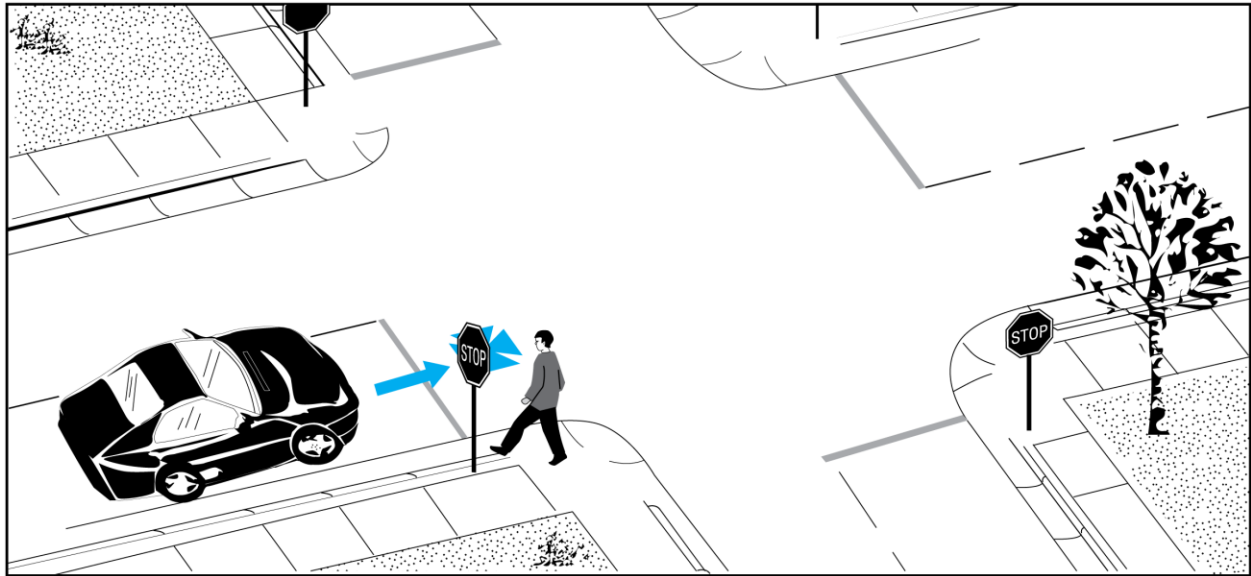


Figure 5. Motorist Failed to Yield to Pedestrian (Source: PBCAT)

Motorist Left Turn – Parallel Paths (10 percent) In the next highest crash type, the motor vehicle was initially traveling on a parallel path with the pedestrian before making a left turn and striking the individual (Figure 6). The pedestrian may have been approaching the intersection from the same or opposite direction as the motor vehicle. Seventy percent of these occurred at signalized intersections, but there is insufficient detail in crash reports to determine more specifics such as whether there was a protected turn phase separate from the pedestrian crossing signal or other treatments. In North Carolina, this crash type was the second most frequent type among injury crashes (killed, disabling injury, or evident injury types), or across rural or urban locations, and was ranked first in the City of Boulder, CO.

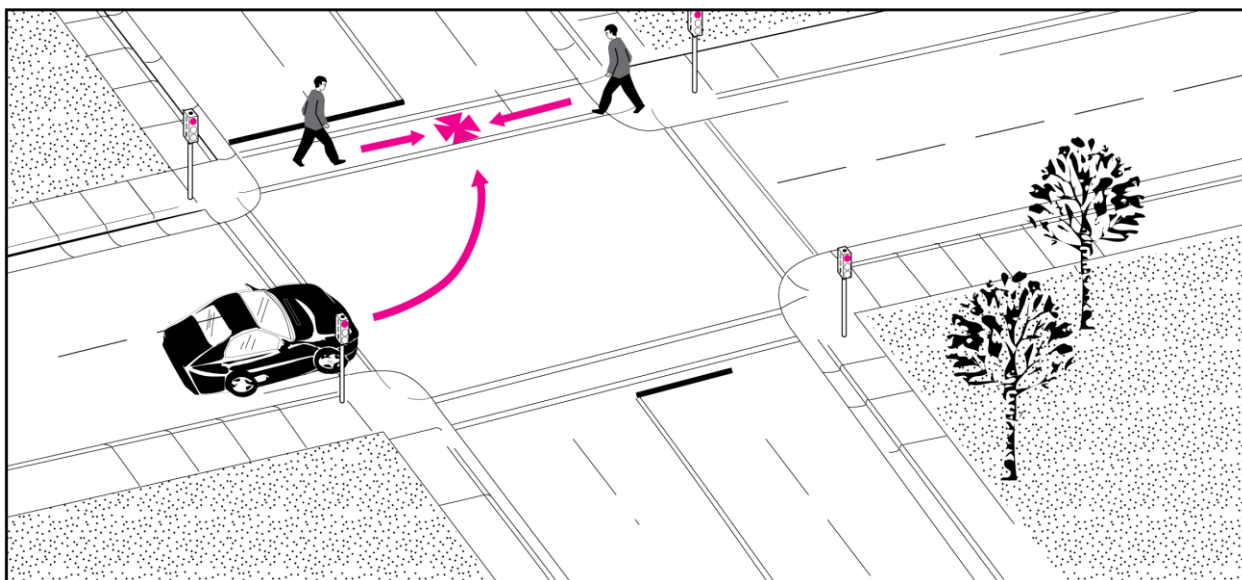


Figure 6. Motorist Left Turn - Parallel Paths (Source: PBCAT)

Pedestrian Dash (8 percent) The pedestrian ran into the roadway and was struck by a vehicle and there is no mention in the crash report material that the motor vehicle's view of the pedestrian was obstructed. In essence, pedestrian Dash crash type is similar to Pedestrian Failed to Yield but suggests that the pedestrian ran suddenly into the street (Figure 7).

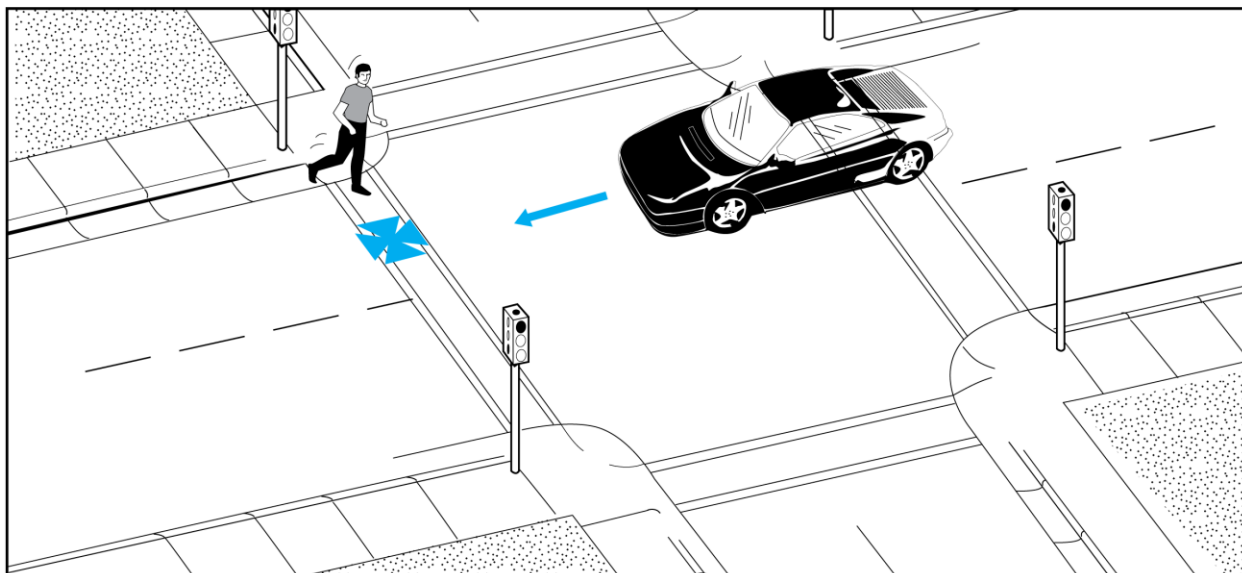


Figure 7. Pedestrian Dash (Source: PBCAT)

One additional intersection related crash type ranked in the top 5 for all crashes (NC and Boulder), but accounted for smaller percentages of fatal (FARS) and injury crashes (NC):

Motorist Right Turn – Parallel Paths (2 percent) The motor vehicle was initially traveling on a parallel path with the pedestrian before making a right turn and striking the individual (Figure 8). The pedestrian may have been approaching the intersection from the same or opposite direction as the motor vehicle turning

right. In this scenario, motorists are often looking left before making a right turn across the path of the pedestrian crossing to the motor vehicles right (the motor vehicles destination leg).

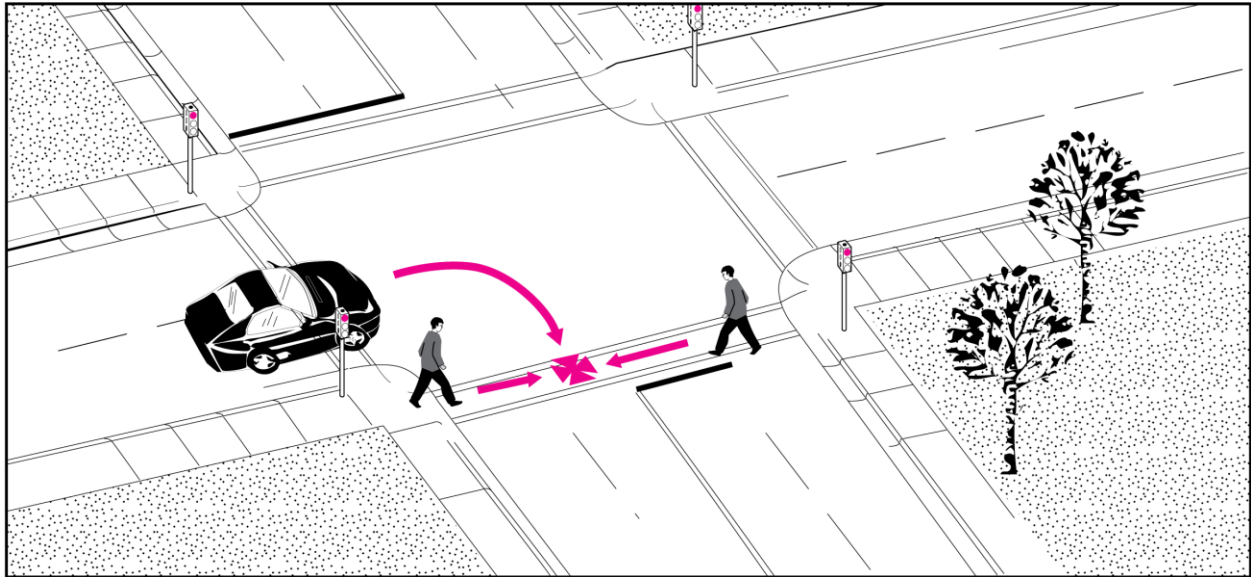


Figure 8. Motorist Right Turn - Parallel Paths, Pedestrians (Source: PBCAT)

The above five pedestrian intersection-related crash types are considered the “high priority” pedestrian-involved crash types for the purposes of this report, since they resulted in the largest portions of fatalities and high proportions of injury crashes in North Carolina, and all-severity pedestrian crashes in Boulder, CO.

Walking/Running Along Roadway With Traffic - From Behind (2 percent) also ranked in the top 10 in FARS but since this crash type is less associated with intersections, it is not included in this discussion. Others are more unusual types – including Vehicle – Vehicle/Object (a motor vehicle strikes another motor vehicle or object which then strikes the pedestrian) and Lying in Roadway, and these types may have little to do with intersection design.

The five pedestrian intersection-related crash types are considered the “high priority” pedestrian-involved crash types for the purposes of this report:

- Pedestrian Failed to Yield
- Motorist Failed to Yield
- Motorist Left Turn – Parallel Paths
- Pedestrian Dash
- Motorist Right Turn – Parallel Paths

3.2 Types of Crashes Involving Bicyclists

The following section includes a summary of the top 10 types of crashes involving bicyclists for bicyclist fatalities for 2014-2016. Data presented are further summarized in Table 8. Bicycle fatalities and crashes are rarer events than pedestrian fatalities and crashes, and the various databases are not as consistent for crashes involving bicyclists as they are for types of crashes involving pedestrians. During the project, 2016 data were added to the FARS database, so in order to have a larger dataset for comparisons and

analysis, we re-analyzed the data to include 2016 bicyclist fatalities. The other datasets, based on more years, were not changed.

Bicyclist Ride Through – Signalized Intersection (15 percent of Bicyclist fatalities nationally) Although not ranked first in any of the other datasets, this top type in FARS is also highly represented (within the top 10 most frequent types) in each of the other datasets. The bicyclist in this scenario rides into a signalized intersection against a red indication without stopping (Figure 9).

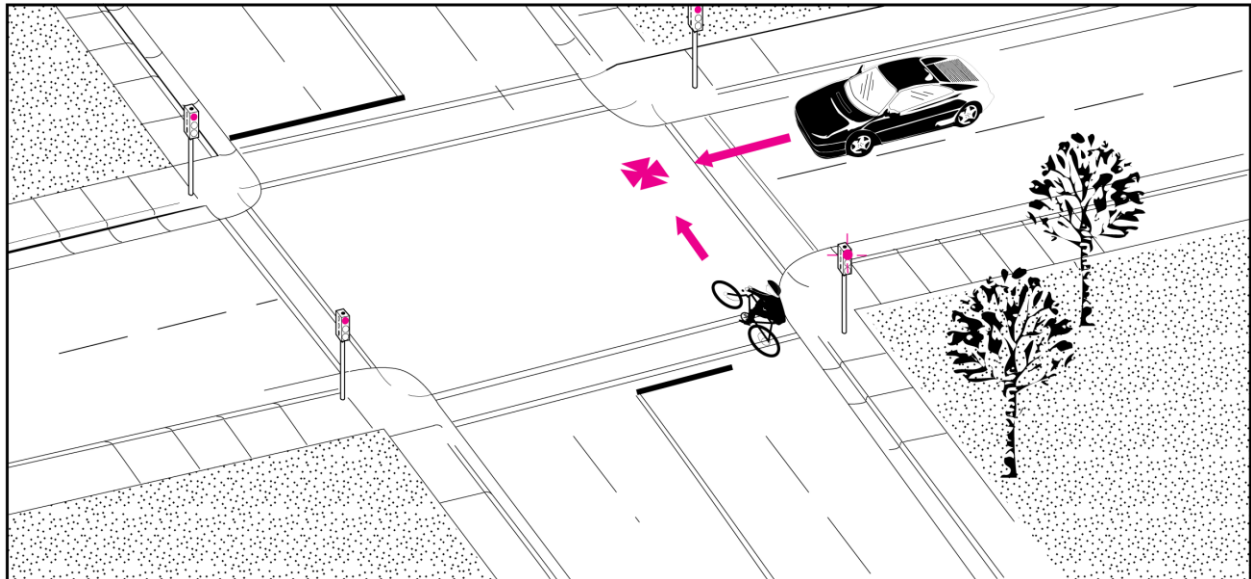


Figure 9. *Bicyclist Ride Through - Signalized Intersection* (Source: PBCAT)

Some of the behaviors mentioned by BIKESAFE that may contribute to this crash type include that the bicyclist may be trying to rush through on an amber indication or fail to see the red signal. The signal may be more difficult to observe if the bicyclist is traveling wrong-way or riding on the sidewalk or on a two-way cycle path against traffic. Data from FARS (Figure 10) suggest that one or both of these factors are often present in these types of fatal crashes. However, none of the fatalities was documented to involve a bicyclist in a bike lane or separated bikeway prior to the crash. The bicyclist may also not want to interrupt momentum or stop for a signal with an excessively long delay or one that does not detect bicyclists' presence. Inexperience could also contribute to this type of crash.

A plurality of these bicyclist fatalities (47 percent) occurred when the motorist was traveling on an undivided, two-way facility (Figure 11). However, combined, 45 percent occurred when the motorist was driving on a trafficway divided by a median of some type (protected/raised or unprotected type) suggesting that bicyclists may often be trying to cross larger roads when these fatalities occur. Information on the bicyclist's roadway type was not available.

Bicyclist ride out at signalized intersections was also 10th most frequent (3 percent) among fatal collision types. In this type, the bicyclist may initially stop, but then may attempt to cross if the signal does not appear to be responding, or if the cyclist perceives a gap that they mistakenly deem suitable for crossing. Related types include a bicyclist failing to clear an intersection before a signal change or being trapped in the intersection. These accounted for about 1 percent.

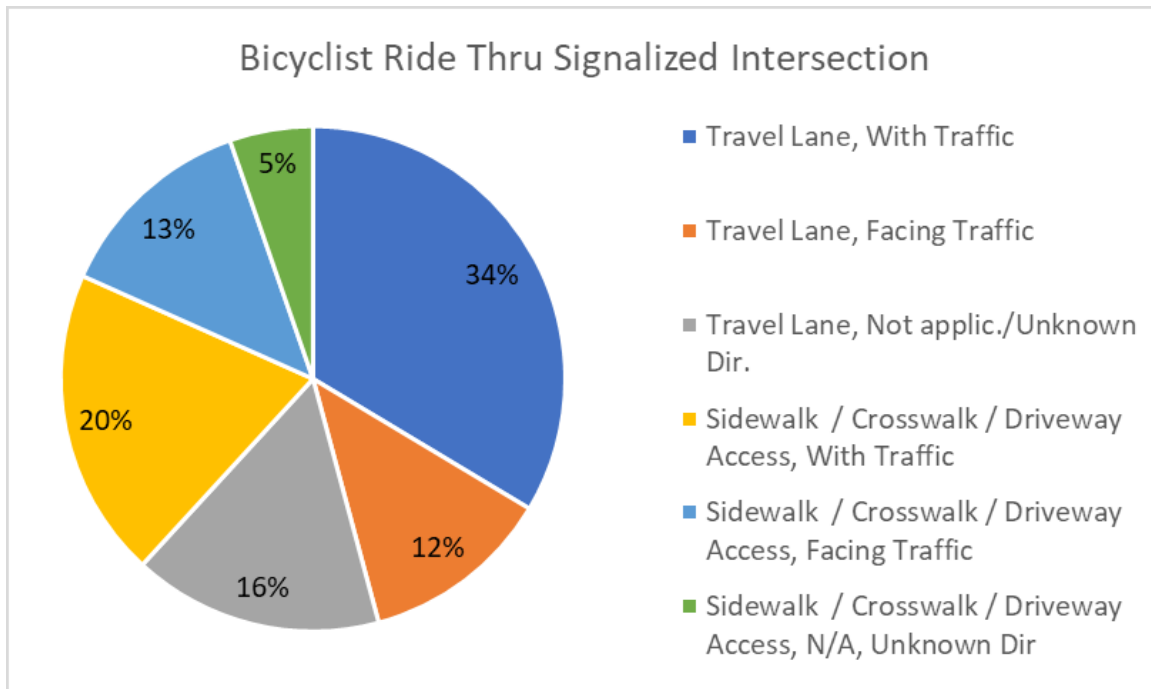


Figure 10. Bicyclist Ride Thru (Red) Signalized Intersection Fatalities (Bicyclist Initial Riding Position and Direction Relative to Motor Traffic)

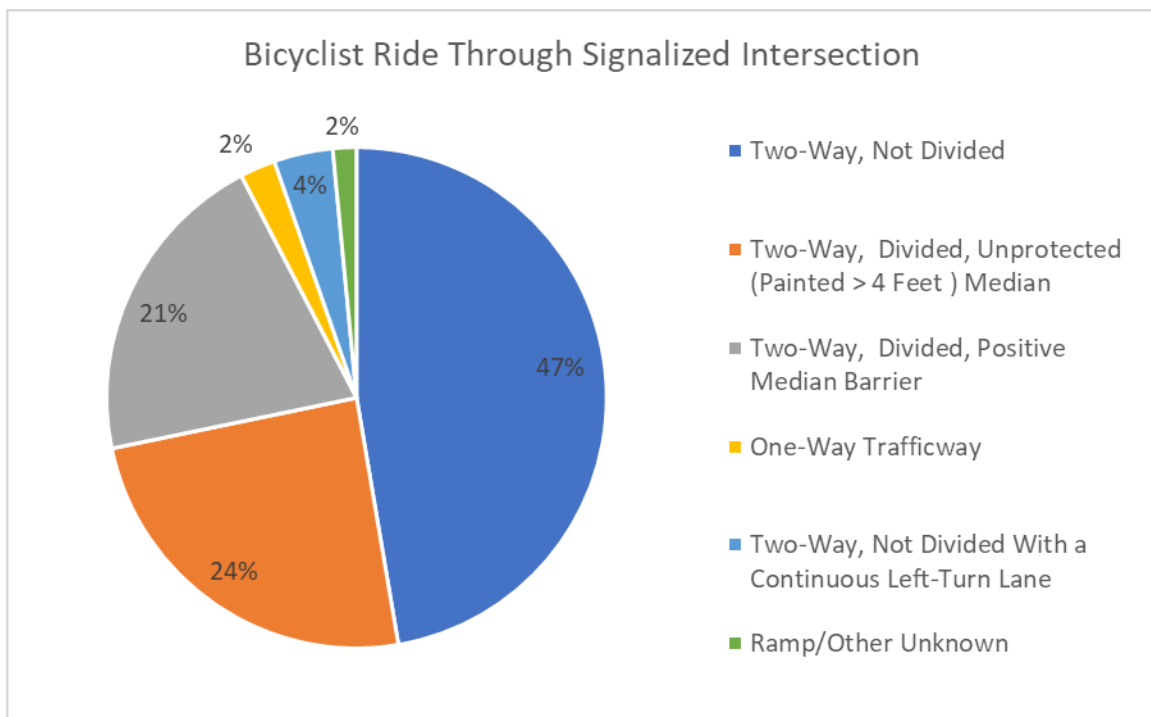


Figure 11. Bicyclist Ride Through Signalized Intersection and Motorist's Trafficway Type

Bicyclist Ride Through – Sign-Controlled Intersection (12 percent, 2nd in the FARS list) and *Bicyclist Ride Out – Sign-Controlled Intersection* (6 percent; 4th in the list) Due to the similarity of these specific types they were combined for further discussion and analysis.

Bicyclist Ride Through or Out – Sign-Controlled Intersection (18 percent) In these two types, the bicyclist enters an intersection and fails to stop or yield at a non-signalized intersection (typically controlled by a stop sign), colliding with a motor vehicle traveling through the intersection (Figure 12). Possible contributing circumstances include a bicyclist not wanting to lose momentum at a stop sign. The bicyclist may also ride out after slowing or stopping and underestimate the time needed to clear the intersection or underestimate the speed of a motor vehicle that is approaching an uncontrolled leg. In most of the fatalities nationwide (85 percent (data not shown)), the motorist was indicated to face no traffic controls, suggesting that most of these collisions occurred at locations with two-way stops where the motor vehicle trafficway had priority.

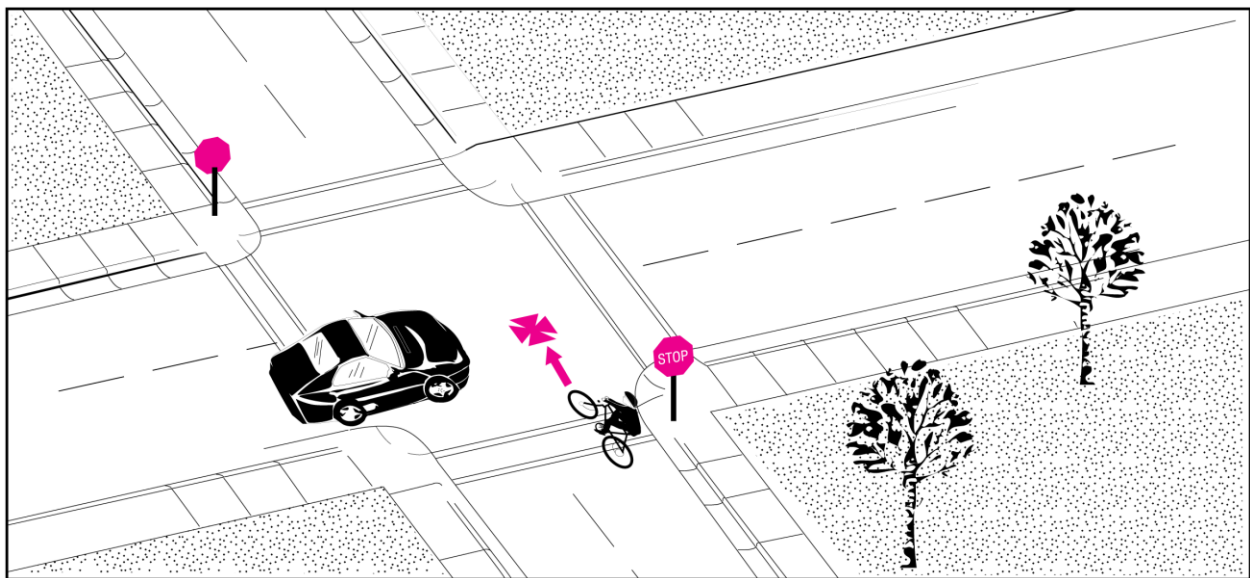


Figure 12. *Bicyclist Ride Through or Out - Sign-Controlled Intersection* (Source: PBCAT)

Sidewalk or wrong-way riding, may contribute to this type by increasing the chances the bicyclist will not notice and obey sign control or that the motorist may fail to detect the bicyclist approaching from an unexpected direction. The data from FARS suggests that these were potential contributing factors in a number of crashes of this type. While nearly half (48 percent) of the bicyclists were riding in a regular travel lane in the same direction as motor vehicle traffic prior to the crash, about 35 percent involved either wrong way riding (25 percent) on either a travel lane (21 percent) or sidewalk (4 percent) or other sidewalk riding (Figure 14). Altogether, 10 percent of bicyclists in this type of crash were cycling on a sidewalk just prior to the crash. Four percent of the total were both cycling on a sidewalk/crosswalk area and riding wrong-way just prior to the crash. Younger bicyclists may tend to be disproportionately involved in this crash type. A bicyclist may also not understand right-of-way rules at four-way stop-controlled locations, but a relatively small percentage (9 percent) of fatalities seemed to occur at four-way stop-controlled locations.

The motorist was indicated to be traveling on a two-way, undivided road in a majority (64 percent) of crashes involving fatalities, with roads divided by a median of some type accounting for 24 percent and 9 percent on roads with a continuous two-way left turn lane.

Crossing Paths - Uncontrolled Intersection (8 percent) This crash type, third for fatalities, which was less frequent in the North Carolina and Boulder databases, was however, 12th most frequent (not shown) among those killed or injured in NC, and seventh among all severity crashes in NC. This type occurs at a junction that is apparently not controlled by signs or signals, and the bicyclist and motor vehicle were initially on perpendicular paths. Obviously, all types of conflicts may occur at uncontrolled junctions. A bicyclist may be trying to cross or access a main road and is struck by a through motor vehicle. In FARS, more than half (52 percent) of these occurred at T-intersections, and 48 percent of the bicyclists killed were struck when trying to cross a road divided by a median of some type.

Motorist Left Turn – Opposite Direction (5 percent) This type is fifth most frequent among fatalities nationally, and also occurs within the top 10 in the other datasets. This type involves a motor vehicle turning left across the path of a bicyclist approaching the intersection from the opposite direction (Figure 13). This was also a frequent type in data the researchers analyzed from the city of Seattle. New York City has also identified left-turning crashes as problematic for both bicyclist and pedestrian injuries and has conducted in-depth examination of scenarios associated with those crash types. Most of these fatalities occurred in daylight and over a range of speed limits from as low as 25 mph. Nationally, in 60 percent of these fatalities, the motorist faced no traffic control, with 25 percent at a signal-controlled location per data in FARS, and a majority (71 percent) occurred when the motorist was initially on a two-way, undivided trafficway before turning. The left-turning motor vehicle may be waiting for a gap in oncoming traffic and fails to look for, see, or yield to the oncoming bicyclist.

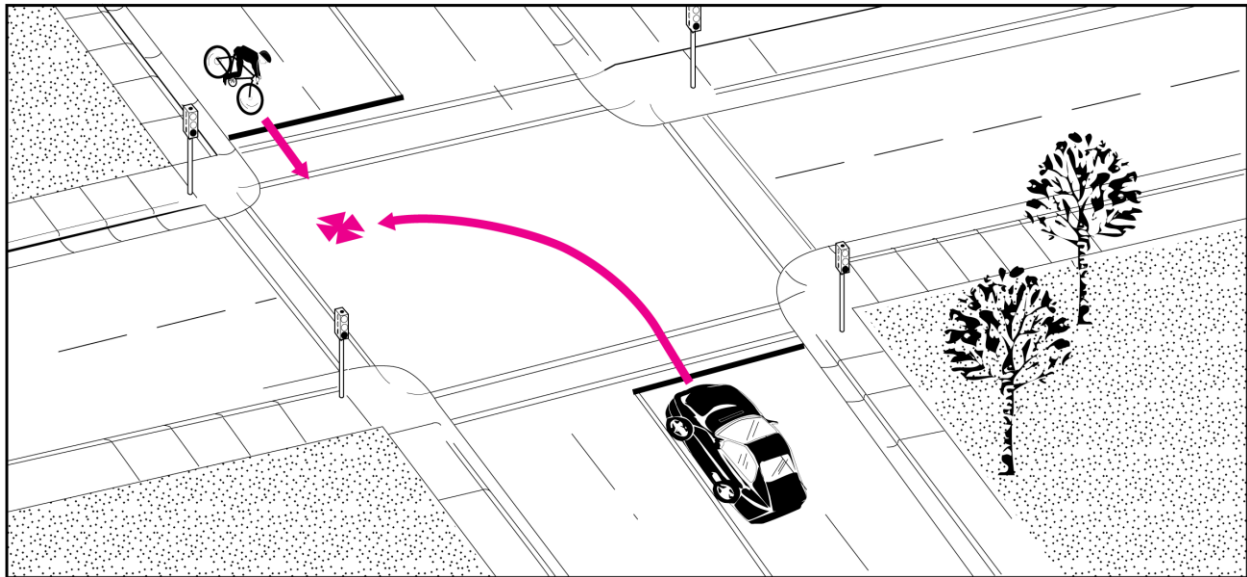


Figure 13. Motorist Left Turn - Opposite Direction (Source: PBCAT)

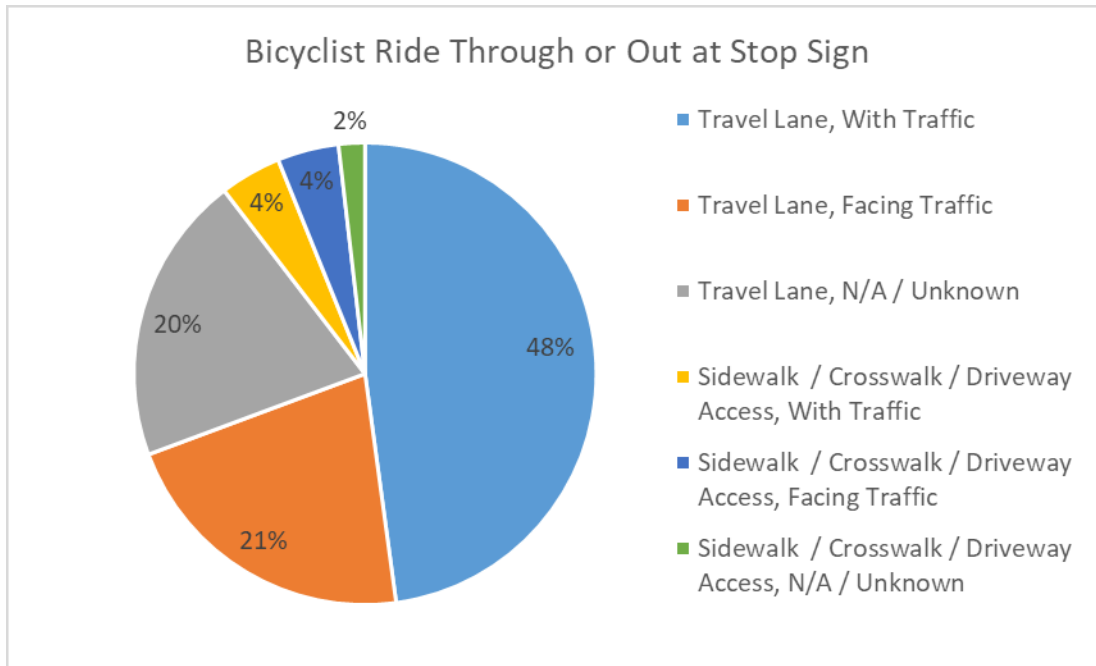


Figure 14. Bicyclist Ride Through or Out at Sign-Controlled Intersection and Bicyclist Initial Riding Position and Direction Relative to Motor Traffic

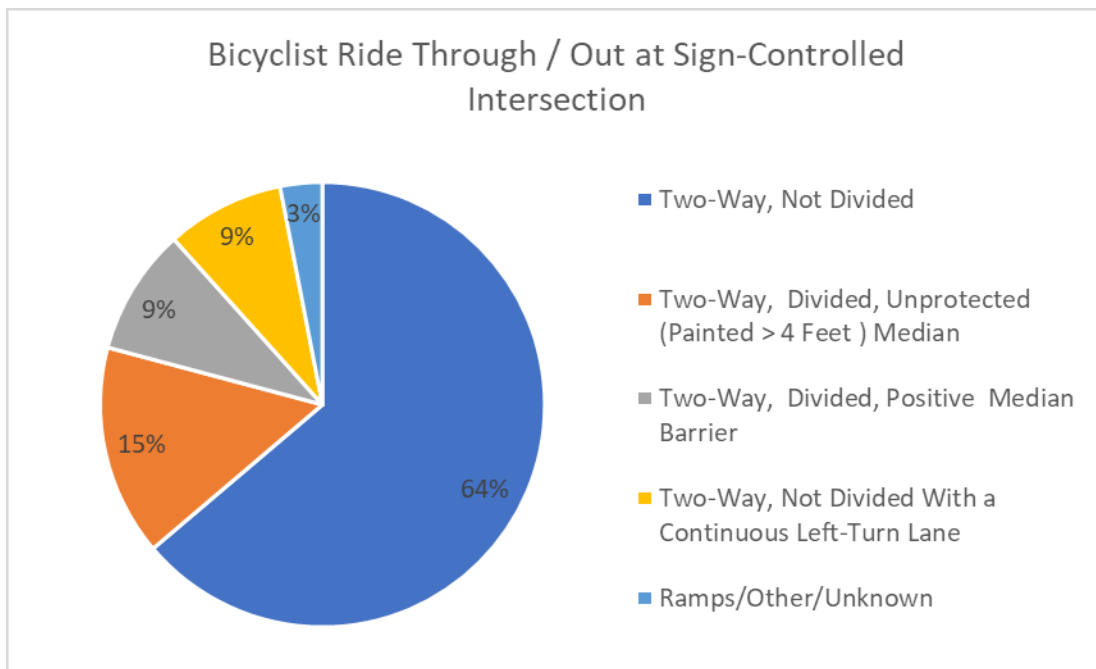


Figure 15. Bicyclist Ride Through or Out at Sign-Controlled Intersection and Motorist's Trafficway Type

The sixth most frequent type, *Crossing Paths - Intersection - Other / Unknown control* (5 percent) doesn't provide much detail about the location characteristics, but this angle-type collision lends additional support for the need to consider intersection safety and design to enable bicyclists to cross areas of conflict.

Motorist Right Turn – Same Direction (4 percent) This type is eighth most frequent in the FARS top 10, and is also found in the top 10 for all severity crashes in NC and in Boulder, although it does not figure in the top 10 for injury crashes in NC. This type can occur at an intersection or merge area. The motor vehicle turns or merges right across the path of a bicyclist who is traveling straight ahead in the same direction (Figure 16). The motor vehicle may not see the bicyclist approaching (if stopped at the intersection), may have passed the bicyclist and misjudged the speed of the bicyclist, or believe (mistakenly) that the bicyclist should wait for them.

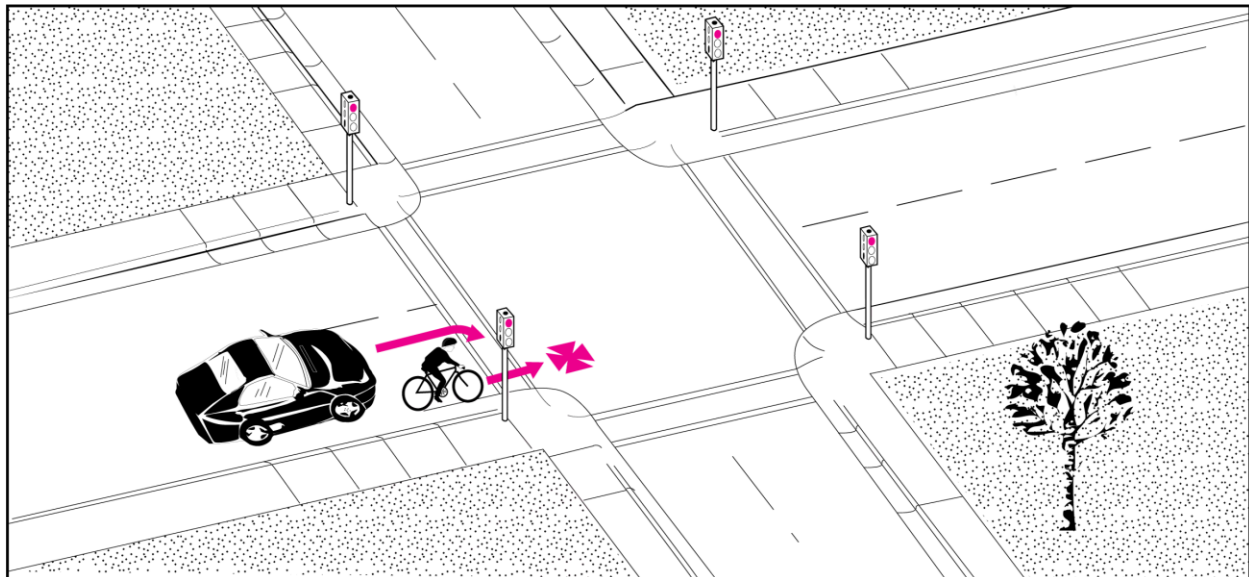


Figure 16. *Motorist Right Turn - Same Direction* (Source: PBCAT)

Motorist Drive Out at Sign-Controlled location The most frequent crash type in the NC and Boulder datasets (both rural and urban in NC, and for severe types involving KAB injuries) may be worthy of further consideration, although it is not ranked very highly in fatal crashes nationally. The direction of riding and facility the bicyclist was on prior to crash were examined for this type using North Carolina and Boulder data. (There were insufficient fatalities involving this type in the FARS database.) In North Carolina, 29 percent were riding facing traffic on a sidewalk before the crash, 23 percent were riding facing traffic in a travel lane, and 4 percent were riding facing traffic on a paved shoulder or bicycle lane, multi-use path or other facility. In total, 57 percent of bicyclists were riding facing traffic in this crash type. When the bicyclist was on a road divided by a median with a positive barrier, the percentage was 68 percent and on other median-divided roads (unprotected median), the percentage was 59 percent. In Boulder, even higher percentages of bicyclists were traveling on sidewalks facing traffic (34 percent) or riding against traffic on a multiuse path (28 percent) when involved in this type, while none were found to be riding facing traffic in a travel lane. Both educational/training measures and engineering countermeasures may be needed to address this crash type where motor vehicles may pull out at a stop-controlled intersection after looking to the left for traffic and not notice a bicyclist approaching from the right (Figure 17).

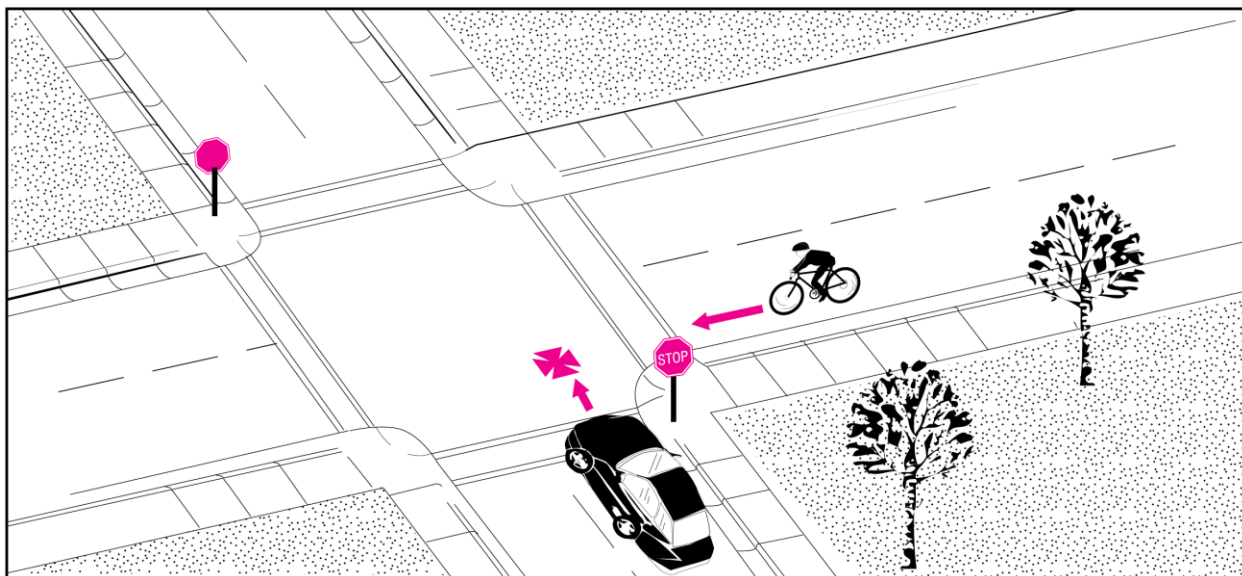


Figure 17. Motorist Drive Out at Sign-Controlled Location (Source: PBCAT)

Several other of the high frequency fatal bicycle crash types (from FARS data) were also over-represented, compared to all types, among bicyclists traveling in the wrong direction. These included:

- Bicyclist Ride Through – Signalized Intersection
- Bicyclist Ride Through – Sign-Controlled Intersection
- Crossing Paths – Uncontrolled Intersection

In all the above types of fatal crashes (FARS data only), the bicyclists were riding on a sidewalk or in a regular traffic lane, not in a bicycle lane or other type of facility. Sidewalk riding may be an indicator of discomfort with the on-road facilities provided, but may also suggest a lack of cycling skill or knowledge. However, bicyclists riding on a sidewalk or sidepath in the opposite direction of motorized traffic may not notice or be able to see traffic control devices.

The following six intersection-related crash types were identified in the top 10 crash-types in multiple databases and are chosen as “high priority” crash types for the purposes of this report:

- Bicyclist Ride Through – Signalized Intersection
- Bicyclist Ride Through – Sign-Controlled Intersection
- Bicyclist Ride Out – Sign-Controlled Intersection
- Motorist Left Turn – Opposite Direction
- Motorist Right Turn – Same Direction
- Motorist Drive Out at Sign-Controlled

Table 8. Top 10 Bicycle Crash Types at Intersections or at Intersection-Related Locations Involving Fatally-Injured Bicyclists

Bicyclist Fatalities U.S. (FARS) 2014-2016			Fatal, Disabling, Evident Injury Bicycle Crashes (NC 2008-14)			All Severity Bicycle Crashes NC 2008-14			All Severity Bike NC 2008-14			All Severity Bike Boulder CO 2008-14		
All Area Types			All Area Types			Urban (inside Municipalities)			Rural (outside Municipalities)			Urban		
Crash Type	Freq	%	Crash Type ¹	Freq	%	Crash Type	Freq.	%	Crash Type	Freq	%	Crash Type	Freq.	%
Bicyclist Ride Through - Signalized Intersection	131	14.5	Motorist Drive Out - Sign- Controlled	240	16.2	Motorist Drive Out - Sign- Controlled	527	19.7	Motorist Drive Out - Sign- Controlled	95	17.7	Motorist Drive Out - Sign-Controlled	145	17.5
Bicyclist Ride Through - Sign- Controlled Intersection	112	12.4	Motorist Left Turn - Opposite Direction	224	15.1	Motorist Left Turn - Opposite Direction	346	12.9	Bicyclist Ride Through - Sign- Controlled	77	14.3	Motorist Left Turn - Opposite Direction	122	14.7
Crossing Paths - Uncontrolled Intersection	65	7.2	Bicyclist Ride Through - Sign- Controlled	183	12.3	Bicyclist Ride Through - Sign- Controlled	232	8.7	Motorist Left Turn - Opposite Direction	61	11.3	Motorist Right Turn - Same Direction	113	13.6
Bicyclist Ride Out - Sign- Controlled Intersection	51	5.7	Motorist Right Turn - Same Direction	87	5.9	Bicyclist Ride Through - Signalized	150	5.6	Bicyclist Left Turn - Same Direction	36	6.7	Motorist Drive Out - Right-Turn-on- Red	103	12.4
Motorist Left Turn - Opposite Direction	49	5.4	Bicyclist Ride Through - Signalized	76	5.1	Motorist Right Turn - Same Direction	141	5.3	Motorist Right Turn - Same Direction	26	4.8	Bicyclist Ride Through - Signalized	37	4.5
Crossing Paths - Intersection - Other / Unknown Control	48	5.3	Bicyclist Ride Out - Sign-Controlled Intersection	60	4.0	Signalized Intersection - Other / Unknown	132	4.9	Motorist Overtaking - Other / Unknown	23	4.3	Motorist Right Turn - Opposite Direction	28	3.4
Motorist Right Turn- Same Direction	39	4.3	Signalized Intersection - Other / Unknown	46	3.1	Motorist Drive Out - Right Turn on Red	99	3.7	Crossing Paths - Uncontrolled Intersection	21	3.9	Motorist Drive Through - Sign- Controlled	23	2.8
Bicyclist Left Turn - Same Direction	33	3.7	Motorist Overtaking - Other / Unknown	45	3.0	Bicyclist Ride Out - Sign- Controlled	77	2.9	Bicyclist Ride Out - Sign-Controlled	18	3.3	Bicycle Only	20	2.4
Bicyclist Ride Out - Signalized Intersection	27	3.0	Bicyclist Left Turn - Same Direction	35	2.4	Crossing Paths - Intersection - Other / Unknown	75	2.8	Bicyclist Ride Through - Signalized	16	3	Motorist Left Turn - Same Direction	19	2.3
Motorist Overtaking - Other / Unknown	26	2.9	Motorist Drive Out - Right Turn on Red	35	2.4	Motorist Overtaking - Other / Unknown	68	2.5	Motorist Turning Error - Left Turn	16	3	Motorist Drive Out - Signalized	16	1.9
Subtotal Top 10	555	61.5	Subtotal Top 10	1031	69.4	Subtotal Top 10	1847	69.1	Subtotal Top 10	389	72.3	Subtotal Top 10	626	75.6
Total	902		Total	1485		Total	2674		Total	538		Total	828	
Crash types were ranked in terms of frequency; the FARS top 10 excluded the type “Unknown Approach Paths” - which was technical tied for 8th, but provides no information about what happened in the collision. Shading indicates crash types that are in the FARS top 10 and at least one other column.														

3.3 Summary of Findings

The analysis of crash types revealed that there are some pedestrian and bicycle crash types that are common across the databases studied. The analysis also showed that while most pedestrian and bicyclist fatalities did not occur at intersections, most crashes in the one city studied (Boulder) did occur at intersections or were intersection-related. The analysis also highlighted that of the intersection and intersection-related pedestrian and bicyclist fatalities most occurred at four-way intersections in urban areas, involved two-way undivided roadways. While the highest percent (46 percent) of intersection or intersection related pedestrian fatalities were at signalized intersections, the highest percent (48 percent) of bicyclist intersection or intersection-related fatalities were at unsignalized intersections. Common conditions for bicycle intersection crashes were daylight, but for pedestrian fatalities dark lighted conditions (nighttime where artificial lighting was in use) were more common. Common speed limits were 30 to 35 mph for both bicycle and pedestrian intersection and intersection-related crashes.

CHAPTER 4: Assessing Gaps in Knowledge and Research

Phase I of this research effort sought to identify priorities for further research and guidance based on three primary inputs: a review of literature and research, interviews with practitioners and assessment of common crash types. Each of these steps was previously summarized in Chapters 1 through 3. This Chapter describes how those three activities were used to identify priorities for performing research and developing guidance during subsequent project phases.

4.1 Literature

During the literature review phase of the project, our team explored existing syntheses of literature as well as more recently completed studies, based on a scan of a wide range of sources. We found that while some treatments already have at least one study evaluating safety effectiveness, most treatments lack sufficient investigation on their safety benefits. Even those treatments for which crash modification factors have been estimated often need additional study concerning under what conditions they are most effective. Our team identified countermeasures where little existing research, or only some completed studies, revealed a gap in understanding regarding the safety effects that might be expected from proper application of those countermeasures.

4.2 Interviews

Our team interviewed practitioners from state, local, and regional agencies to better understand the needs of transportation professionals with respect to guidance on intersection safety countermeasures. In those interviews, the project team discovered that agencies around the country desire additional research and guidance on pedestrian and bicycle safety at intersections and have specific interest in the following treatments: PHBs/HAWK, RRFB, LPI, protected intersections, protected left turn phasing, crossing treatments at freeway ramps, marked crosswalks, right-turn-on-red, and intersection curb radius as well as guidance on pedestrian refuge islands. Interviewees desired guidance on how to address right and left turn crashes, mid-block crashes, and speed and distraction related crashes. Cities expressed specific interest in traffic calming measures and more systemic city-wide safety improvements. Counties expressed interest in suburban conditions including understanding the impacts of channelized right-turn lanes and raised crosswalks. States are interested in HSIP funding and crash-based safety analysis to identify locations to treat.

4.3 Crash Types

We also performed an assessment of common crash types involving pedestrians and bicyclists at intersection locations using state, city, and national-level data. The analysis of crash types revealed that there are some pedestrian and bicycle crash types that are common across the databases studied. Based on the analysis of crash types in three crash databases coded using the Pedestrian and Bicycle Crash Analysis Tool, the team identified common crash types which we recommend as high priority for this project.

For most crash types, 80 to 90 percent or more occurred in urban areas. The analysis also showed that while most pedestrian and bicyclist fatalities did not occur at intersections, most crashes in the one city studied (Boulder) did occur at intersections or were intersection-related. Nationally, in urban areas, 44 percent of bicyclist fatalities and 32 percent of pedestrian fatalities occurred at intersections or were intersection-related.

A majority of bicycle and pedestrian fatalities involved through movements with one or the other party failing to yield or comply with traffic controls. Left turn crash types were more common than right-turning types for both pedestrian and bicyclist fatalities. The analysis also highlighted that of intersection and intersection-related pedestrian and bicyclist fatalities, mostly occurred at four-way intersections in urban areas and involved motorists that were initially traveling on two-way undivided roadways. However, when the pedestrian failed to yield, the most frequent pedestrian crash type by far, the fatally-injured pedestrians were most often struck while attempting to cross a road divided by a median of some type (including painted medians). While the highest percent (46 percent) of intersection or intersection-related pedestrian fatalities were at signalized intersections, the highest percent (48 percent) of bicyclist intersection or intersection-related fatalities were at unsignalized intersections. In crashes where neither party faced traffic control, fatally-injured bicyclists were also commonly struck at intersections with median-divided roads.

Common conditions for bicycle intersection crashes were daylight, whereas for pedestrian fatalities dark but lighted conditions were more common. Common speed limits were 30 to 35 mph for both bicycle and pedestrian intersection and intersection related crashes.

4.4 Other Sources

As our team prepared its recommendations for the research plan, we also checked several sources that documented priority research areas and topics. Our goal was to make sure our recommendations aligned, or at least did not conflict, with research road maps and recommendations laid out by other organizations. In particular, we reviewed the following sources:

- TRB Research Needs Statements Database (TRB n.d.)
- FHWA Strategic Agenda for Pedestrian and Bicycle Transportation (Twaddell et al. 2016)
- FHWA Strategic Plan (FHWA n.d.)

Our scan of these sources revealed no specific recommendations for performing research on the safety effects of intersection treatments for pedestrians and bicyclists that would change our proposed research plan. The Strategic Agenda for Pedestrian and Bicycle Transportation, while supportive of more research into these topics, did not offer other specific recommendations that would change our team's priorities. The document includes a listing of joint research priorities supported by the AASHTO Technical Committee on Nonmotorized Transportation and the TRB Standing Committees on Pedestrians and Bicycle Transportation (Page 27). The following topics on that AASHTO list are most relevant to our recommended research plan:

- *Getting smart on protected intersections*
- *Designing two-way separated bike lanes at intersections*

Other relevant priorities in this table include topics related to transitions between bicycle facility types and signal timing considerations for accessibility. Other topics on the list were not as related to the focus of our project.

The FHWA Strategic Plan is a more high-level document that did not make specific recommendations for safety research. The Research Needs Statements (RNS) database managed by TRB was useful in assessing other research priorities, but none of the topics uncovered specifically aligned with the priorities we identify in our research plan. Some related RNS included the following:

- In-Service Performance Evaluations of Alternative Intersections (TRB 2016)

- Traffic Control at Shared-Use Path Road Crossings (TRB 2015)
- Comprehensive Review and Synthesis of Emerging Urban Street and Intersection Design Guides (TRB 2016)
- Design Options to Reduce Turning Motor Vehicle / Bicycle Conflicts at Intersections (TRB 2010)

While these statements are related to the overall topic of bicycle and pedestrian safety at intersections, none include elements that would require our team to rethink our priorities or the recommended approach to developing our research plan recommendations.

4. 5 Overall Findings

Considering the inputs from Phase I of this research effort, our team developed a series of tables to demonstrate overlap between gaps in the literature, priorities gleaned from practitioner interviews, and the ability of countermeasures to address common crash types. These tables present countermeasures that address pedestrian and bicyclist safety at signalized locations, unsignalized locations, and both types of locations.

Table 9 describes the symbols used in Table 10 to identify low, medium, and high priorities based on findings from these three components of our Phase I research.

Table 9. Explanation of symbols used to denote levels of priority in following table

Research Need	Practitioner Interest	Address Crash Types
 Well-covered in current literature	 Little interest in further guidance	 Addresses few common crash types
 Some studies exist but more work needed	 Some interest in more guidance	 Addresses some common crash types
 Little to no existing research	 High interest in further info/guidance	 Addresses several common crash types

Table 10 illustrates the large gaps in our knowledge of the effectiveness of pedestrian and bicycle safety treatments especially for bicycle treatments. For example, some treatments, such as RRFBs and PHBs, have been evaluated for their safety benefit to pedestrians, but similar evaluation has not been conducted for bicyclist safety. Some pedestrian treatments applicable for both signalized and unsignalized intersections also have little study of their effectiveness. It was clear that treatments applicable for both signalized and unsignalized intersections have the most potential to address multiple common crash types.

The team identified a list of pedestrian and bicycle intersection treatments used in the U.S., detailed the research gaps, crash types and interviewee interest for each. From the list, five treatments were identified as high priority for future study based on whether the treatment has not been previously studied, addresses high priority crash types, and was of interest to interviewees. For most crash types, 80 to 90 percent or more occurred in urban areas, so urban areas are a focus of the evaluation.

Criteria for selecting treatments to study

- Addresses high priority crash types
- Is of interest to interviewees

- **Has not already been studied**
- Applies to common intersection types
- Is feasible to study (available data, number of sites, timeframe, study design, etc.)
- Provides balance between pedestrian and bicycle-specific treatments
- Is low cost (easy to implement no matter the local budget)

Treatments that are prioritized based on the first three criteria:

- Curb extensions and curb radius reduction
- Four-way stop
- Pedestrian hybrid beacon for bicycles
- Protected intersections
- Speed management

Only one of these could be studied with the existing budget. The NCHRP panel selected protected intersections due to the lack of study of this intersection type in the U.S. in current literature, interest from practitioners and its potential to address some prominent crash types. At the time of the project initiation only ten protected intersections had been installed in the U.S., making it an important intersection type to study and evaluate as its use grows.

Table 10. Priorities generated from Phase I research to inform research and guidance

Signalized Intersections				Unsignalized Intersections				Both (Signalized/Unsignalized) Intersections			
Pedestrian Treatments	Research Need	Practitioner Interest	Addresses Crash Types	Pedestrian Treatments	Research Need	Practitioner Interest	Addresses Crash Types	Pedestrian Treatments	Research Need	Practitioner Interest	Addresses Crash Types
Leading Pedestrian Interval (LPI)				Four-Way Stop				Curb Extension			
Passive Detection - Pedestrian				Median Islands				Curb Radius Reduction			
Protected Phases				Roundabouts				Speed Management (Traffic Calming)			
Signal Timing and Delay				Interchange Treatments	[1]			Number of Through Lanes and Turn Lanes			
No Turn on Red Signs				Driveway Treatments				Crosswalk Yellow Warning Sign			
Pedestrian Countdown Signals				Pedestrian Hybrid Beacon				Crosswalk Marking			
Flashing Yellow Arrow				Rectangular Rapid Flash Beacon				Raised Crossing			
Barnes Dance				In-Pavement Warning Lights				Lighting			
				In-Road Signs				Access Management	[1]		
								Advance Stop/Yield Lines			
								Parking Restrictions			
								Gateway Treatment			
Bicyclist Treatments	Research Need	Practitioner Interest	Addresses Crash Types	Bicyclist Treatments	Research Need	Practitioner Interest	Addresses Crash Types	Bicyclist Treatments	Research Need	Practitioner Interest	Addresses Crash Types
Protected intersections				Pedestrian Hybrid Beacon				Speed Management (Traffic Calming)			
Bicycle Signals				Roundabouts				Number of Through Lanes and Turn Lanes			
Signal Timing and Delay				Rectangular Rapid Flash Beacon				Curb Extension			
Passive Detection - Bicycle				Median Islands				Raised Crossings			
Protected Phases				In-Road Signs				Lighting			
Two Stage Bicycle Turn Queue Boxes				Interchange Treatments	[1]			Access Management	[1]		
Leading Bicycle Interval (LBI)								Advance Stop/Yield Lines			
Bike Box								Colored Pavement Markings Through Intersections			
No Turn on Red Signs								Mixing Zone Treatments			

[1] See individual treatments

CHAPTER 5: Analysis of Protected Intersection Treatments

5.1 Introduction and Purpose

To better understand the safety impacts of protected intersections, the project team conducted a limited research effort of two protected intersections in the U.S., each compared to a nearby untreated intersection with similar characteristics. The study used observations of road user behavior to investigate whether protected intersection design reduces risk of crashes involving motorists turning right or left across the path of parallel pedestrians and bicyclists, which are among the most frequent intersection crash types. This chapter summarizes data collection, reduction, and analysis, and presents the results of the analysis.

Protected intersections represent one of many potential countermeasures to improve bicyclist safety at intersections. Compared to treatments such as roundabouts and road diets, protected intersections are relatively new to the U.S. However, emerging findings from local and international studies point to the potential for protected intersection design to improve safety for walking and bicycling. A key to safety for pedestrians and bicyclists at intersections is slowing the speed of right turning vehicles as they cross the pedestrian and bicycle crossing movements. Figure 18 shows how elements of a protected intersection slow motor vehicle turning speeds through a tighter turning radius, increase the visibility of pedestrians and bicyclists to drivers using forward waiting zones, and provide safe, buffered waiting zones for non-motorized users.

A protected intersection is a Dutch concept that leverages intersection geometry to better protect bicyclists. It can be used with separated bike lanes and other pavement markings approaching the intersection but has a few key intersection features: introduction of physical separation at intersection such as islands or bollards to reduce the available turning radius for cars and re-aligning bike lanes and paths away from travel lanes to better position through-bicyclists to avoid conflict with right-turning cars (Gilpin et al. 2015). These realignments are sometimes referred to as “bend-outs.” The treatment creates a recessed crossing for bicyclists, and with islands or bollards, provides space for motorists to begin a right turn before crossing the bicyclist or pedestrian path. This scenario provides greater ability to see and react to approaching bicyclists and provides a zone where motorists can “pause” and yield out of the path of through motor vehicles, which theoretically encourages drivers to yield to bicyclists crossing the intersection. The bend out also changes the trajectory of and reduces the speed of bicyclists.

The protected intersection can also reduce pedestrian crossing distance exposed to motor vehicles, an important consideration to reducing risk at the intersection. However, pedestrians who wait at the curb behind the bicycle facility may include the vision impaired, who would have a longer crossing distance at the protected intersection, though with a shorter distance exposed to motor vehicles. Bicyclists should yield on red to pedestrians crossing perpendicular to them before proceeding to the bicyclist stop bar.

While protected intersections are just beginning to be implemented in cities in the United States, the protected intersection design has been implemented across the Netherlands and codified in Dutch CROW guidance (CROW 2017). The “Dutch Intersection” includes traffic islands or bollards in the roadway to sharpen the turning radius of motor vehicles (thus slowing motor vehicle turning speeds) and realigning the bicycle crossing to providing one car-length of space between the parallel traffic lane (Pucher and Beuhler 2008).

A Danish study of bicycle facility designs in signalized intersections suggests that the protected intersection design offers key safety advantages (Kidholm et al. 2017). The study found that the protected

intersection design had significantly fewer right-hook conflicts compared to a standard intersection approach with a shared bicycle lane and through-right travel lane for automobiles. Out of five intersection layouts evaluated, the protected intersection was deemed to provide the highest level of safety for bicyclists.

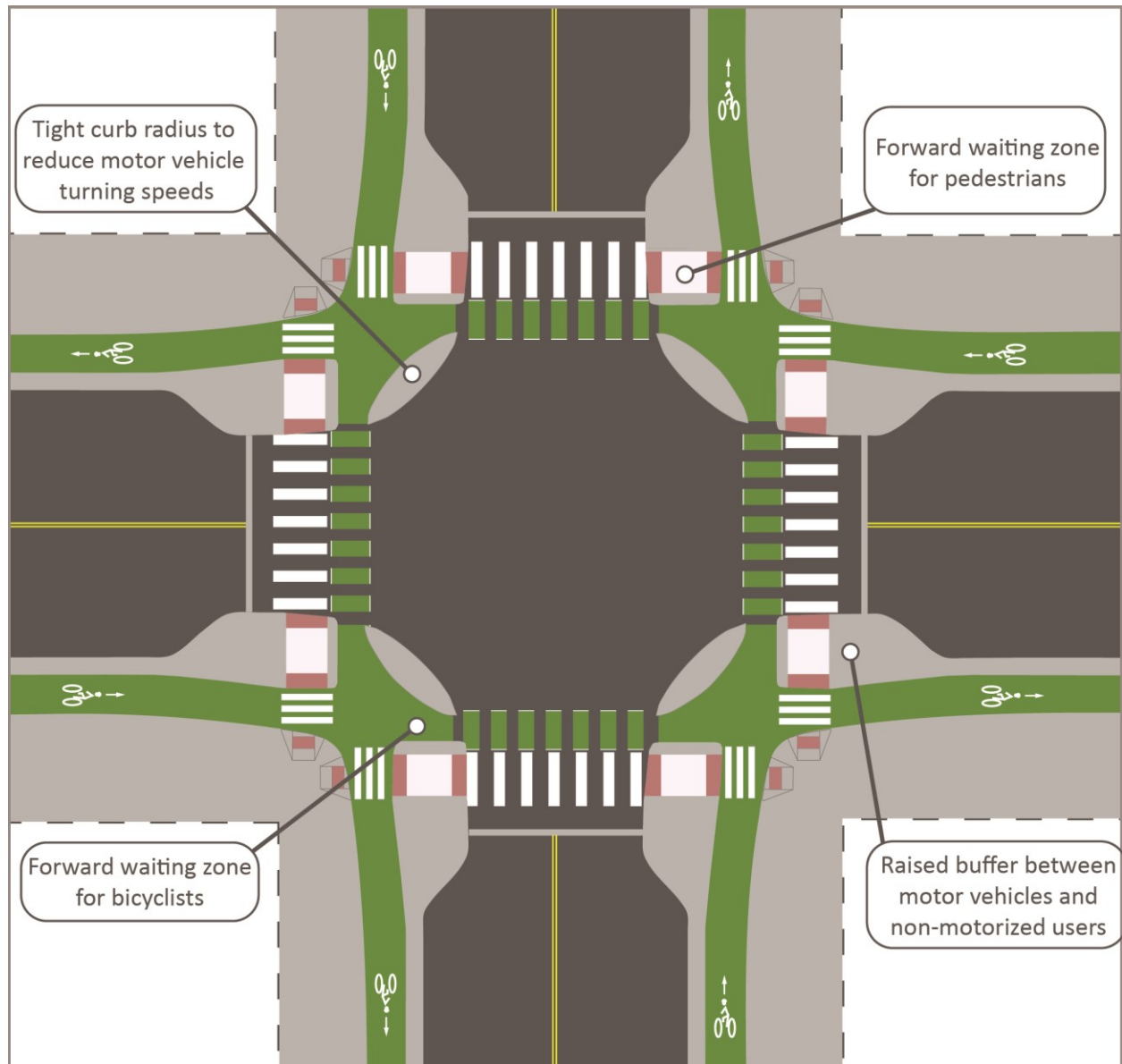


Figure 18. Generalized Protected Intersection Diagram

The Dutch focus on slowing motor vehicle turning speeds and introducing separation between motor vehicles and non-motorized users to improve safety is not limited to protected intersections. FHWA's 2010 Report *Roundabouts: An Informational Guide* recommends providing set-back pedestrian crossings to "separate vehicle-vehicle and vehicle-pedestrian conflict points" (National Academies of Sciences, Engineering, and Medicine 2010). Existing bicycle guidance in the United States, such as Massachusetts Department of Transportation's (MassDOT) 2015 *Separated Bike Lane Planning & Design Guide*, also

points to intersection design that slows motor vehicle turning movements (MassDOT 2015). MassDOT specifically recommends that intersections with separated bike lanes be designed to ensure slow-speed turning movements of 10 mph or less.

Protected intersections have been implemented in varying forms across the United States, including Utah, Illinois, Texas, New York, and California (Alta Planning and Design 2015, San Francisco Municipal Transportation Agency 2018). A 2018 New York City DOT study and 2016 San Francisco Municipal Transportation Agency study both studied protected intersection design in the United States. The New York City study examined two pilot protected intersections at Columbus Avenue and W 70th Street and Amsterdam Avenue and W 85th Street. The study found that the intersections' turning radii did not satisfactorily reduce motor vehicle turning speeds. The study acknowledged that using a design with a tighter radius will likely slow speeds at which motor vehicles turn (New York City Department of Transportation 2018). The San Francisco study examined an unsignalized protected intersection at 9th and Division Street. The study found that the protected intersection design reduced speeds and improved motor vehicle yielding rates, with 96 percent of drivers approaching a bicyclist yielding and 98 percent of vehicles turning at speeds at or below the speed limit (San Francisco Municipal Transportation Agency 2018). It should be noted that neither of these studies have been published in academic literature. The San Francisco study shows a reduction in motor vehicle turning speeds, while the New York City DOT study provides a reminder that continued implementation and refinement of protected intersection designs is necessary to achieve the safest design for all roadway users.

5.2 Study Overview

Because protected intersections have been implemented relatively recently in the U.S., and many locations and more years of data are likely needed to examine safety using crash and injury outcomes, the study focused on safety at protected intersections, using surrogate measures including user behaviors and interactions to assess safety.

To understand the effect of the protected intersection design on safety, this study investigated yielding behaviors and avoidance maneuvers. Events between road users were coded qualitatively in terms of observed behavior and severity and quantitatively in terms of Post Encroachment Time (PET), the difference in time between when one user and the next occupy the same space (Hurwitz et al., 2015, Puscar et al. 2018, Zangenehpour 2015). Both were used because quantitative assessment alone does not necessarily indicate that road users made any effort to avoid each other or were even aware of each other even when PET is less than two seconds. This is why others have sometimes relied primarily on qualitative coding of avoidance behavior (Monsere et al. 2015).

In addition, the study investigated whether the deflection of the bicyclist traveling through the protected intersection combined with the smaller turning radius for motor vehicles reduces the speeds of both bicyclists and motor vehicles. Such a speed reduction is likely to reduce risk of fatal or severe injury pedestrian and bicyclist crashes.

The study design involved observing pedestrian, bicyclist, and driver behavior at treated and untreated comparison intersections by collecting volume and behavior observations from video and speed from radar. One treated intersection was compared to one untreated intersection in two cities, for a total of four study sites. The research team identified safety criteria shown in Table 11, and the first four required data to be collected in the field using non-conspicuous video-taping. The fifth criterion, speed, required data to be collected by radar in the field. The sixth criterion, crash history, was evaluated using data from state crash databases. However, the sample size of pedestrian and bicycle crashes, as alluded to above,

was insufficient for before-after analysis or any formal effectiveness analysis at the two treated sites and two comparison sites.

Table 11. Summary of Safety Evaluation Data Collection

Data Type	Source
Road User Compliance	Video Recorded with Manual Data Reduction
Yielding Rates	Video Recorded with Manual Data Reduction
Maneuvers	Video Recorded with Manual Data Reduction
Pedestrian and Bicyclist Waiting Position	Video Recorded with Manual Data Reduction
Speeds through Crosswalks	Radar Speed Survey
Crash History	Public Agency Data

5.3 Locations Selected for Evaluation

To the research team’s knowledge, there were 10 protected intersections in the U.S at the time of this study. These were located in Atlanta, GA; Austin, TX; Berkeley, CA; Chicago, IL; College Station, TX; Davis, CA; Salt Lake City, UT; and San Francisco, CA. In reviewing these intersections for potential inclusion in the study, the team’s criteria for candidate intersections included:

- Four-leg intersection
- Two-way streets
- Signal controlled
- Presence of bike lanes on at least two approaches (but not a two-way separated bike lane)
- Availability of comparable intersections along same corridor (or nearby)

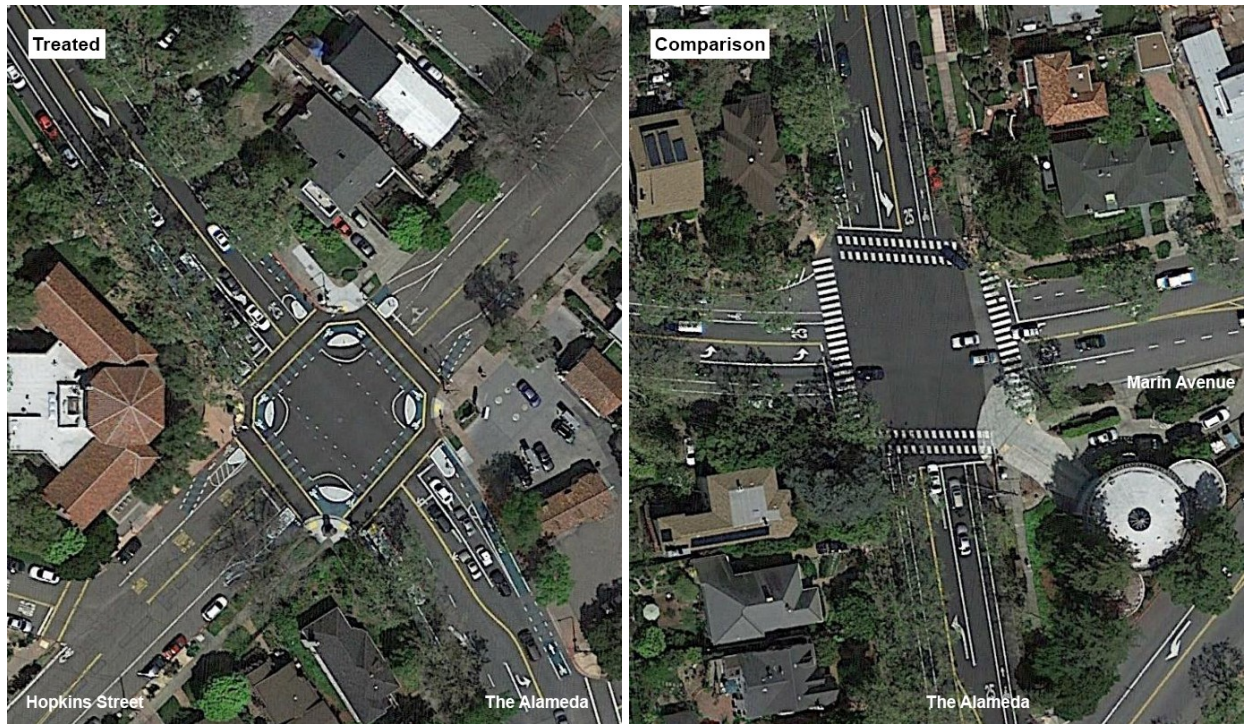
Some protected intersections were considered but not selected because of the presence of two-way cycle tracks or the meeting of one-way streets, which make those sites more difficult to compare. Based on responses from candidate jurisdictions and intersection characteristics, the research team selected to study intersections in Berkeley and Salt Lake City (SLC). The “treated” and “comparison” intersections are identified in Table 12.

Table 12. Study Sites

City	Treated Intersection	Comparison Intersection
Berkeley, CA	The Alameda / Hopkins Street	The Alameda / Marin Avenue
Salt Lake City, UT	300 South (Broadway) / 200 West	300 South (Broadway) / 200 East

5.3.1 BERKELEY, CALIFORNIA

The treated and comparison intersections in Berkeley are shown in Figure 19. The comparison site was selected based on its similarity to the treated intersection, as shown in Table 13. The north-south cross street is the same street (The Alameda). The two intersections are separated by a few blocks. They were expected to have similar vehicle, bicycle, and pedestrian volumes.



Source: Map data ©2018 Google Earth

Figure 19. Berkeley Study Intersections

Table 13. Berkeley Study Site Characteristics

Characteristics	Treated Intersection	Comparison Intersection
Approach Configuration	Major: Two-lane with left turn pockets; Minor: Shared lanes	Two-lane with left turn pockets on 4 approaches
Crossing Distances	Range between 58 and 66 feet	Range between 64 and 94 feet
Median on Major Street	No	No
Bike Lanes	All approaches	3 approaches
Bike Turning Treatment	Raised curb protection on all approaches	None
Marked Crosswalks	✓	✓
Driveways within 250 feet	North leg: 2 driveways; South leg: 4 driveways, 1 intersecting street;	North leg: 5 driveways; South leg: 3 driveways, 1 intersecting street;

Characteristics	Treated Intersection	Comparison Intersection
	East leg: 7 driveways; West leg: 3 driveways, 1 intersecting street	East Leg: 5 driveways, 1 intersecting street; West Leg: 4 driveways
Land Use	Single- and two-family residential	Single-family residential
Grade	Negative grade on southeast-bound approach (The Alameda)	Negative grade on northbound approach (The Alameda)
Transit Facilities	Far-side bus stops on 2 approaches	Mid-block bus stop on adjacent block
Traffic Signal	✓	✓
Permissive Lefts	✓	✓
No Bike Phases	✓	✓
Do Not Prohibit Right Turn on Red	✓	✓

5.3.2 SALT LAKE CITY, UTAH

The treated and comparison intersections in Salt Lake City are shown in Figure 20. The comparison site was selected based on its similarity to the treated intersection, as shown in Table 14. The east-west cross street is the same street (300 South). The two intersections are separated by four blocks. They were expected to have similar vehicle, bicycle, and pedestrian volumes.

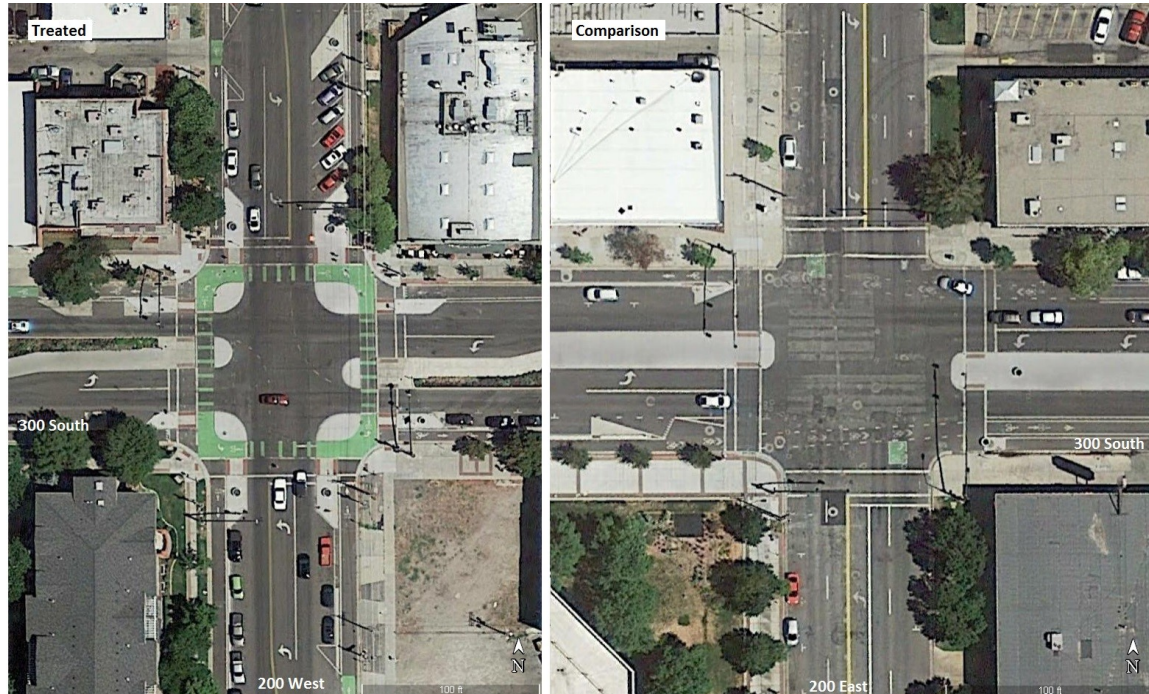


Figure 20. Salt Lake City Study Intersections

Source: Map data ©2016 Google Earth

Table 14. Salt Lake City Study Site Characteristics

Characteristics	Treated Intersection	Comparison Intersection
Approach Configuration	Two-lane with left turn pockets, all approaches	Major: two-lane with left turn pockets; Minor: four-lane with left turn pockets
Crossing Distances	Range between 85 and 96 feet	Range between 70 and 90 feet
Median on Major Street	✓	✓
Bike Lanes	All approaches	2 approaches: Protected on west leg, buffered on east leg
Bike Turning Treatment	Raised curb protection on all approaches	Two-stage left turn box for major street
Marked Crosswalks	✓	✓
Driveways within 250 feet	North leg: 2 driveways; South leg: 2 driveways; East leg: 1 driveway; West leg: 1 driveway	North leg: 4 driveways; South leg: 3 driveways; East leg: 1 driveway; West leg: No driveways
Land Use	Central Business District, Downtown Warehouse/Residential District	Central Business District, Residential Mixed Use, Urban Center Transit Station
Grade	Flat	Flat
Transit Facilities	None	Light rail stop located one block south
Traffic Signal	✓	✓
Permissive Lefts	✓	✓
No Bike Phases	✓	✓
Do Not Prohibit Right Turn on Red	✓	✓

5.4 Contextual Data

For each study site, general site characteristics were collected and documented in addition to data on safety treatments specifically. Though every effort was made to match treated sites with comparison sites, some variability in their characteristics could not be avoided. These contextual data were used to understand whether and how each protected site varies from its comparison site. These data also informed the generalizability of the results.

5.4.1 PEDESTRIAN CROSSING VOLUME

Pedestrian crossing volumes were collected during peak hours across three days of data collection at the treated and comparison intersections (Table 19). In Berkeley, the treated intersection had four times higher pedestrian crossing volumes (398 pedestrians observed in the 40 hours of video studied) than the comparison intersection (105 pedestrians observed in the 40 hours of video studied). In Salt Lake City, the treated intersection had slightly lower, but roughly comparable, pedestrian crossing volumes (1,595

pedestrians observed in the 56 hours of video studied) than the comparison intersection (1,802 pedestrians observed in the 56 hours of video studied).

5.4.2 BICYCLE CROSSING VOLUME

Bicycle volumes were collected during peak hours across three days of data collection at the treated and comparison intersections (Table 19). In Berkeley, the treated intersection had higher, but roughly comparable, bicycle volumes (169 bicycles observed in the 40 hours of video studied) than the comparison intersection (114 bicycles observed in the 40 hours of video studied). In Salt Lake City, the treated intersection had slightly lower, but roughly comparable, bicycle volumes (444 bicycles observed in the 56 hours of video studied) than the comparison intersection (545 bicycles observed in the 56 hours of video studied).

5.4.3 MOTOR VEHICLE VOLUME

State agency annual average daily traffic (AADT) volumes were only available for two of the studied roads. Local agencies provided data on motor vehicle volumes. Table 15 summarizes the motor vehicle traffic volumes available from local and state agencies at the studied intersection. In addition, counts of turning motor vehicles were collected during the data collection process which will be discussed in the Safety Evaluation Results section (see Table 19).

Table 15. Motor Vehicle Volumes from Local and State Agencies

Location	AADT	Source
Berkeley Treated		Berkeley 2009 Turning Movement Counts (September 23, 2009)
The Alameda	13,000-14,000*	
Hopkins	3,000-6,000*	
Berkeley Comparison		
The Alameda	8,000-11,000*	
Marin Av.	12,000-16,000*	
Salt Lake City Treated		
E 300 Street S	6,500-8,000	Salt Lake City
S 200 Street W	12,000	Utah DOT 2016 data
Salt Lake City Comparison		
E 300 Street S	6,500-8,000	Salt Lake City
S 200 Street E	17,000	Utah DOT 2016 data
*Value estimated from peak hour turning movement count assuming peak hour is 8% of AADT		

5.4.4 INTERSECTION GEOMETRY

As shown in Figure 19 and Table 13, the treated intersection in Berkeley has two lanes with left turn pockets on the northbound and southbound approaches, and single shared lanes on the eastbound and westbound approaches. The comparison intersection in Berkeley has two-lanes with left turn pockets on all four approaches.

As shown in Figure 20 and Table 14, the treated intersection in Salt Lake City has two lanes with left turn pockets on all four approaches. The comparison intersection in Salt Lake City has two lanes with left turn pockets on the major road approach, and four lanes with left turn pockets on the minor road approach. A median is provided on the major street (300 South) at both treatment and comparison intersections.

Because the median connects to parking, pedestrians sometimes cross the center of the intersection from median to median.

5.4.5 PRESENCE OF SELECTED TREATMENTS

The treated intersection in Berkeley has bike lanes and raised curb protection on all approaches. The comparison intersection has bike lanes on three approaches and no bicycle turning treatments.

The treated intersection in Salt Lake City has bike lanes and raised curb protection on all approaches. The comparison intersection has bike lanes on two approaches: protected on the west leg and buffered on the east leg. It has a two-stage left-turn box for the major street. This is different from the Berkeley comparison intersection, which has no such bicycle left-turn box. For this reason, as well as other differences between the two cities and specific intersections, analysis of each city was considered separately.

5.4.6 PRESENCE OF MARKED CROSSWALK

The treated intersection in Berkeley has a yellow marked crosswalk on all approaches. It also has a green paint ladder crosswalk on all approaches for bicycles. The comparison intersection in Berkeley has a Continental crosswalk (also known as, a zebra crosswalk) on both Marin Avenue approaches and a double zebra crosswalk on both Alameda approaches.

The treated intersection in Salt Lake City has marked crosswalks on all approaches. It also has green dashed crossing markings paralleling the crosswalk on all approaches for bicycles. The comparison intersection in Salt Lake City has a standard marked crosswalk on all approaches.

5.4.7 TURNING RADIUS

One of the features of protected intersections is tighter turning radius to reduce right turning motor vehicle speeds. Table 16 documents the effective turning radius at each intersection studied and shows that the treated intersections have tighter turning radii than the comparison intersection, as expected.

Table 16. Effective Turning Radius

	Effective Turning Radius (Feet)			
	Berkeley		Salt Lake City	
Corner	Treated	Comparison	Treated	Comparison
North or Northeast	40	58	50	58
West or Northwest	45	53	45	53
East or Southeast	40	50	45	50
South or Southwest	40	55	40	55
Average	41	54	45	54

5.4.8 PRESENCE AND LOCATION OF DRIVEWAYS

The treated intersection in Berkeley has four driveways and one intersecting street (Yolo Avenue) located within 250 feet of the intersection on the south leg. It also has two driveways within 250 feet of the intersection on the north leg, and seven driveways within 250 feet of the intersection on the east leg. It has three driveways and one intersecting street (Sonoma Avenue) located within 250 feet of the intersection on the west leg.

The comparison intersection in Berkeley has three driveways and one intersecting street (Monterey Avenue) located within 250 feet of the intersection on the south leg. It has five driveways located within 250 feet of the intersection on the north leg, and five driveways and one intersecting street (Monterey Avenue) within 250 feet of the intersection on the east leg. The southeast corner of the intersection includes a large driveway that provides access to the Berkeley Fire Department Station #4. It has four driveways located within 250 feet of the intersection on the west leg. Though fire events did occur during the video times periods processed, this did not affect many observations.

The treated intersection in Salt Lake City has one driveway located within 250 feet of the intersection on each major road approach. It also has two driveways (one on either side of the road) within 250 feet of the intersection on the north leg, and two driveways (both on the east side of the road) within 250 feet of the intersection on the south leg.

The comparison intersection in Salt Lake City has one extended driveway (approximately 130 feet in length) located within 250 feet of the intersection on the east leg (major road approach), three driveways on the south leg, and four driveways on the north leg. There was not enough activity at these driveways to significantly impact the results of the study.

5.4.9 LAND USE AND TOPOGRAPHY

The properties surrounding the treated intersection in Berkeley are zoned single-family and restricted two-family residential. The North Branch Berkeley Public Library is located in the northwest corner of the intersection. The terrain on all four legs approaching the intersection is rolling, with a moderate downward slope on The Alameda from north to south and a moderate downward slope on Hopkins Street from east to west.

The properties surrounding the comparison intersection in Berkeley are zoned single-family residential. The Berkeley Fire Department Station 4 is located on the southeast corner of the intersection. The terrain on all four legs approaching the intersection is rolling, with a substantial downward slope on The Alameda from south to north and a substantial downward slope on Marin Avenue from east to west. The downward slope is likely to increase bicyclist speeds which may impact safety.

The treated intersection in Salt Lake City is located within the Salt Lake City Warehouse Historic District. The properties surrounding the intersection are zoned Central Business District and Downtown Warehouse/Residential District. The terrain on all four legs approaching the intersection is relatively flat, with a moderate downward slope on 300 South from east to west.

The properties surrounding the comparison intersection in Salt Lake City are zoned Central Business District, Residential Mixed Use, and Urban Center Transit Station. A Utah Transit Authority (UTA) TRAX light rail line and station is located one block south of the intersection. The terrain on all four legs approaching the intersection is relatively flat.

5.4.10 SIGNAL TIMING

Both treated and untreated intersections in Berkeley and in Salt Lake City are two-phase signals without bicycle phases. The intersections have pedestrian traffic signals on each approach. None of the intersections prohibit right turns on red.

The pedestrian signal timing was consistent with the MUTCD at each location. Countdown timers were present at each crossing. Yellow and Red Clearance intervals were also consistent among study intersections to the casual observer. The cycle length was 65 seconds at most times and 70 seconds on weekday afternoons at both the Berkeley intersections. Similarly, cycle length in Salt Lake City was 60

seconds on morning and evening weekday peak hours and 72 seconds at off peak hours and on weekends at both intersections.

5.4.11 DATA COLLECTION

This section summarizes the general collection method and equipment installation for each data type. Table 17 shows the collection method and preliminary collection timing for each data attribute.

Table 17. Data Collection Summary

Criteria	Data Collection
General Intersection Traffic Characteristics	Public agency data; Video with manual counts for coded periods
Compliance with Traffic Controls	Video collected: three mid-week days and one weekend day; 24 hours per day Not all collected video was reduced - see below for hours coded.
Yielding Rates	
Avoidance Maneuvers and Conflicts	
Pedestrian and Bicyclist Waiting Behavior	
Speeds through Crosswalks	Radar gun; one weekday; recorded from 7am to 9am and 4pm to 6pm (midweek) and 11:30 am to 1:30 pm (weekend)
Crash History	Obtain crash history from local agency or state agency crash databases

5.5 Study Design

5.5.1 EQUIPMENT INSTALLATION

Four cameras were installed at each intersection on traffic signal poles, utility poles, or, where possible, street trees. Cameras were placed on corners with a view of one crossing. An effort was made to identify mounting locations that allowed for an approach view, but scarce mounting options across all four sites meant that corner angles needed to be used for the sake of comparability. However, the trade-off with this location choice was that it was hard to observe the slowing of bicyclists and motor vehicles approaching the intersection. Figure 21 and Figure 24 show the camera mounting locations for the four intersections studied. Figure 22, Figure 23, Figure 25, and Figure 26 show street-level view of the mounting locations, for example, camera locations. Camera locations for the Berkeley sites were somewhat more challenging, given the grade and skew of the intersection approaches, as shown in Figure 21.



Source: Map data ©2018 Google Earth

Figure 21 Berkeley Video Camera Mounting Locations



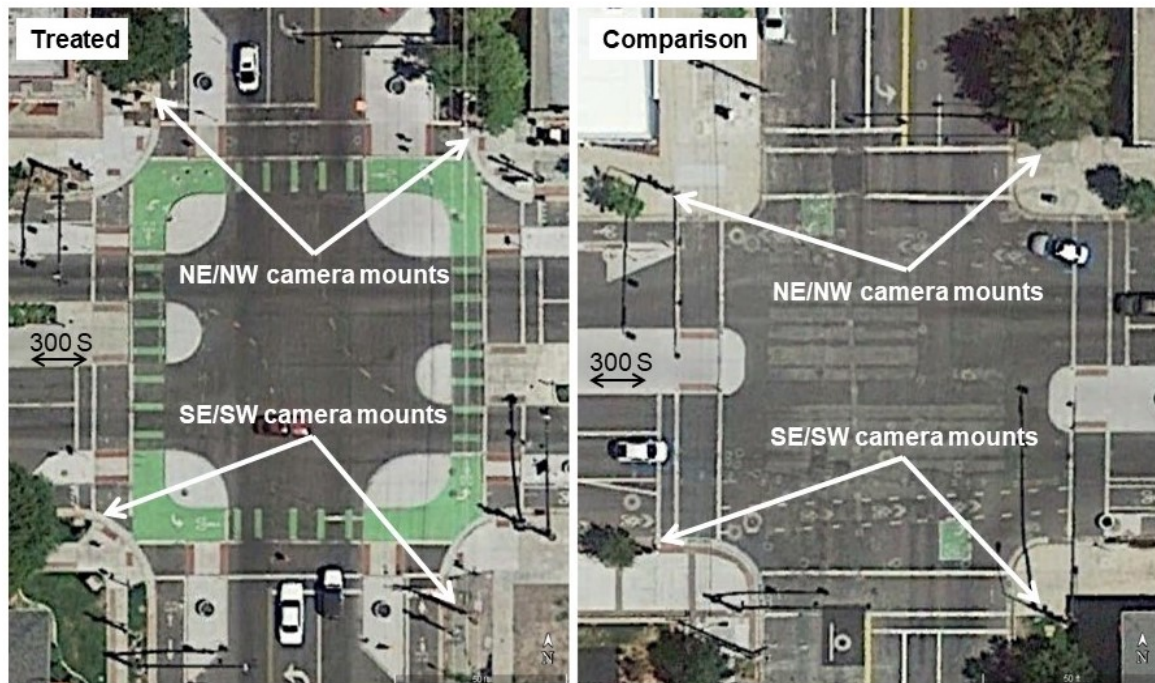
Source: Map data ©2017 Google Street View

Figure 22. Berkeley Example Treated Intersection Camera Placement (West Corner)



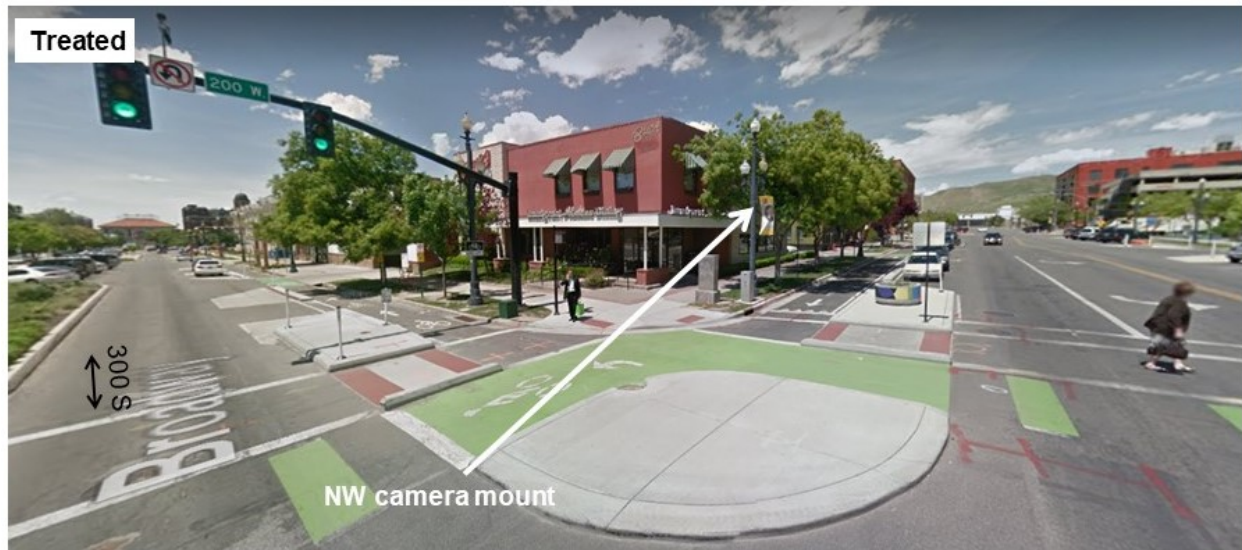
Source: Map data ©2017 Google Street View

Figure 23. Berkeley Example Comparison Intersection Camera Placement (West Corner)



Source: Map data ©2016 Google Earth

Figure 24. Salt Lake City Video Camera Mounting Locations, Treated and Comparison Intersections.



Source: Map data ©2016 Google Street View

Figure 25. Salt Lake City Example Treated Intersection Camera Placement (Northwest Corner)



Source: Map data ©2016 Google Street View

Figure 26. Salt Lake City Example Comparison Intersection Camera Placement (Northwest Corner)

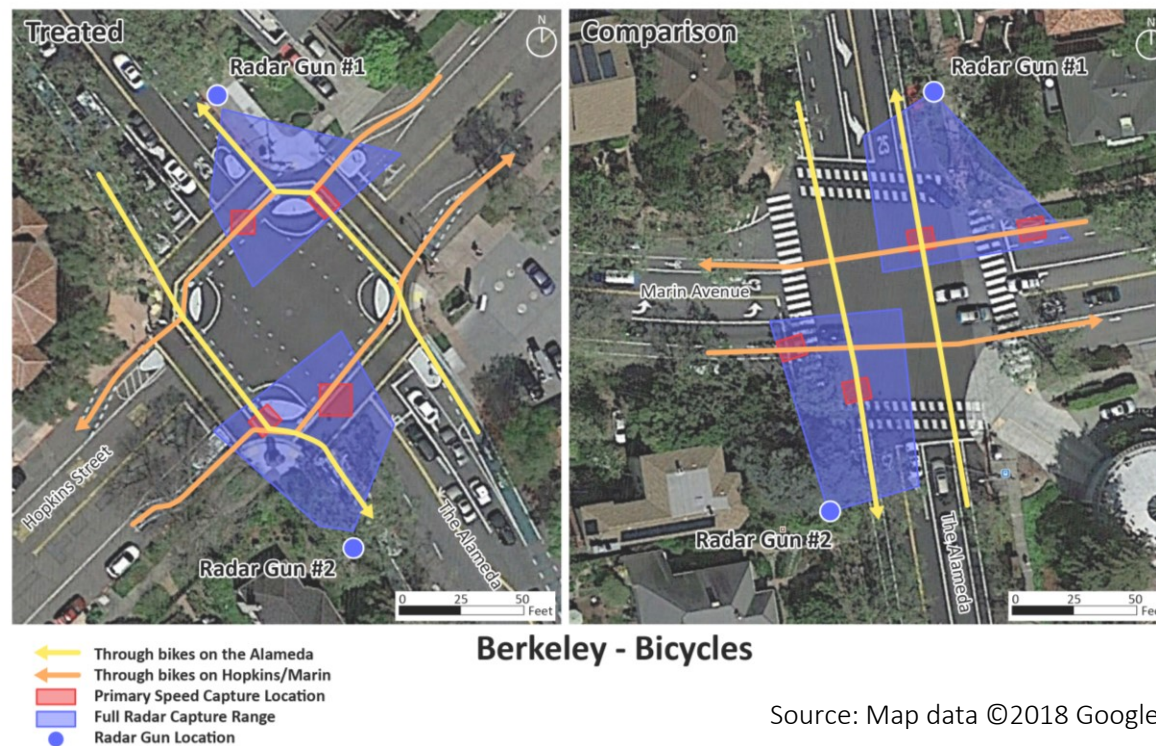
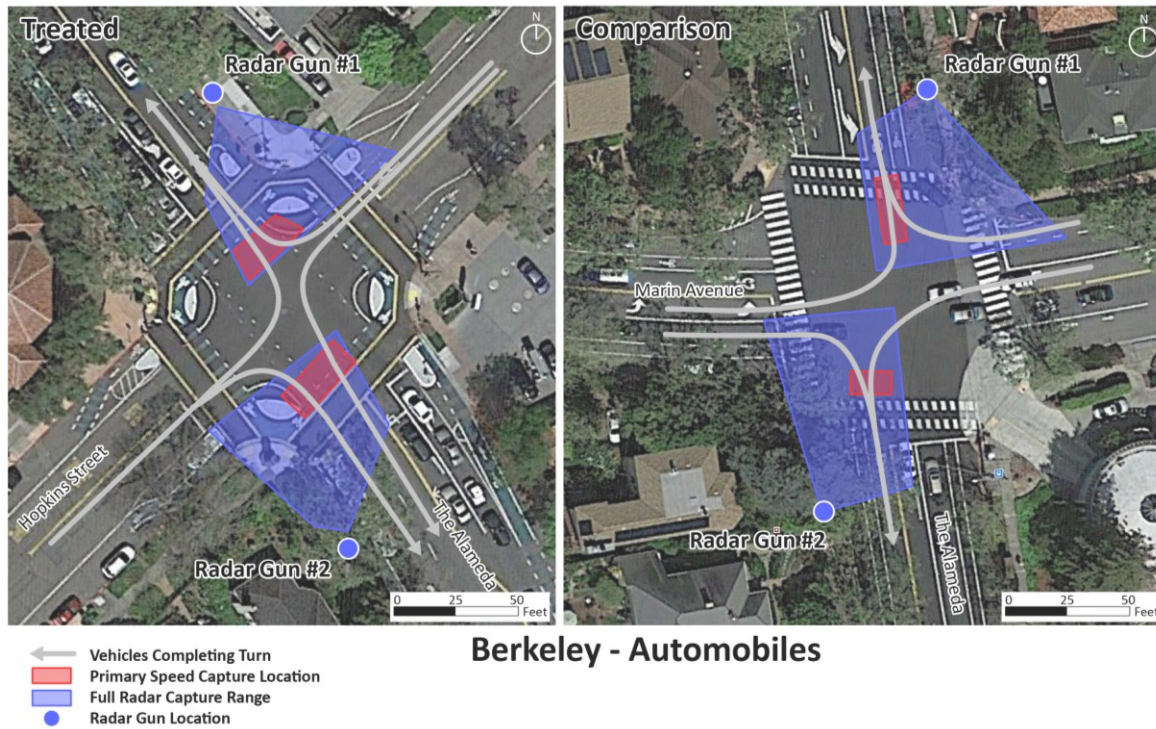
The radar gun survey was conducted by two technicians per session, located at diagonal corners. Technicians captured two hours of turning speeds for vehicles and one hour of turning speeds for bicycles. Speeds were recorded for left- and right-turning vehicles and for bicycles that arrived during the green phase. For vehicles, speeds of turning vehicles were captured as they passed the crosswalk of the receiving leg, as shown in Figure 27 and Figure 28. Due to limited budget, the technicians only captured vehicle turning speeds from the minor street onto the major street.

For Berkeley, technicians stood on near the north and south or northeast and southwest corners of The Alameda (considered the major street) approximately 50 feet downstream from the oncoming traffic to capture the speed at the crosswalk as vehicles completed right or left turns onto The Alameda (Figure 27).

For Salt Lake City, technicians stood downstream of the southeast corner (30 feet east at 300S and 200W; 50 feet east at 300S and 200E) and in the west leg median (50 feet west at 300S and 200W; 60 feet west at 300S and 200E) to capture speeds of vehicles turning from each corresponding minor street onto 300 S (Broadway) as shown in Figure 28.

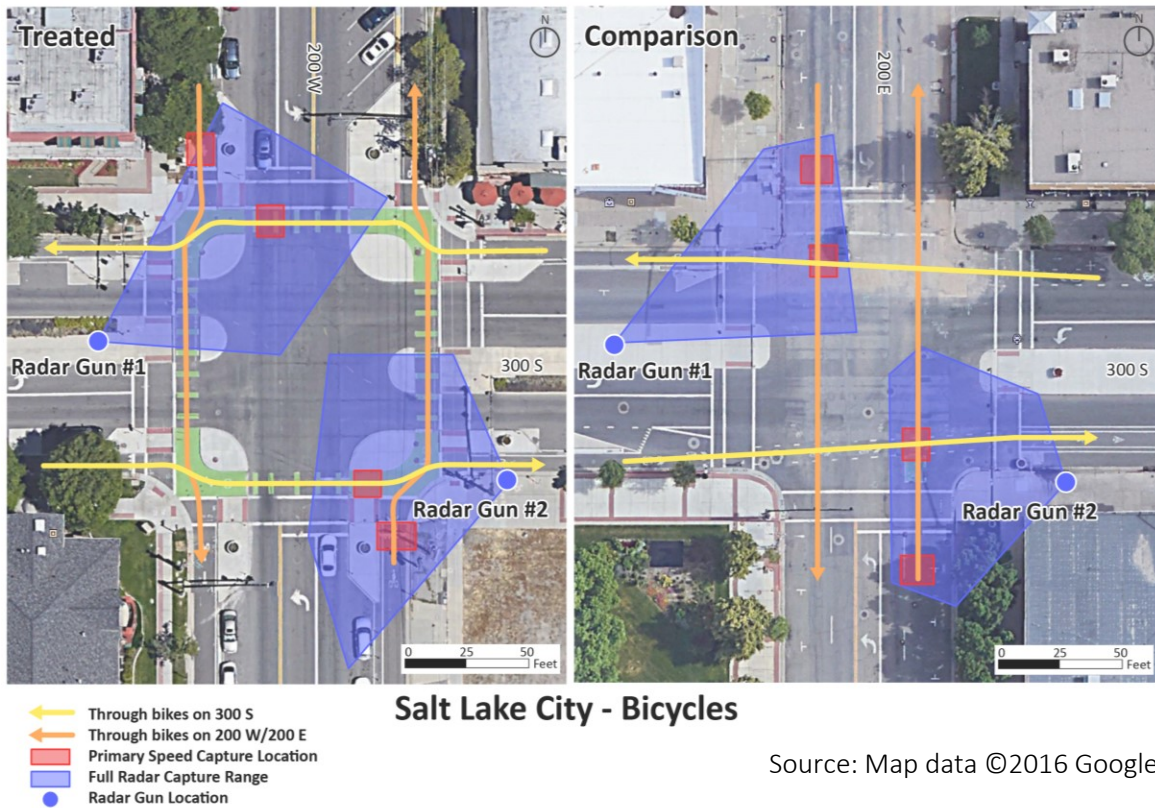
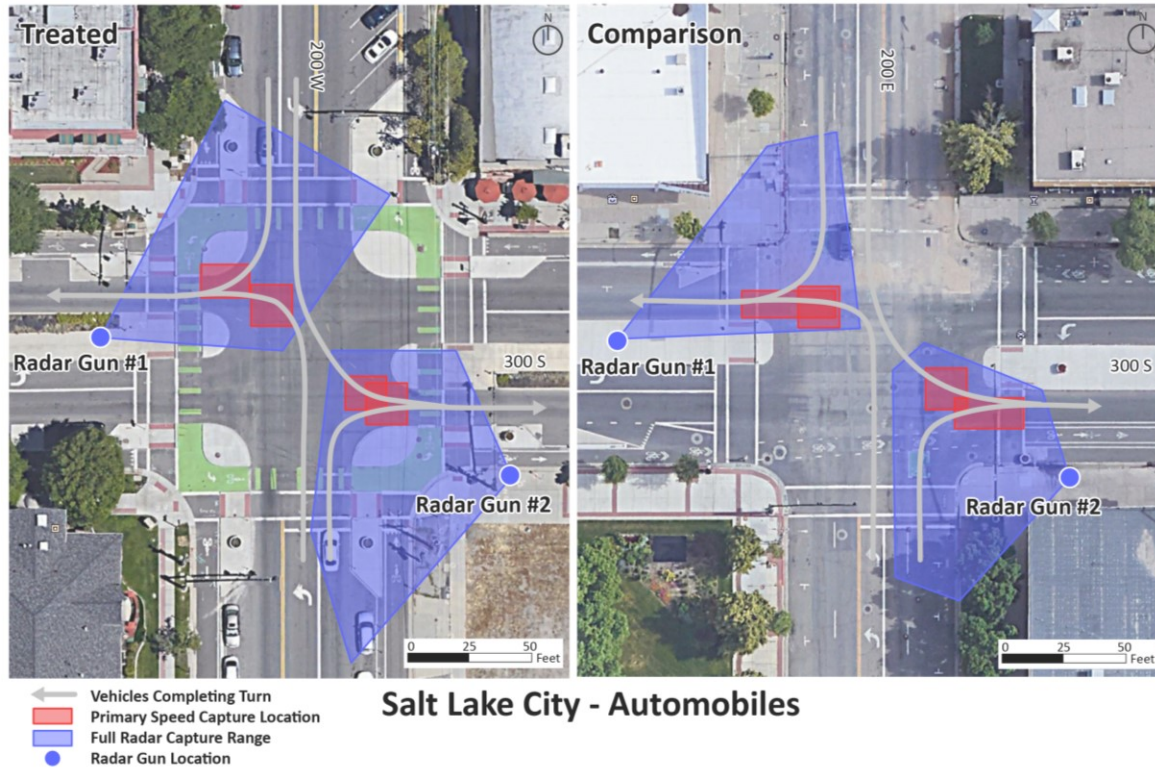
Since radar gun technology works by capturing the changing frequency of an object moving toward or away from it, the position and angle shown in the figures was meant to maximize accuracy of the speed recording in spite of the technician's inability to stand in the middle of the lane facing oncoming traffic. The maximum speed observed by the radar gun was recorded for each vehicle. The speeds were adjusted based on the angle between the direction in which the vehicle is moving and the direction in which the radar gun is pointed. The calculation is presented in the Safety Evaluation Results Section of this report.

A similar approach was used to capture speeds of bicycles traveling through the "bend-outs" and across a comparable location at untreated intersections during a different hour. In anticipation of potentially low bicycle volumes, the radar gun survey for bicyclists captured the speed of both through movements on the major street and on the minor street. For bicyclists riding through the intersection on the minor street, the technician, standing as indicated by the black dot in Figure 27 and Figure 28, aimed the radar gun as the bicyclist was moving away into the "bend-out" at the location shown by the boxes in the figures.



Source: Map data ©2018 Google Earth

Figure 27. Berkeley Radar Gun Positioning



Source: Map data ©2016 Google Earth

Figure 28. Salt Lake City Radar Gun Positioning

5.5.2 DATA REDUCTION

This section summarizes the video data reduction process. Details of video coding are provided in Appendix B.

For each pedestrian or group of pedestrians crossing the intersection the following factors were coded:

- the location of the pedestrian or group of pedestrians while waiting (curb, or within the area protected by the island or in the bike lane for treated sites, or beyond the curb, in the bike lane at untreated sites);
- whether the pedestrian or group of pedestrians physically activated the crosswalk signal by pushing the button or not (when this could be observed);
- the pedestrian signal display when the pedestrian or group of pedestrians stepped into the crosswalk (walk, don't walk, flashing); and
- whether the pedestrian or group of pedestrians stayed in the marked area of the crosswalk or veered out during crossing.

For both pedestrians and bicyclists crossing the intersection, the following factors were coded:

- the crossing action as the individual arrived and exited the intersection on green,
- arrived and exited on yellow,
- arrived on green and exited on red,
- arrived on yellow and exited on red,
- arrived on red and waited for green,
- turned right on red,
- passed through the light when it was red (ran the light), and
- started through the intersection on a red which soon turned green (jumped the green).

The position of the pedestrian or bicyclist when they entered the intersection was coded as bike lane, crosswalk, motor vehicle through lane, motor vehicle left lane, or sidewalk. "Sidewalk" refers to bicyclists who entered the intersection from the sidewalk, but in general the riding position on approach to the intersection could not be observed. All pedestrians with their position coded as "crosswalk" entered the intersection from the sidewalk unless noted otherwise.

For each bicyclist or group of bicyclists passing through the intersection, the direction of travel relative to motor vehicle traffic (with traffic on same side of roadway or facing traffic) and the type of crossing action was also coded (through, right, regular left, two-stage left, two-stage through, or U-turn). "Regular left" refers to the bicyclist merging into the motor vehicle lanes and turning left like a motor vehicle would. "Two-stage left" is when a bicyclist first crosses one leg of the intersection and then waits for the light to cross the other leg with traffic. Protected intersections are designed to better allow bicyclists to make two-stage left turns. A few "Two-stage through" movements were also observed in Salt Lake City. This movement refers to the bicyclist entering the intersection facing traffic (wrong-side), crossing their initial leg at the crosswalk, and crossing another leg at the crossing area and continuing with their initial direction, traveling with traffic.

Manual reduction of video data focused on identifying events between different modes. PET was used as a metric to determine which events would be coded. For this project, **PET was defined as "the time difference between when one road user passes out of an area of possible collision and when the conflicting**

road user arrives at the same area” (Puscar et al. 2018). The area where the paths of the two road users intersect and where the two road users would have collided had they both been there at the same time is called the potential conflict area. Any intersection of road user travel paths between pedestrians, bicyclists, and motor vehicles with less than five seconds PET was coded (Zangenehpour et al. 2015). Events with less than two seconds PET were of particular interest since these suggest higher potential for conflict (Hurwitz et al. 2015). For the remainder of this report, the term **“event” refers to the situation when road users cross paths within less than five seconds PET**. Especially in lower speed and busy environments, there may be many events that are not necessarily unsafe. For some events, no interactions were observed.

All events were coded by type of observable behavior on approach to the potential conflict area. Interactions between vehicles and bicycles, between vehicles and pedestrians, and between bicycles and pedestrians were documented. Every event was coded by indicating which party passed through the potential conflict area first and second. If noticeable behaviors including slowing (gradual or sudden), stopping, or trajectory changes were observed by either or both parties, these interactions were coded:

- MV Gradual – Gradual change in motor vehicle speed or trajectory
- Bike Gradual – Gradual change in bicycle speed or trajectory
- Ped Gradual – Gradual change in pedestrian speed or trajectory
- MV Sudden – Sudden change in motor vehicle speed or trajectory
- Bike Sudden- Sudden change in bicycle speed or trajectory
- Ped Sudden – Sudden change in pedestrian speed or trajectory
- Both slow/stop – Both parties in the interaction slow or stop

These observations, and the nearness of interaction (PET) were used to make an additional determination of yielding (no need to yield, first party yields, second party yields, or near collision (hereafter near-miss) or collision). In some cases, a close call or “near miss” was judged to have occurred, even though neither party clearly reacted by slowing or stopping or changing direction.

Yielding definition. If noticeable change in speed or stopping by one or both parties occurred, this information was used to determine that an intentional yield of right-of-way was made, which as mentioned may be a narrow definition.

If one party crossed in front of the other with no apparent adjustments in speed or trajectory or obvious conflict, these were denoted as *Motor Vehicle Cross Bike* (motor vehicle arrived and proceeded first through the potential conflict area), *Bike Cross Motor Vehicle* (the bicycle arrived and proceeded first), *Pedestrian Cross Bike*, and so forth, and the conclusion was that an intentional yield was not necessarily required (that the parties had sufficient space and distance, given their speed). In these cases, it is, however, also possible that the second party slowed on the approach, outside of the observation zone of the cameras, and so, in a broad sense, could also have constituted “yielding” but in any case, was an apparently safe interaction.

Near miss definition. Interactions were noted as “near misses,” if a very close interaction in space or time was observed (even if no party noticeably reacted, as happened in a few cases), or, as noted above, with sudden stopping or slowing or other types of avoidance maneuvers such as the following:

- Sudden swerving (changes in trajectory) by motorists and/or bicyclists
- A quick side- or back-stepping by pedestrians to avoid a turning vehicle or bicyclist – excluding instances of distraction, such as a pedestrian looking at a cell phone (where identifiable)

- Sudden stopping by one or both to avoid other road user
- Avoidance maneuver by a bicyclist to pass a slower moving bicyclist

Finally, turning movement and crossing counts for motor vehicles, bicycles, and pedestrians were manually reduced from the video for the coded periods.

Table 18 shows the dates and times for which collected video was reduced. The team chose only to code peak daylight hours when more people, drivers, and interactions were expected. The corresponding legs were coded for the comparison intersection for the same periods. The same person coded the video for both treated and the comparison intersection for a given intersection approach and time period. At the Berkeley site, two intersection legs (north and east) could not be coded due to issues with the video angle or quality, so additional hours were coded for the other legs. Sun glare also obscured the video, or other problems prevented video from being coded for some portion of the intended coding periods. In some cases, events were coded, but not all of the factors could be observed, which resulted in missing data for some observations. Additional information on how the video was coded are available in Appendix B.

Table 18. Dates and Times Treated and Comparison Intersections Were Coded

City	Date	Times Coded
Salt Lake City all intersection approaches	Wednesday, May 23, 2018	7:00AM to 9:00AM Noon to 1:00PM 4:00PM to 6:00PM
	Thursday, May 24, 2018	7:00AM to 9:00AM Noon to 1:00PM 4:00PM to 6:00PM
	Saturday, May 26, 2018	7:00AM to 9:00AM (east and west legs only) Noon to 2:00PM 4:00PM to 6:00PM (east and west legs only)
Berkeley (south and east legs only)	Wednesday, May 9, 2018	7:00AM to 9:00AM Noon to 6:00PM
	Thursday, May 10, 2018	Noon to 6:00PM
	Saturday, May 12, 2018	10:00AM to 4:00PM

5.6 Safety Evaluation Results

Safety at treated and untreated intersections were compared using six criteria:

1. Road user compliance
2. Yielding rates, and/or near misses or crashes between turning vehicles and bicyclists or pedestrians, and between bicyclists and pedestrians.
3. Maneuvers (who proceeded first, second; gradual or sudden changes in trajectory or speed of either party)
4. Pedestrian and bicyclist waiting position at protected intersection corners
5. Speed of vehicles and bicycles passing through crosswalks
6. Crash history

Data for the first four criteria listed above were obtained by using video to observe road user behavior. For the purposes of this analysis as previously mentioned, **an event was defined as any instance where two or more road users' paths crossed each other within a five second interval (PET).**

As noted above, the evidence of slowing or changes in trajectory, along with the nearness of the interaction were used to code whether a conflict or a normal, apparently safe “yield” occurred during these interactions (next sections). Analysis results for each criterion are provided in the following subsections.

5.6.1 VIDEO OBSERVATION SUMMARY

Table 19 summarizes the number of observations coded from the video. Compared to Berkeley, Salt Lake City had roughly twice as many bicyclists per hour per approach and three times as many pedestrians per hour per crosswalk. Bicyclist volumes per hour per approach are comparable between treated and comparison intersections in each city. Pedestrian and motorist left turn volumes per approach are similar between intersections in Salt Lake City, but motorist right turns per approach are about twice as high at the comparison intersection. In Berkeley, pedestrian volumes per approach are four times higher at the treated intersection compared to the comparison intersection.

Motor vehicle interactions with non-motorized travelers (bicyclists and pedestrians) were collected during peak hours across three days of data collection at the treated and comparison intersections (Table 19). In Berkeley, the treated intersection had higher motor vehicle interactions (196 total motor vehicle interactions observed in 40 hours of video studied) than the comparison intersection (69 motor vehicle interactions observed in hours of video studied). In Salt Lake City, the treated intersection had higher motor vehicle interactions (833 motor vehicle interactions observed in 56 hours of video studied) than the comparison intersection (678 motor vehicle interactions observed in 56 hours of video studied).

Table 19. Summary of Observations Coded from Video Reduction

Observations	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Number of Partial Days of Reduced Video	3	3	3	3
Number of Total Camera Hours Reduced (separate camera for each crosswalk)	40	40	56	56
Number of Bicycles Recorded	169	114	444	545
Number of Pedestrians Recorded	398	105	1,595	1,802
Bicyclists per Hour per Approach	4	3	8	10
Pedestrians per Hour per Crosswalk	10	3	28	32
Motorist Left Turns per Hour per Approach*	51	92	31	32
Motorist Right Turns per Hour per Approach*	51	82	33	59
Percent of turning vehicles that are trucks or buses	3.9%	1.3%	1.5%	1.0%
Percent of turning vehicles that are motorcycles	0.3%	0.3%	1.2%	0.7%
Number of Interactions of Bicyclists with other Bicyclists (Bike-Bike Interactions)	0	1	6	6
Number of Bicyclist-Motorist (Bike-MV) Interactions	43	22	123	119
Number of Bicyclist-Pedestrian (Bike-Ped) Interactions	3	0	23	22
Number of Pedestrian-Motorist (Ped-MV) Interactions	153	47	555	714

*Motor vehicle volume counts are based on 44 hours of video for each Salt Lake City intersection and 36 hours of data for each Berkeley intersection.

5.6.1.1 Bicycle Interaction Summary

To place the findings into further context, it helps to see the overall numbers of bicyclists, their maneuvers through the intersections, and any corresponding motor vehicle maneuvers for the interactions coded. These data are shown in Table 20 and Table 21.

At both the treated and untreated intersections in Berkeley, a majority of bicyclists were traveling through the intersection (no turns). A majority of interactions with motorists involved a “right hook” type movement, motorists turning right across the path of through bicyclists (49 percent of interactions at the treated intersection, and 55 percent at the comparison location) (data in Table 20).

In Salt Lake City, a majority of bicyclists were also traveling straight through the intersection, but there was a much wider range of interactions with both left turning, right turning, and through motorists (Table 21). At the treated intersection, there was a somewhat higher percentage, 25 percent of interactions, involved motorists turning right across the path of through bicyclists, than at the comparison intersection (21 percent).

Additionally, in Salt Lake City, 19 percent of bicyclist interactions with motorists were with “through” motorists, and 18 percent with left turning motorists. At the comparison intersection, there was a slightly higher percentage (26 percent) of interactions involved motorists turning left across the path of through bicyclists than at the treated intersection. Twenty-four percent of all the interactions at the comparison location involved through motorists and through bicyclists, suggesting issues with signal compliance.

5.6.1.2 Pedestrian Interaction Summary

Table 22 summarizes the observed interactions between pedestrians and motor vehicles by motorist maneuver in relation to the pedestrian's crossing direction.

When "parallel path" is indicated this means that the initial direction of the motorist was parallel to the pedestrian when starting across the intersection, but that then the motorist turned right or left into the path of the pedestrian crossing the intersection. For example, "Motorist Left Turn, Parallel path – same direction" means that the motorist and pedestrian started crossing the intersection traveling the same direction, but that then the motorist (who likely had a permissive left) turned into the path of the pedestrian crossing the intersection.

When "perpendicular paths" are indicated, the motorist and pedestrian started crossing the intersection in perpendicular directions. Either the motorist made a right turn on red to create an interaction with a pedestrian or the pedestrian was crossing against the light and thus interacted with a left turning motorist. Similarly, "Thru Vehicle" usually refers to the situation where pedestrians have a stop hand (red light) and crossing traffic (perpendicular to pedestrian path) has the green, in which case interactions occur when the pedestrian crosses against the light.

For all four intersections, parallel path interactions are more common than perpendicular path interactions. This makes sense since these interactions generally occur when motorist and pedestrian both have the green/walk signal, while perpendicular path interactions occur when one of the road users has a red/stop hand signal. The Berkeley intersections had a higher percentage of parallel path interactions (80 to 91 percent) compared to the comparison intersection (59 and 69 percent).

The Berkeley treated intersection had a higher percentage of perpendicular path interactions (15 percent) compared to only 9 percent at the comparison intersection. In Salt Lake City, the treated intersection also had a higher percentage of perpendicular path interactions (39 percent) than the comparison intersection (31 percent).

Right turn perpendicular path crashes, the kind commonly seen when a motorist turns right on red, were a low percent of total interactions, but were more common in Salt Lake City (8 to 15 percent of interactions) than in Berkeley (3 to 4 percent of interactions).

Left-turn and through motorist perpendicular interactions with pedestrians, those usually associated with pedestrians running the red light, were more common at the treated intersections in both cities (12 and 30 percent of interactions in Berkeley and Salt Lake City, respectively) than at the comparison intersection (4 and 15 percent of interactions in Berkeley and Salt Lake City, respectively).

In Berkeley, left turn motorist parallel path interactions with crossing pedestrians were a much lower percent of total interactions at the treated intersection (13 percent) than at the comparison intersection (49 percent) and a lower percent of pedestrians crossing at the intersection were involved with left turn parallel path interactions (5 percent) than at the comparison intersection (22 percent).

Salt Lake City saw a different trend, with less difference between treated and comparison intersections when it comes to left turning parallel path motorist interaction with pedestrians, and with the treated intersection having a slightly higher percent of interactions of this type than the comparison (24 percent).

Table 20. Bicyclist-Motorist Interactions in Berkeley.

Motorist Maneuver:	Treated					Comparison				
	Left	Right	Thru	Total Bike- MV Interactions	Bicycle Volume	Left	Right	Thru	Total Bike- MV Interactions	Bicycle Volume
Two-Stage Left	1	3	1	5	11	0	0	0	0	0
Regular Left	1	0	8	9	24	0	1	2	3	9
Right	0	0	0	0	29	0	0	0	0	10
Thru	4	21	2	27	100	4	12	2	18	91
U-turn	0	0	0	0	0	0	0	0	0	1
Not Coded	0	2	0	2	6	1	0	0	1	3
Total	6	26	11	43	170	5	13	4	22	114

"Not Coded" indicates that the coder could not determine the bicyclist turning movement.

Table 21. Bicyclist-Motorist Interactions in Salt Lake City.

Motorist Maneuver:	Treated						Comparison					
	Left	Right	Thru	U-turn	Total Bike- MV Interac- tions	Bicycle Volume	Left	Right	Thru	U-turn	Total Bike- MV Interac- tions	Bicycle Volume
Two-Stage Left	4	9	6	1	20	45	1	3	1	1	6	15
Two-Stage Thru	4	1	1	0	6	9	0	0	0	0	0	2
Regular Left	1	5	9	0	15	33	2	6	11	0	19	64
Right	1	0	0	0	1	92	2	0	2	0	4	72
Thru	23	33	25	0	81	254	32	26	29	0	87	367
U-turn	0	1	0	0	1	2	0	0	0	0	0	1
Not Coded	2	5	0	0	7	9	3	2	2	0	7	24
Total	35	54	41	1	131	444	40	37	45	1	123	545

Table 22. Pedestrian-Motorist Interaction Summary

Motorist Maneuver	Motorist Path Relative to Pedestrian Path	Berkeley		Salt Lake City	
		Treated	Comparison	Treated	Comparison
Motorist Left turn	Parallel Paths - Same Direction	10	12	83	110
	Parallel Paths - Opposite Directions	10	11	79	66
	Perpendicular Paths	4	0	17	11
Motorist Right turn	Parallel Paths - Same Direction	48	6	84	157
	Parallel Paths - Opposite Directions	55	14	89	165
	Perpendicular Paths	5	2	48	108
Thru Vehicle	Perpendicular Paths	14	2	154	96
Other / Unusual	Unclear Paths of Motorist or Pedestrian	7	0	12	11
		no U-turns	no U-turns	no U-turns	includes U-turns
	Total Interactions	153	47	566	724
	Total Pedestrians Observed	397	105	1,595	1802

5.6.2 ROAD USER COMPLIANCE

To understand the effect of the treatment on road user compliance with traffic laws and controls, rates were assessed for each of the following behaviors:

- Bicyclist compliance
 - Ran Light – Percentage of bicyclists that ran a red light during the main red phase
 - Arrive Green Exit Red – Percentage of bicyclists that ran a red light at the end of the yellow phase
 - Jump Green – Percentage of bicyclists that entered the intersection early, just before the green signal, “jumping” the green light
- Bicyclist position and direction of movement
 - Bicyclist riding position when entering the intersection, whether in the bike lane, a regular traffic lane, or entered from the sidewalk into crosswalk area
 - Direction of bicyclist travel when entering intersection, whether riding with or facing (wrong way/wrong side) adjacent traffic
 - Bicyclist two-stage or regular left turns
- Pedestrian crossing behavior
 - Walkman – Percentage of pedestrian crossing during the walk indication
 - Flashing – Percentage of pedestrians crossing during the flashing red hand indication
 - Stop Hand – Percentage of pedestrians crossing during the red hand indication

The research team distinguished between *running a red light as yellow turns to red* and *jumping a red light before red turns to green (early departure)*. While the former is unsafe behavior for motorists and bicyclists alike, the latter is often an intentional though non-compliant attempt to increase safety by traversing the intersection before motor vehicle traffic. Treated intersections were expected to have a lower rate of each violation type compared to untreated intersections.

Motorist compliance with red-lights was not specifically coded consistently, so it will not be discussed in this section.

5.6.2.1 Bicyclist Compliance

Table 23 show compliant and non-compliant bicyclist behavior at the Berkeley and Salt Lake City study intersections regardless of interactions with motor vehicles. They show the types of maneuvers as a percentage of total observed bicycle trips through each intersection. In Berkeley, the percentage of bicyclists jumping the green light at the treated intersection was 2 percentage points higher than at the comparison intersection. In Salt Lake City, the percentage of bicyclists jumping the green light was 1.3 percentage points lower at the treated intersection than at the comparison intersection. The treated intersection had 3.8 percentage points fewer bicyclists running the red light at the end of the yellow phase (Arrive Green Exit Red) than the comparison intersection in Berkeley but a higher percentage in Salt Lake City (0.7 percentage points higher). Finally, the treated intersection had a slightly higher percentage of bicyclists running the red light during the main part of the red phase than the comparison intersection in both cities (1.9 percentage points higher in Berkeley, 4.0 percentage points higher in Salt Lake City).

Table 23. Summary of Bicycle Compliance at Intersections

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Bicyclists (n)	162	112	436	536
Percent Bicyclists Jump Green	5.6%	3.6%	1.8%	3.2%
Percent Bicyclists Arrive Green Exit Red	0.6%	4.5%	1.8%	1.1%
Percent Bicyclists Ran Light	3.7%	1.8%	14.2%	10.3%
Percent Bicyclists Other Legal Behavior	90.1%	90.2%	82.1%	85.4%

5.6.2.2 Bicyclist Position and Direction of Movement

Bicyclist riding position on entering the intersection is shown in Table 24 and Figure 29 and Figure 30. Riding position may be influenced by the bicyclists' perceived comfort with the roadway sections leading to the intersection as well as the intersection itself. Motorists may better notice bicyclists who are not riding on the sidewalk. A majority of bicyclists at all locations - treated and comparison - in both cities were using the bike lane. However, in Berkeley, a larger percentage were using the bike lane at the comparison location (91 percentage) compared to the protected intersection (65 percent), and relatively more bicyclists were using the sidewalk or crosswalk area or the regular motor vehicle travel lanes at the treated intersection. There was also a higher percentage of vehicular-style lefts (12 percent) at the treated intersection (data shown below) than at the comparison location (7 percent).

In Salt Lake City, these trends were reversed, with a higher percentage of bicyclists (86 percent) using the bike lane at the treated intersection, and 58 percent at the comparison intersection. Consistent with this finding, more vehicular style left turns were made at the comparison intersection in Salt Lake City than the treated intersection.

Table 24. Bicyclist Riding Position when Entering Intersection

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Bicyclists (n non-missing)	166	112	444	541
Percent entering in bike lane	65.1%	91.1%	86.3%	57.5%
Percent of bicyclists in motor vehicle lane	24.7%	6.3%	6.5%	24.2%
Percent sidewalk/crosswalk	10.2%	2.7%	7.2%	18.3%

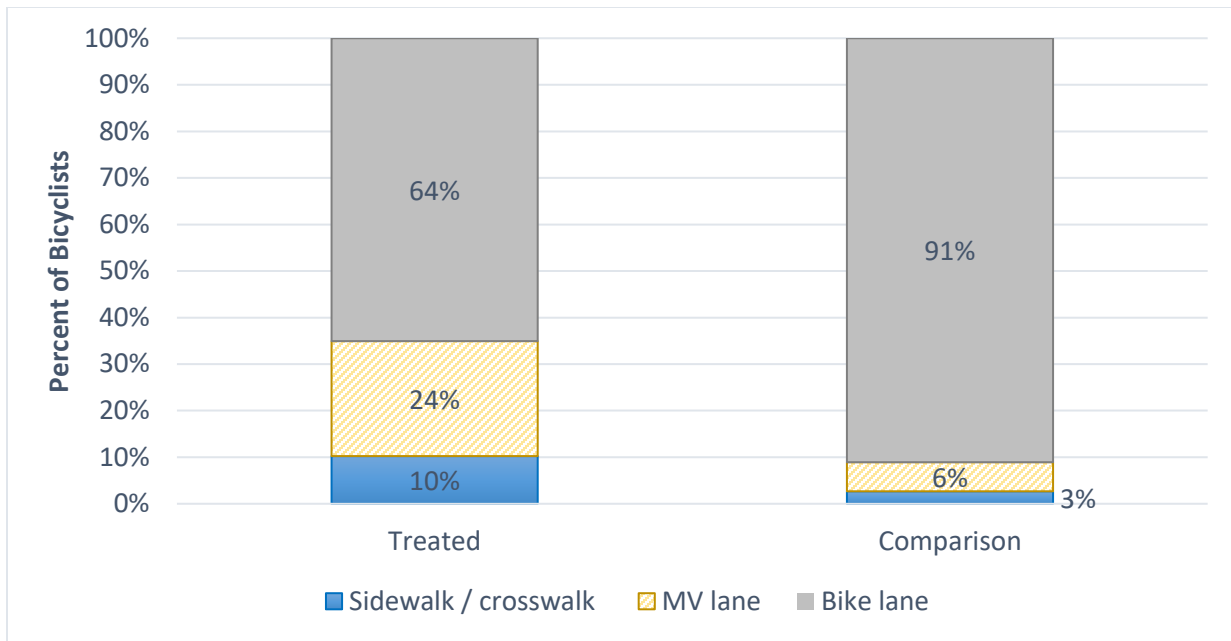


Figure 29. Berkeley - Bicyclist Riding Position Entering the Intersection

These factors could have safety ramifications if bicyclists in the bike lane are more noticeable to motorists than those cycling on the sidewalk. Since we lack before period data, we cannot say with certainty that the intersection design contributes to these differences, which may also be influenced by intended movement through the intersection.

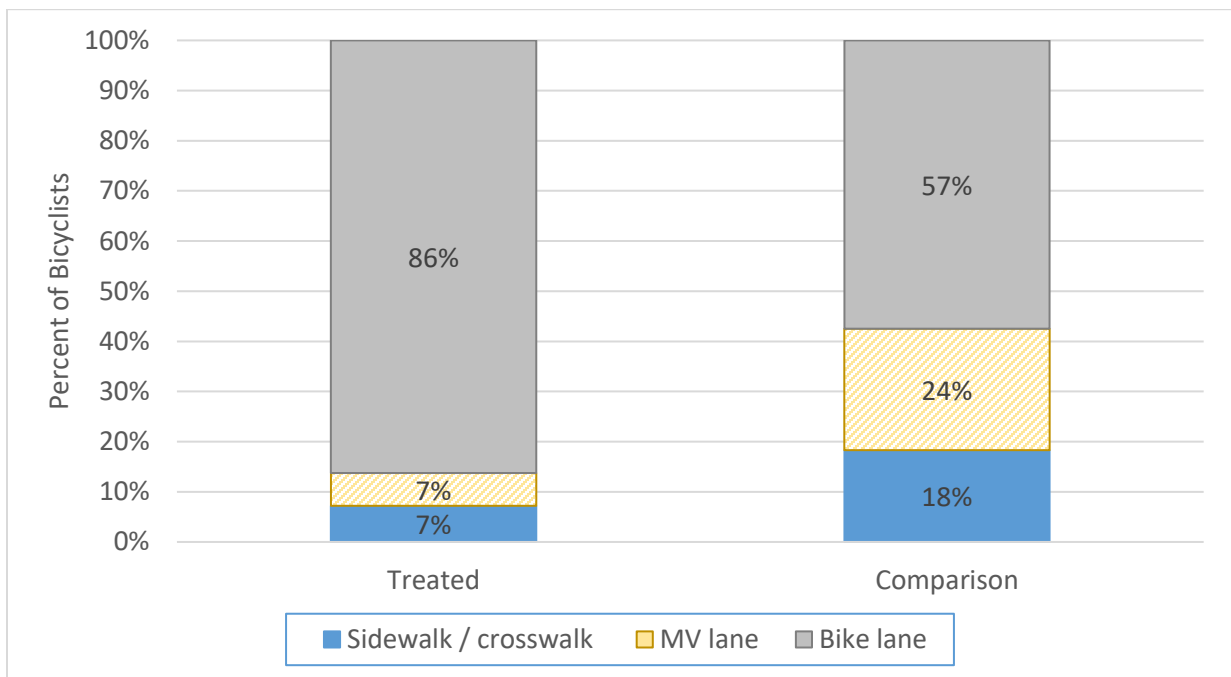


Figure 30. Salt Lake City - Bicyclist Riding Position Entering the Intersection

Riding direction may also influence bicyclist safety, since bicyclists approaching from an unexpected direction may be less likely to be observed before the motorist makes a maneuver (such as a right-turn-on-red). Here again, results are inconsistent across the two cities. In Berkeley, a higher percentage of bicyclists were riding against the direction of adjacent traffic at the protected intersection, whereas, the proportions were nearly identical for the treated and comparison locations in Salt Lake City (Table 25).

Table 25. Bicyclist riding direction entering the intersection

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Bicyclists (n non-missing)	165	112	446	545
Percent riding facing traffic	7.3%	1.8%	11.7%	11.6%
Percent riding with traffic	92.7%	98.2%	88.3%	88.4%

5.6.2.3 Bicyclist Left Turns

Another feature of protected intersection design that may affect bicyclist safety is whether it alters the propensity of bicyclists to make two-stage left turns, and whether that reduces the number of conflicts and potential crashes. When bicyclists must make left turns across busy on-coming traffic where separate turn phases are not provided, the bicyclist may conflict with opposite direction traffic, as well as with same direction traffic when there is a need to merge across same direction lanes to enter a left turn lane. In the case when no left turn lane is present, this turning movement is especially hazardous to bicyclists as it exposes them to possibly being hit from behind by a motorist as they wait to turn left from the vehicle lane.

The protected intersection design may encourage bicyclists to make two-stage left turns more often, thereby potentially reducing their exposure to these kinds of conflict, but potentially increasing bicyclists' exposure to motor vehicle turning traffic, since the bicyclist must cross two legs of the intersection. The data suggest that bicyclists at the protected intersections in both Berkeley and Salt Lake City made two-stage style left turns a higher percentage (31 percent in Berkeley; and 58 percent in Salt Lake City) of the time than those at the comparison intersections (Table 26). Note that in Salt Lake City, there were painted green boxes to encourage bicyclist two-stage lefts at the comparison intersection, and 19 percent of bicyclists making left turns there did make two-stage lefts, whereas, none made two-stage lefts at the comparison location in Berkeley, which lacked such a marking.

Table 26. Summary of Bicyclist Left Turn Maneuvers at Treated and Comparison Intersections

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
2-Stage Left (% of all lefts)	11	0	45	15
	31%	0	58%	19%
Vehicular Left (% of all lefts)	24	9	33	64
	67%	100%	42%	81%
Percentage of vehicular left Bikes entering intersection from BL, SW or MV thru lane	25%	44%	48%	42%
Total Left turns	36	9	78	79

5.6.2.4 Pedestrian Crossing Behavior

Table 27, Figure 31 and Figure 32 compare the percentage of pedestrians crossing during the red hand or flashing red hand at the Berkeley and Salt Lake City study intersections. They show pedestrian crossing behavior as a percentage of total observed pedestrian trips through each intersection. In Berkeley, 22 percentage points more observed pedestrians crossed during the stop hand at the treated intersection than the comparison intersection. In Salt Lake City, 6 percentage points more observed pedestrians crossed during the stop hand at the treated intersection than at the comparison intersection.

Of those who cross during the stop hand, some jump the green and others simply run the red light. For example, at the Salt Lake City treated intersection, of the 336 pedestrians who started crossing during the stop hand, 66 jumped the green (20 percent of those who started across during the stop-hand).

In Berkeley, the treated intersection had seven second pedestrian walk time on all approaches, but at the comparison intersection pedestrian walk time was seven seconds across one road but double (14 seconds) across the other road. In Salt Lake City, pedestrian walk time was seven seconds at the treated intersection and five seconds at the comparison intersection. Walk time is in addition to pedestrian clearance time which varied. Cycle length for both intersections was the same (65 to 70 seconds in Berkeley and 60 to 72 seconds in Salt Lake City). See Appendix C for signal timing details.

Table 27. Summary of Pedestrian Crossing Behavior at Intersections

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Pedestrians (n)	397	82	1,581	1,792
Percent Pedestrians Walkman	65.7%	74.4%	63.4%	63.6%
Percent Pedestrians Flashing	4.8%	18.3%	15.3%	21.1%
Percent Pedestrians Stop Hand	29.5%	7.3%	21.3%	15.2%

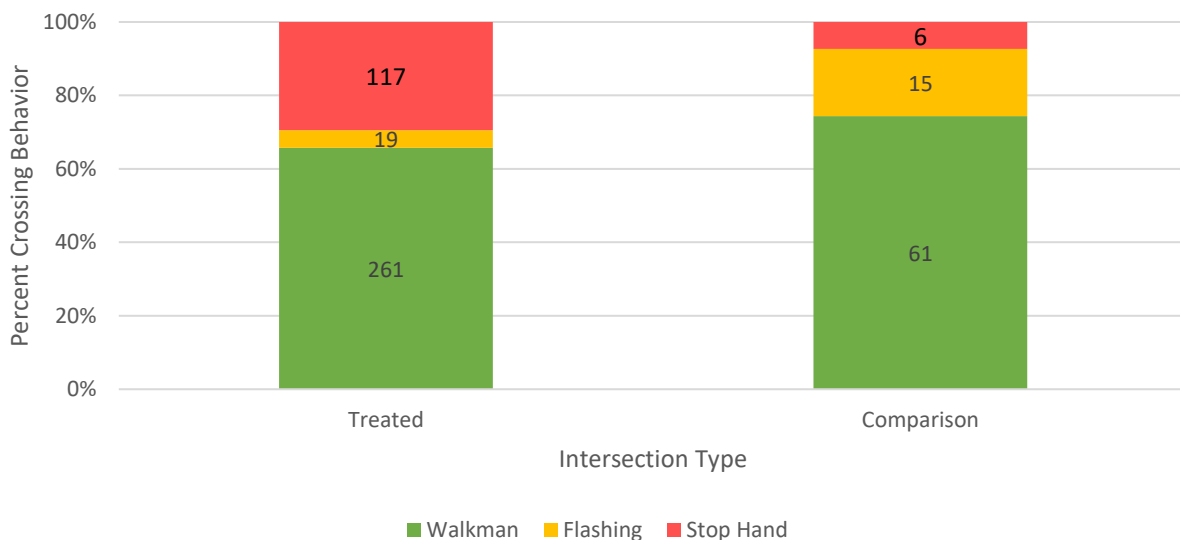


Figure 31. Berkeley | Pedestrian Crossing Behavior at Intersections

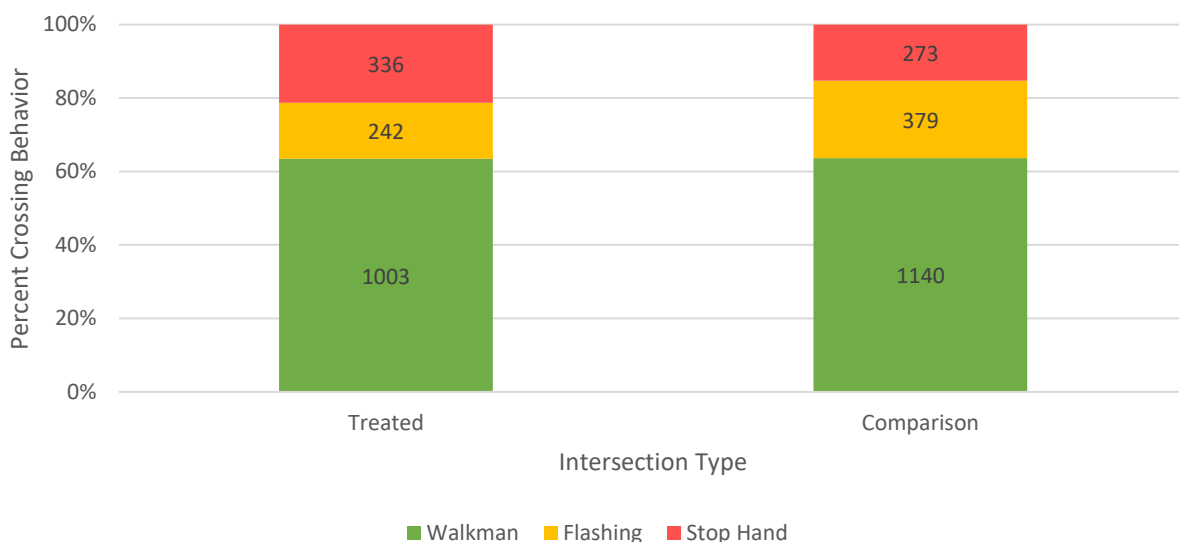


Figure 32. Salt Lake City | Pedestrian Crossing Behavior at Intersections

5.6.2.5 Pedestrian Crossing Position

Table 28, Figure 33 and Figure 34 compare the percentage of pedestrians crossing outside of the crosswalk at the Berkeley and Salt Lake City study intersections. They show the number of pedestrians who veer out of the painted crosswalk as a percentage of total observed pedestrians through each intersection. In Berkeley, 4 percentage points fewer “veer-out” crossing events were observed at the treated intersection than at the comparison intersection. The opposite was observed in Salt Lake City, where 9 percentage points more “veer-out” crossing events were observed at the treated intersection than at the comparison intersection. These results may be influenced by desire lines or other unknown factors that vary for the treated and comparison locations in the two cities.

Table 28. Summary of Pedestrian Crossing Position

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Pedestrians (n)	390	101	1563	1760
Uses Crosswalk	81.3%	77.2%	80.7%	89.7%
Veers Out	18.7%	22.8%	19.3%	10.3%

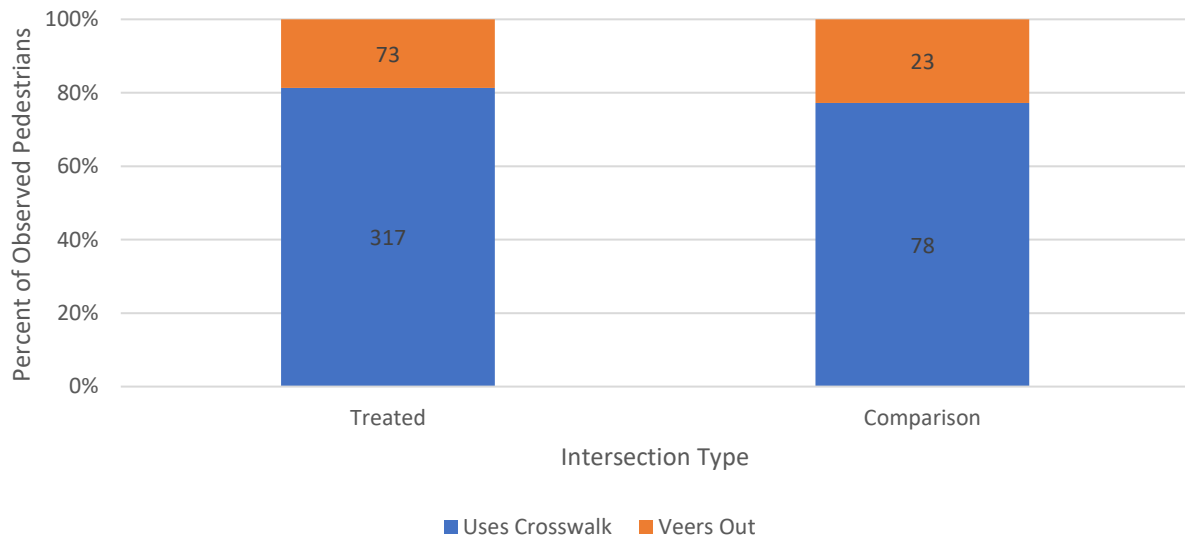


Figure 33. Berkeley | Pedestrian Crossing Position

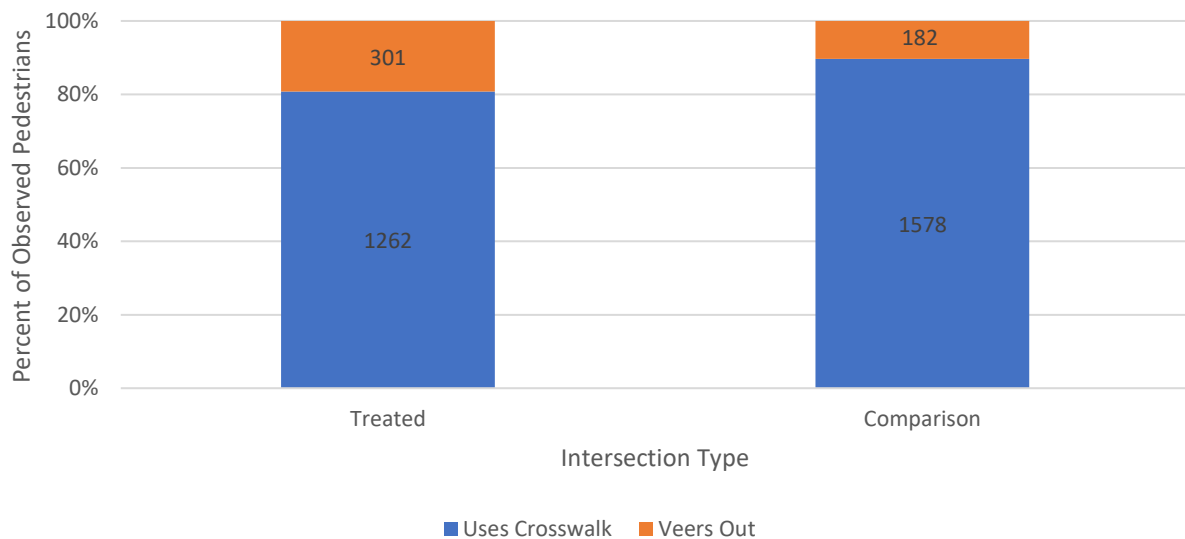


Figure 34. Salt Lake City | Pedestrian Crossing Position

5.6.3 YIELDING RATES

The research team assessed yielding rates to bicyclists and pedestrians, including in the following situations:

- Right- and left-turning motorists yielding to bicyclists (per total number of interactions between turning vehicles and bicycles)
- Right- and left-turning motorists yielding to pedestrians (per total number of interactions between turning vehicles and pedestrians)

- Right- and left-turning bicyclists yielding to pedestrians (per total number of interactions between turning bicycles and pedestrians)

The objective was to compare whether the treated intersections have a higher rate of motorists yielding to bicyclists and pedestrians and bicyclists yielding to pedestrians.

5.6.3.1 Driver Yielding to Bicyclists

Table 29, Figure 35 and Figure 36 compare total yielding rates at the Berkeley and Salt Lake City study intersections. They show the number and type of yielding interactions as a percentage of total observed interactions between left- and right-turning motorists and bicyclists. Table 29 also shows the yielding types as a percent of number of bicyclists and number of total interactions. Comparing the two intersections in Berkeley, motorists yielded to bicyclists less at treated intersections than at comparison intersections, though the low numbers of yielding interactions make the results less reliable. At the two sites in Salt Lake City, motorists yielded to bicyclists less in terms of percent of bicyclist-motorist yielding interactions at treated intersections than at comparison intersections.

In terms of percent of bicyclist-motorist turning yielding interactions, motorists were observed yielding to bicyclists less at the treated intersection (80 percent in Berkeley, 68 percent of interactions in Salt Lake City), and much more often at the comparison intersection (100 percent in Berkeley, 77 percent in Salt Lake City). Similarly, for all the metrics, bicyclists yielded to turning motorists at a higher percentage at the treated intersection than at the comparison intersection.

In Berkeley, the near miss at the treated intersection involved a left-turning motorist and a bicycle. In Salt Lake City, five near misses were observed between a turning motorist and a bicycle at the treated intersection (four with right turning motorists and one with a left turning motorist), and one near miss was observed at the comparison intersection.

Table 29. Summary of Bicyclist – Motorist Yielding Interactions | Left- and Right-Turning Motor Vehicles

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Number of Bicyclists Observed	169	114	444	545
Total Bike-MV Interactions	43	22	123	119
Bicyclist-MV Turning Yielding Interactions (n)	15	9	50	35
Bicyclist Yield to Motorist Interactions	2	0	11	7
Motorist Yield to Bicyclist Interactions	12	9	34	27
Near Miss Interactions	1	0	5	1
Bicyclist Yield to Motorist (% of bicyclists)	1%	0%	2%	1%
Motorist Yield to Bicyclist (% of bicyclists)	7%	8%	8%	5%
Near Miss (% of bicyclists)	1%	0%	1%	0%
Bicyclist Yield to Motorist (% of bike-MV interactions)	5%	0%	9%	6%
Motorist Yield to Bicyclist (% of bike-MV interactions)	28%	41%	28%	23%
Near Miss (% of bike-MV interactions)	2%	0%	4%	1%
Bicyclist Yield to Motorist (% of n)	13%	0%	22%	20%
Motorist Yield to Bicyclist (% of n)	80%	100%	68%	77%
Near Miss (% of n)	7%	0%	10%	3%

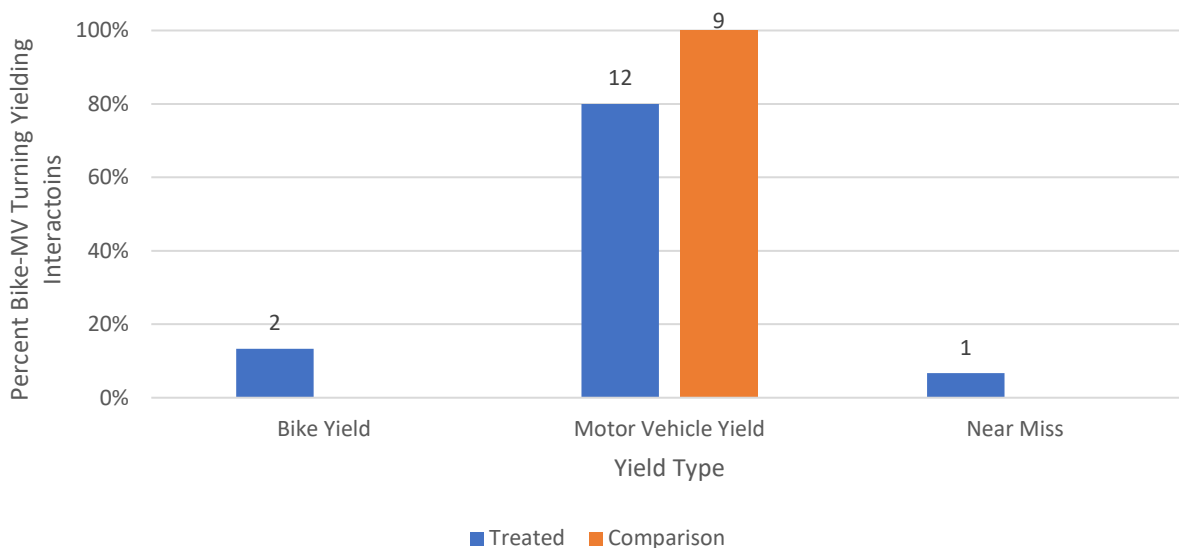


Figure 35. Berkeley | Bicyclist – Motorist Yielding Interactions | Left- and Right-Turning Motor Vehicles

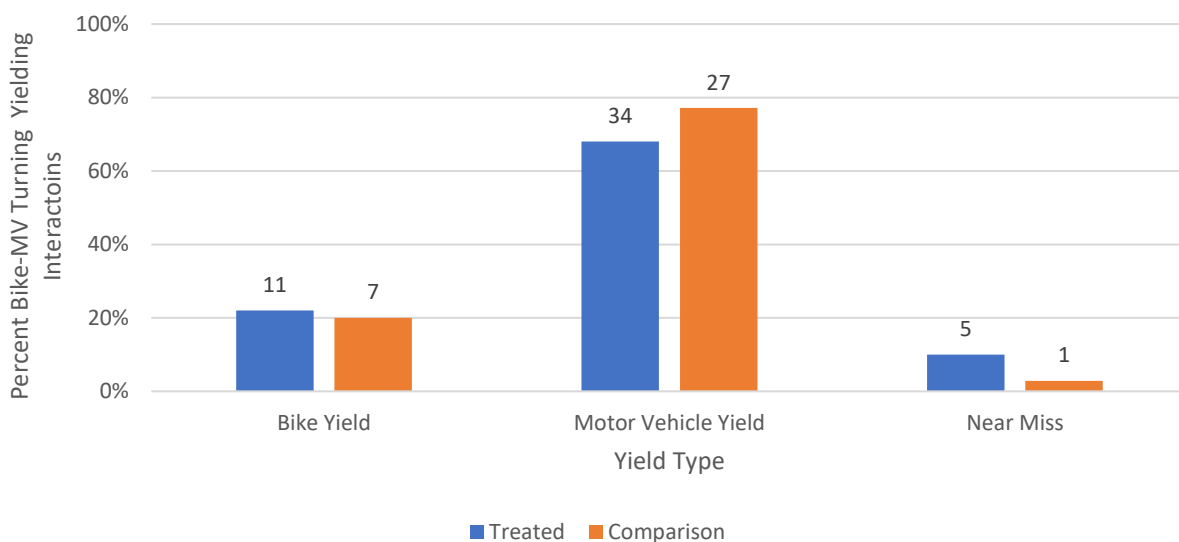


Figure 36. Salt Lake City | Bicyclist – Motorist Yielding Interactions | Left- and Right-Turning Motor Vehicles

Table 30 and Table 31 further divide the driver yielding shown in Table 29, Figure 35 and Figure 36 by motor vehicle turning movement for Berkeley and Salt Lake City study intersections. These tables show the number and type of yielding interactions between left- and right-turning motorists and bicyclists. Percentages are not shown because the total number of interactions is low.

In Berkeley, only five total yielding interactions were observed between a bicyclist and a motorist. Both of the left turning motorists at the comparison intersection yielded, while only one of the three left-turning

motorists yielded at the treatment intersection. For right turning vehicles at the Berkeley intersections, all seven of the motorist interactions with bicyclists at the comparison intersection involved yielded to bicyclists, and in all but one of the motorist interactions with bicyclists at the treated intersection the right turning motorist yielded to the bicyclist. This shows a high rate of yielding at both intersections

In Salt Lake City left-turning motorists yielded more frequently to bicyclists at the comparison intersection than at the treated intersection, and right-turning motorists yielded more frequently to bicyclists at the treated intersection than at the comparison intersection.

Table 30. Summary of Bicyclist – Motorist Yielding Interactions | Left-Turning Motor Vehicles

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Bike-MV Left-turn Yielding Interactions (n)	3	2	19	22
Bicyclist Yield to Motorist Interactions	1	0	7	1
Motorist Yield to Bicyclist Interactions	1	1	11	20
Near Miss Interactions	1	0	1	1

Table 31. Summary of Bicyclist – Motorist Yielding Interactions | Right-Turning Motor Vehicles

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Bike-MV Right-turn Yielding Interactions (n)	12	7	31	13
Bicyclist Yield to Motorist Interactions	1	0	4	6
Motorist Yield to Bicyclist Interactions	11	7	23	7
Near Miss Interactions	0	0	4	0

5.6.3.2 Driver Yielding to Pedestrians

Table 32, Figure 37 and Figure 38 compare total yielding rates at the Berkeley and Salt Lake City study intersections. They show the number and type of yielding interactions as a percentage of total observed interactions between left- and right-turning motorists and pedestrians.

In general, motorists yielded to pedestrians at about the same rate at the treated intersection as at the comparison intersection in Berkeley and in Salt Lake City, though there was slightly more yielding in Salt Lake City comparison intersection than at the treated intersection (3 percentage points more motorists yielding at the comparison intersection). Motorist yielding pedestrians at high rates in both cities. In Berkeley, no near misses were observed between a turning motorist and a pedestrian. In Salt Lake City, two near misses were observed at the comparison intersection and one at the treated intersection.

Table 32. Summary of Pedestrian-Motorist Yielding Interactions | Left- and Right-Turning Motor Vehicles

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Number of Pedestrians Observed	398	105	1,595	1,802
Total Pedestrian-MV Interactions	153	47	555	714
Pedestrian-MV Turning Yielding Interactions (n)	92	26	257	426
Pedestrian Yield to Motorist Interactions	2	1	18	15
Motorist Yield to Pedestrian Interactions	90	25	238	409
Near Miss Interactions	0	0	1	2
Pedestrian Yield to Motorist (% of pedestrians)	1%	1%	1%	1%
Motorist Yield to Pedestrian (% of pedestrians)	23%	24%	15%	23%
Near Miss (% of pedestrians)	0%	0%	0%	0%
Pedestrian Yield to Motorist (% of pedestrian-MV interactions)	1%	2%	3%	2%
Motorist Yield to Pedestrian (% of pedestrian-MV interactions)	59%	53%	43%	57%
Near Miss (% of pedestrian-MV interactions)	0%	0%	0%	0%
Pedestrian Yield to Motorist (% of n)	2%	4%	7%	4%
Motorist Yield to Pedestrian (% of n)	98%	96%	93%	96%
Near Miss (% of n)	0.0%	0.0%	0.4%	0.5%

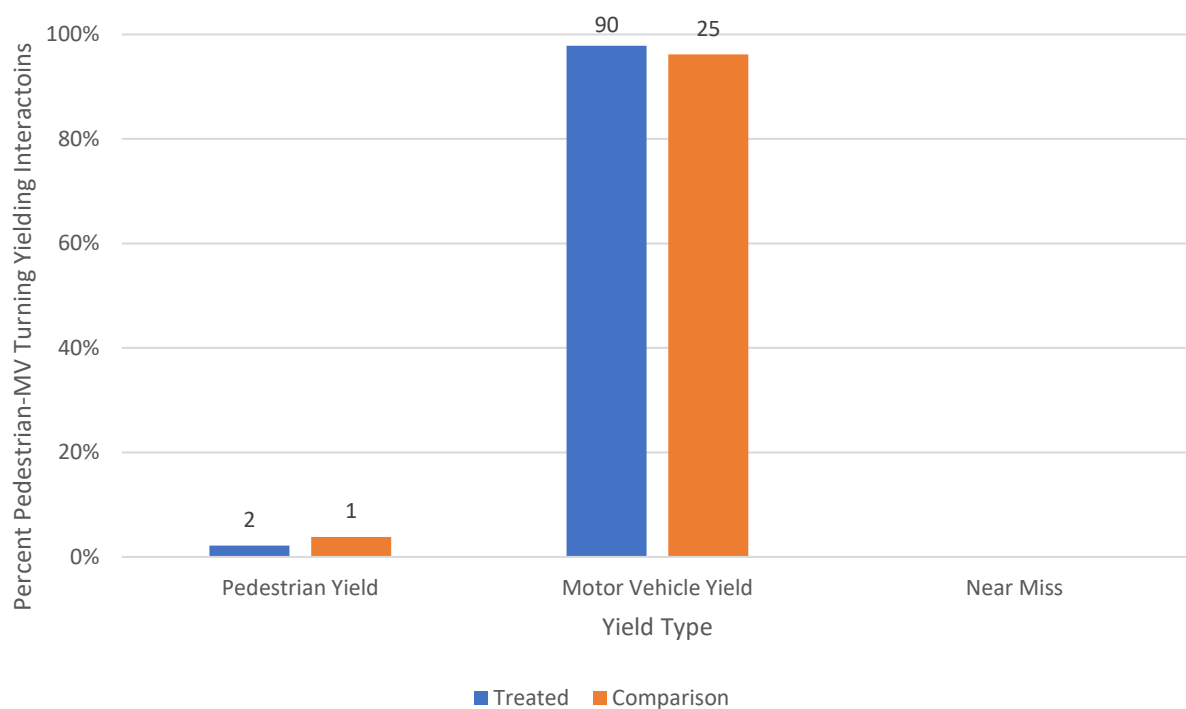


Figure 37. Berkeley | Pedestrian-Motorist Yielding Interactions | Left- and Right-Turning Motor Vehicles

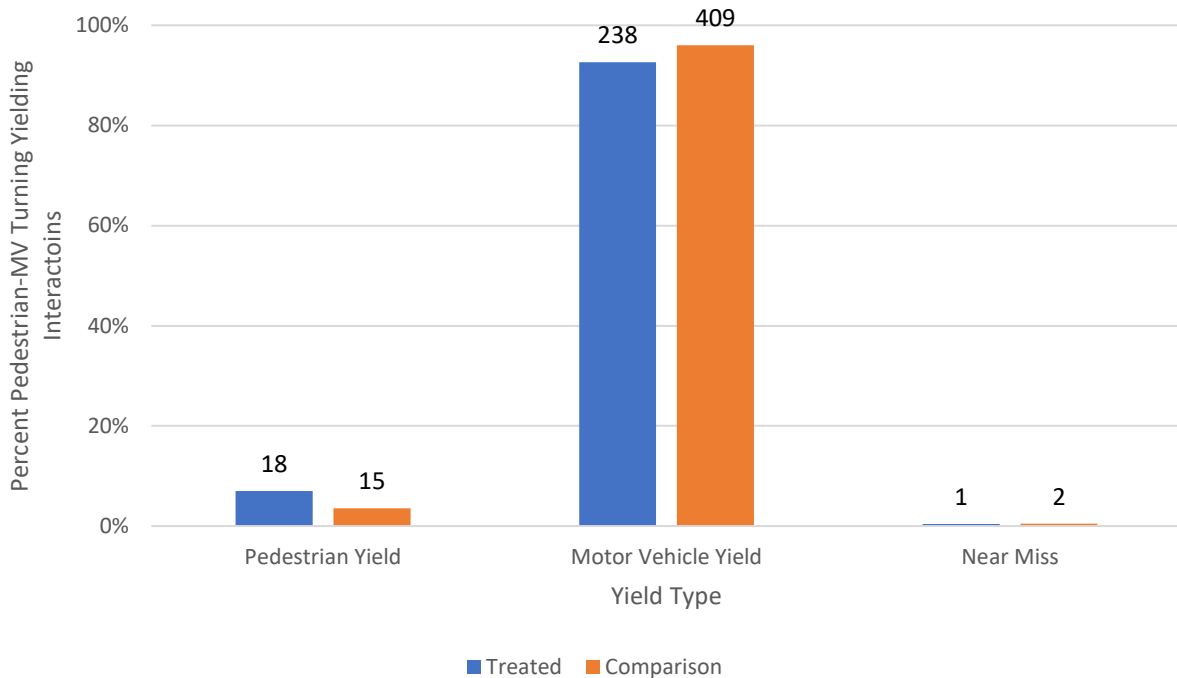


Figure 38. Salt Lake City | Pedestrian-Motorist Yielding Interactions | Left- and Right-Turning Motor Vehicles

Table 33 and Table 34 further divide the driver yielding to pedestrian numbers shown in Figure 37 and Figure 38 by motor vehicle turning movement for Berkeley and Salt Lake City study intersections. These tables show the number and type of yielding interactions as a percentage of total observed interactions between left- and right-turning motorists and pedestrians.

In Berkeley, yielding behavior for left- and right-turning motorists was relatively similar, with a slightly higher percentage of right-turning motorists yielding at the treated intersection than at the comparison intersection (4 percentage points more right-turning motorists yielding at the treated intersection). In Salt Lake City, yielding behavior for left- and right-turning motorists was also relatively similar, but with a higher percentage of left- and right-turning motorists yielding at the comparison intersection rather than at the treated intersection (6 percentage points more left-turning motorists yielding at the comparison intersection and 2 percentage points more right-turning motorists yielding at the comparison intersection).

Table 33. Summary of Pedestrian-Motorist Yielding Interactions | Left-Turning Motor Vehicles

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Pedestrian-MV Left Turning Yielding Interactions (n)	12	12	105	133
Pedestrian Yield to Motorist (% of n)	0%	0%	7%	1%
Motorist Yield to Pedestrian (% of n)	100%	100%	92%	98%
Near Miss (% of n)	0%	0%	1%	1%

Table 34. Summary of Pedestrian-Motorist Yielding Interactions | Right-Turning Motor Vehicles

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Pedestrian-MV Right Turning Yielding Interactions (n)	80	14	143	290
Pedestrian Yield to Motorist (% of n)	3%	7%	7%	5%
Motorist Yield to Pedestrian (% of n)	97%	93%	93%	95%
Near Miss (% of n)	0%	0%	0%	0.3%

5.6.3.3 Bicyclist Yielding to Pedestrians

Only one interaction was observed between a left-turning bicyclist and a pedestrian at either Berkeley intersection. In this case, the bicyclist yielded to the pedestrian at the treated intersection. No interactions between right-turning bicyclists and pedestrians were observed at either Berkeley study intersection.

Only three total bicyclist-pedestrian interactions were observed at the Salt Lake City intersections. Two were yielding interactions between right-turning bicyclists and pedestrians at the comparison intersection in which the bicyclist yielded to the pedestrian. And the other bicyclist-pedestrian interaction in Salt Lake City was a near miss between a left-turning bicyclist and a pedestrian at the treated intersection.

Due to this limited data, little can be learned from this study, but it does demonstrate that bicyclists did yield to pedestrians in the study.

5.6.4 AVOIDANCE MANEUVERS AND CONFLICTS

The third safety criterion focused on interactions among road users, including any observable conflicts, as evidenced by avoidance maneuvers (simple maneuvering) and conflicts (close calls or near misses that could be observed). One can think of the avoidance maneuver as a measure of yielding severity.

These maneuvers were documented for interactions between vehicles and bicycles, between vehicles and pedestrians, and between bicycles and pedestrians. Every interaction was coded by indicating which party passed through the potential conflict area first and second. If noticeable slowing (gradual or sudden), stopping, or trajectory changes were observed by either or both parties, these were also coded:

- MV Gradual – Gradual change in motor vehicle speed or trajectory
- Bike Gradual – Gradual change in bicycle speed or trajectory
- Ped Gradual – Gradual change in pedestrian speed or trajectory
- MV Sudden – Sudden change in motor vehicle speed or trajectory
- Bike Sudden- Sudden change in bicycle speed or trajectory
- Ped Sudden – Sudden change in pedestrian speed or trajectory
- Both slow/stop – Both parties in the interaction slow or stop

If noticeable change in speed or stopping by one or both parties occurred, this information was used to determine if an intentional yield of right-of-way was made. If no changes in speed or trajectory by either party were observed, these were denoted as Motor Vehicle Cross Bike (motor vehicle arrived and proceeded first through the potential conflict area), Bike Cross Motor Vehicle (the bicycle arrived and proceeded first), Pedestrian Cross Bike, and so forth. In these cases, it is also possible that the second party slowed on the approach, outside the observation zone of the cameras, and so, in a broad sense,

could also have constituted “yielding.” Avoidance maneuvers labeled “Sudden” are considered severe avoidance maneuvers.

More “severe” interactions (labeled “Sudden”) were noted if any of the following circumstances were observed. If one or more road users executed a quick maneuver (sudden change in speed or trajectory as noted above) to avoid a collision with another road user, for example:

- Sudden swerving (changes in trajectory) by motorists and/or bicyclists
- A quick side- or back-stepping by pedestrians to avoid a turning vehicle or bicyclist – excluding instances of distraction, such as a pedestrian looking at a cell phone (where identifiable)
- Sudden stopping to avoid another road user
- Avoidance maneuver by a bicyclist to pass a slower moving bicyclist
- Very close interaction in space or time (even if no party noticeably reacted) was deemed a near miss

5.6.4.1 Bicyclist-Motorist Avoidance Maneuvers

Table 35 compares interactions between bicyclists and motor vehicles at both the Berkeley and Salt Lake City study intersections. Figure 39. provides a comparison of interactions between bicyclists and motor vehicles at the Berkeley study intersections. Only one severe avoidance maneuver was observed (MV Sudden) at the treated intersection, and no severe avoidance maneuvers were observed at the comparison intersection.

Across both Berkeley intersections, treated and comparison, there were 65 interactions involving motor vehicles and bicyclists. In 35 of the interactions, there was no noticeable change in speed or trajectory (reaction) of either the motor vehicle or bicyclist at and near the conflict area. Of the 65 interactions, there were only 30 where one party noticeably increased the rate of slowing (or changed trajectory) as they approached the conflict area, seemingly in response to the other party being present. All except one of these, involved gradual slowing/stopping.

Figure 40. compares interactions between bicyclists and motor vehicles at the Salt Lake City study intersections. Two severe avoidance maneuvers were observed (MV Sudden) at the treated intersection, and one severe avoidance maneuver was observed (Bike Sudden) at the comparison intersection.

The lack of severe avoidance maneuvers between motorists and bicyclists means that we cannot compare the treated and comparison intersections for this metric.

Table 35. Summary of Bicyclist – Motorist Avoidance Maneuvers

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Maneuvers (n)	42	23	126	122
Bike cross MV	31.0%	26.1%	19.0%	21.3%
Bike Gradual	21.4%	4.3%	13.5%	15.6%
Bike Sudden	0.0%	0.0%	0.0%	0.8%
Both slow/stop	2.4%	0.0%	4.8%	2.5%
MV cross Bike	19.0%	34.8%	34.1%	36.1%
MV Gradual	23.8%	34.8%	27.0%	23.8%
MV Sudden	2.4%	0.0%	1.6%	0.0%

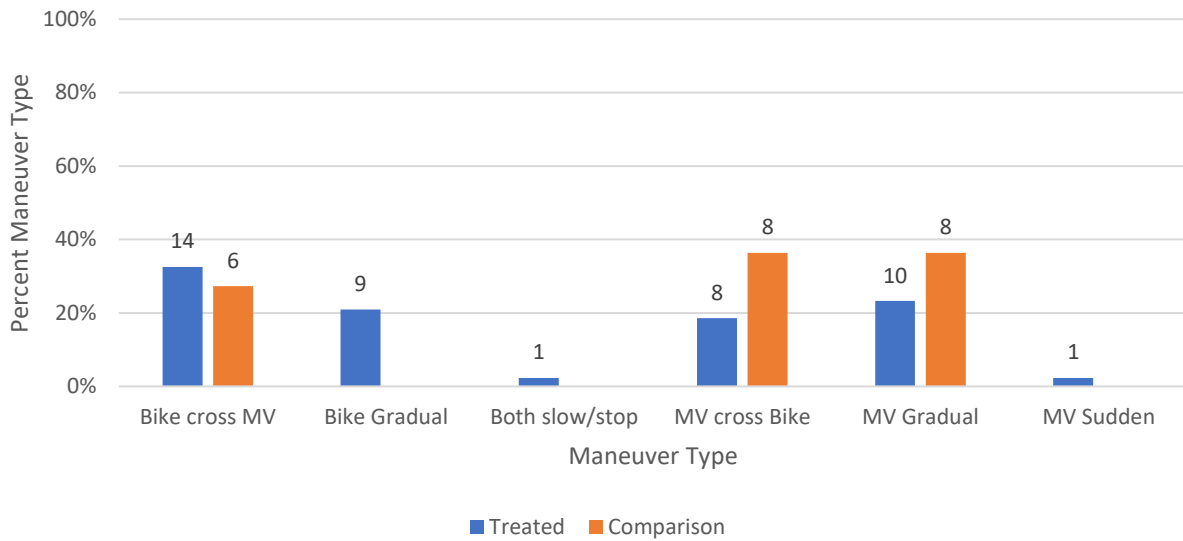


Figure 39. Berkeley | Bicyclist-Motorist Interactions | Maneuvers and Avoidance Maneuvers

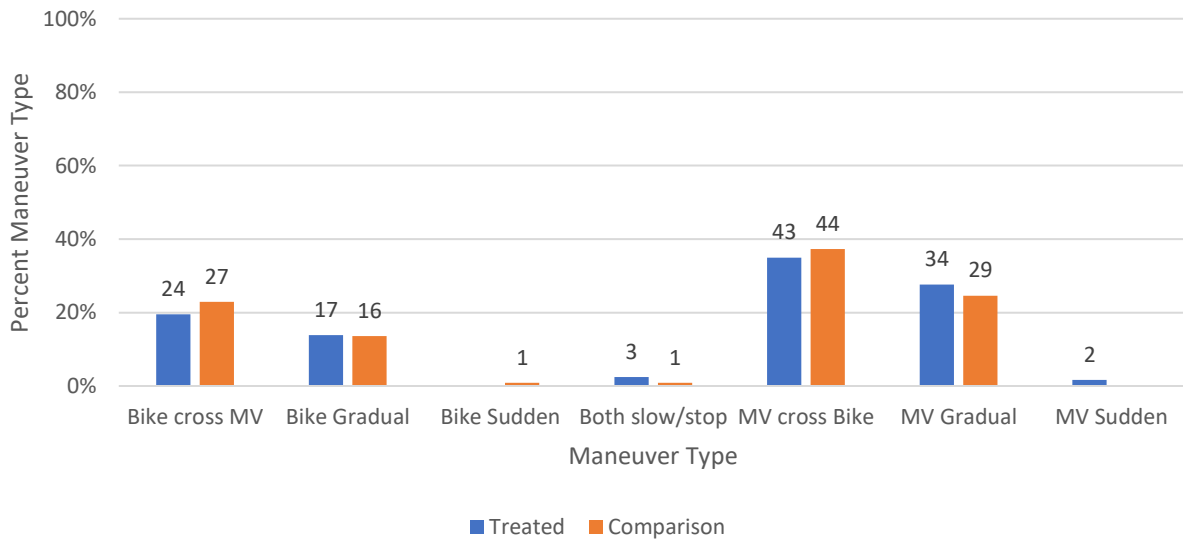


Figure 40. Salt Lake City | Bicyclist-Motorist Interactions | Maneuvers and Avoidance Maneuvers

5.6.4.2 Pedestrian-Motorist Avoidance Maneuvers

Table 36 and Figure 41 compares interactions between pedestrians and motor vehicles at the Berkeley study intersections. Only three severe avoidance maneuvers were observed (MV Sudden) at the treated intersection, and no severe avoidance maneuvers were observed at the comparison intersection.

Table 36 and Figure 42 compares interactions between pedestrians and motor vehicles at the Salt Lake City study intersections. Five severe avoidance maneuvers were observed (three MV sudden and two Ped

Sudden) at the treated intersection, and six severe maneuvers were observed (five MV sudden and one Ped Sudden) at the comparison intersection.

Table 36. Summary of Pedestrian – Motorist Avoidance Maneuvers

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Ped-MV Maneuvers (n)	154	47	553	711
Both slow/stop	0.6%	2.1%	2.4%	1.0%
MV cross Ped	11.7%	17.0%	30.6%	19.1%
MV Gradual	55.2%	51.1%	42.3%	57.1%
MV Sudden	1.9%	0.0%	0.5%	0.7%
Ped cross MV	27.9%	27.7%	19.3%	19.4%
Ped Gradual	2.6%	2.1%	4.9%	2.5%
Ped Sudden	0.0%	0.0%	0.4%	0.1%

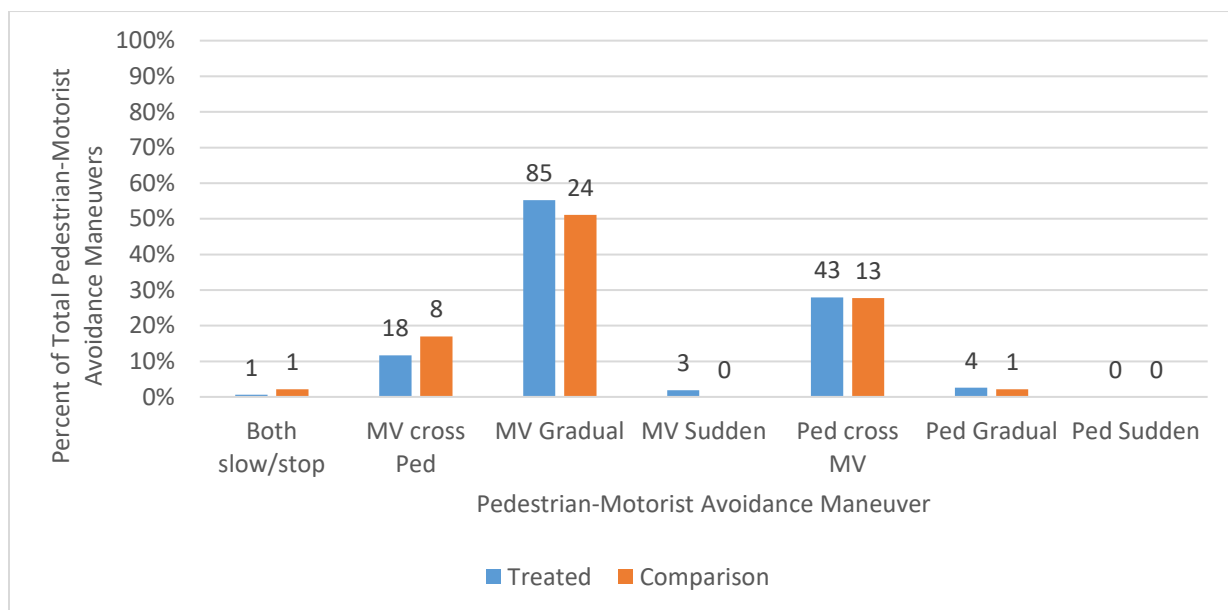


Figure 41. Berkeley | Pedestrian – Motorist Interactions | Maneuvers and Avoidance Maneuvers

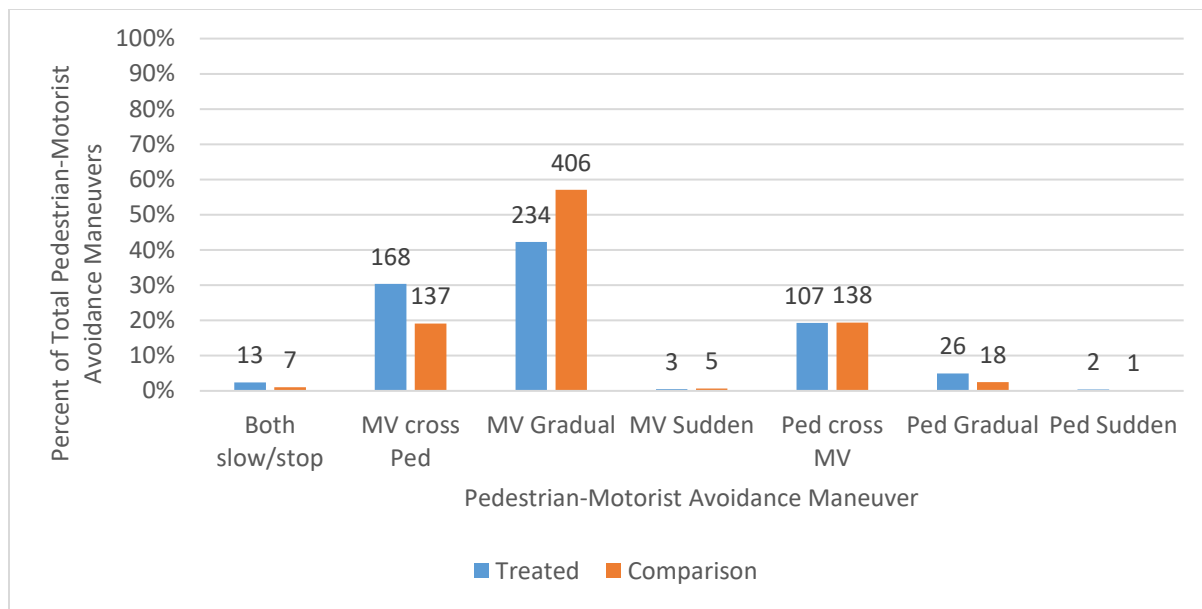


Figure 42. Salt Lake City | Pedestrian – Motorist Interactions | Maneuvers and Avoidance Maneuvers

No avoidance maneuvers were observed in interactions between bicyclists and pedestrians at any of the study intersections in either city.

5.6.5 PEDESTRIAN AND BICYCLIST WAITING POSITION AT PROTECTED INTERSECTIONS

The fourth safety criterion focused on pedestrian waiting behavior at intersection corners and medians. The team observed whether pedestrians wait in the protected part of the treated intersection beyond the “original” curb. The hypothesis was that if pedestrians wait in the protected part of the intersection beyond the curb, also known as the “pedestrian crossing island,” then the decreased crossing distance and increased visibility may reduce potential for interactions or conflicts with turning vehicles (MassDOT 2015).

Table 37 compares pedestrian waiting positions at the treated intersections in Berkeley and Salt Lake City, respectively. In Berkeley, very few pedestrians were observed waiting in the protected portion of the treated intersection (5.7 percent). This presents a stark contrast to the treated intersection in Salt Lake City, where over half of pedestrians were observed waiting in the protected portion of the intersection (53.1 percent). This difference in waiting positions could be due to the fact that the Salt Lake City protected intersection was implemented approximately one year prior (October 2015) to the Berkeley protected intersection (December 2016). Pedestrians have had more time to become accustomed to the Salt Lake City treated and could be more comfortable waiting in the protected portion of the intersection.

Table 37. Summary of Pedestrian Waiting Position

	Berkeley	Salt Lake City
Total Pedestrians (n)	246	431
Pedestrian Crossing Island	5.7%	53.1%
Original Curb	94.3%	46.9%

5.6.6 SPEED OF VEHICLES AND BICYCLISTS

The fifth safety criterion was intended to understand the effect of the intersection design, specifically the turning radius and “bend out” geometry, on motor vehicle and bicycle speeds. The team collected vehicle speeds as a turning movement was completed as shown in Figure 27 and Figure 28. These data address the question of whether protective intersection measures encourage motorists to slow down when turning right or left through a protected intersection compared to a conventional intersection.

Speeds of left and right turning motor vehicles and through bicyclists were collected using radar guns as previously described. The speeds were adjusted based on the angle between the vehicle trajectory and the direction the radar gun was pointed using the following relationship:

$$v_o = \frac{v_m}{\cos \beta}$$

where

v_o = adjusted speed of vehicle

v_m = measured speed

β = the angle between the vehicle trajectory and the direction the radar gun was pointed

Angles with β greater than 30 degrees were excluded from analysis because a small change in angle could result in a large adjustment to the speed. Since the angle, β , was not known precisely, such small variations in the angle could lead to misleading results.

Analysis of the difference between speeds at treated and comparison intersections for a given city was performed using a one-tailed T-test for distributions with unequal variances to determine if speeds at treated intersection were significantly lower or higher than speeds at the comparison intersection. Corresponding p-values are reported in the tables below. P-values less than 0.05 are considered significant for this analysis.

5.6.6.1 Right-Turning Motor Vehicles Speeds

Table 38 shows right-turning motor vehicle speeds through the Berkeley and Salt Lake City intersections. The average right-turning motor vehicle speed through the treated intersection was 14 mph in Berkeley and 12 mph in Salt Lake City.

In Berkeley, applying the criterion that speed readings from radar guns with angles, β , to vehicle trajectory greater than 30 degrees should not be used in analysis, left only data from the treated intersection for analysis. Radar gun to right turning vehicle trajectory angle for the comparison intersection was 32 to 37 degrees. For this reason, we are not reporting findings for right-turning vehicle speeds in Berkeley.

The Salt Lake City protected intersection had significantly lower southbound right-turning motor vehicle speeds than the comparison intersection. Northbound right turns were also measured, but the data were not included in the analysis because the radar gun operator’s angle with respect to the turning traffic was too high (37 degrees), and thus likely to result in misleading measurements.

The two Salt Lake City intersections are just about perfectly square with no differences between the north/south approaches in terms of angle or grade (or at least any grade differences should be consistent at both intersections). As a result, we would think the northbound and southbound right-turn speeds would be similar.

Table 38. Summary of Right-Turning Motor Vehicle Speeds

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Right-turning Motor Vehicles (n)	179	-	15	42
< 10 mph	9%	-	20%	0%
10 mph >= and < 15 mph	62%	-	67%	74%
15 mph >= and < 20 mph	26%	-	13%	26%
>= 20 mph	2%	-	0%	0%
Average	14	-	12	13
T-test p-value	Not enough data		0.0260	

“-” indicates no data available.

5.6.6.2 Left-Turning Motor Vehicle Speeds

As shown in Table 39, the Berkeley protected intersection had significantly higher left-turning motor vehicle speeds at the protected intersection than at the comparison intersection. The Salt Lake City protected intersection had significantly lower left-turning motor vehicle speeds at the protected intersection than at the comparison intersection.

These conflicting findings indicate that more research is needed to understand if protected intersections are associated with lower or higher motor vehicle left-turning speeds. However, protected intersections are not designed to specifically reduce left-turning vehicles speeds.

Table 39. Summary of Left-Turning Motor Vehicle Speeds

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Left-turning Motor Vehicles (n)	85	59	43	84
< 10 mph	1%	10%	5%	
10 mph >= and < 15 mph	13%	34%	63%	52%
15 mph >= and < 20 mph	69%	46%	30%	42%
>= 20 mph	16%	10%	2%	6%
Average Speed	17	15	14	15
T-test p-value	0.0004		0.0034	

5.6.6.3 Bicyclist Speeds

Table 40 shows bicyclist speeds passing through the intersections.

Speeds at the Berkeley intersection are for north and southbound bicyclists on The Alameda at both treated and comparison intersections. In the treated intersection, southbound bicyclists are traveling downhill, while at the comparison intersection northbound bicyclists are traveling downhill, though the magnitude of the grade may differ between intersections. Adjusted speeds in Berkeley vary from zero to 15 mph and average 8 mph at the treated intersection and 6 mph at the comparison intersection.

In Berkeley, there was no significant difference in bicycle speeds between treated and comparison intersections. However, the bicycle speed data were sparse. Four of these (two at the treated and two at the comparison) were recorded as traveling at zero speed, a reading that is likely to be misrepresenting the actual speed since the radar guns are not recommended for use on objects traveling less than 10 mph. Thus, there was insufficient data for valid comparison.

In Salt Lake City, bicyclist speeds at the treated intersection were higher than at the comparison intersection, but when high angle (β) measurements are removed, the difference was not significant. Only east and westbound bicyclists traveling on 300 S in downtown Salt Lake City are included in the analysis due to high angle (β) measurements for speeds collected in the other directions. In Salt Lake City, bicyclist eastbound speeds (no other direction was included in analysis) varied from 11 to 16 mph and averaged 14 mph at the treated intersection. And west and eastbound bicyclist speed varied from zero to 20 mph and averaged 11 mph at the comparison intersection. These intersections are both relatively flat, so grade is unlikely to impact speeds. When only eastbound bicyclists are compared between intersections, the results are similar to those reported in Table 40.

Table 40. Summary of Bicycle Speeds

	Berkeley		Salt Lake City	
	Treated	Comparison	Treated	Comparison
Total Bicycles (n)	5	5	2	11
< 10 mph	40%	60%	0%	27%
10 mph $\geq$ and < 15 mph	40%	20%	50%	55%
15 mph $\geq$ and < 20 mph	20%	20%	50%	18%
$\geq$ 20 mph	0%	0%	0%	0%
Average Speed	8	6	14	11
T-test p-value	0.3364		0.2621	

5.6.7 CRASH HISTORY

Since protected intersections are only recently being constructed in cities across the U.S., data are limited for crashes involving a motorist with a bicyclist or pedestrian before and after installation of protective measures. Under-reporting of crashes involving bicyclists and pedestrians results in further limitation of available datasets. Despite these challenges, the research team reviewed the crash history for each intersection. This evaluation focused on:

- Types of crashes (specifically those involving a motorist with a bicyclist or pedestrian)
- Severity of crashes (minor, property damage, injury, fatal)
- Changes in crash patterns before and after treatment

Berkeley crash data was available from January 2006 to November 2018. Table 41 shows crash types and severity at the treated intersection, and Table 42 shows crash types and severity at the comparison intersection. The protected intersection treatment was implemented in December 2016, and both tables indicate whether each crash occurred before or after treatment implementation.

Nineteen crashes occurred at Berkeley's treated intersection in the 11 years before the treatment installation and five in the approximately two years after the protected intersection was implemented. All crashes involving a bicyclist or pedestrian occurred prior to implementation. Thirty-nine crashes occurred at Berkeley's comparison intersection from 2006 to 2018. Three of the 39 crashes involved a bicyclist or pedestrian, and one crash involving a pedestrian resulted in a fatality.

Table 41. Berkeley | Treatment Intersection | Crash Data

Date	Type	Severity	Before/After
3/16/2007	Broadside	Property Damage Only	Before
4/29/2007	Rear End	Injury (Complaint of Pain)	Before
8/20/2007	Broadside	Injury (Complaint of Pain)	Before
11/26/2007	Broadside	Property Damage Only	Before
12/29/2007	Broadside	Property Damage Only	Before
9/17/2008	Sideswipe	Property Damage Only	Before
3/21/2009	Rear End	Property Damage Only	Before
5/9/2009	Vehicle/Pedestrian	Injury (Complaint of Pain)	Before
8/26/2009	Vehicle/Pedestrian	Injury (Other Visible)	Before
7/15/2010	Sideswipe	Property Damage Only	Before
9/1/2010	Sideswipe	Property Damage Only	Before
12/15/2010	Sideswipe	Property Damage Only	Before
1/25/2013	Broadside	Injury (Complaint of Pain)	Before
3/23/2013	Rear End	Property Damage Only	Before
4/22/2013	Vehicle/Pedestrian	Injury (Other Visible)	Before
10/3/2013	Vehicle/Pedestrian	Injury (Complaint of Pain)	Before
10/23/2013	Head-On	Injury (Other Visible)	Before
4/11/2014	Head-On	Property Damage Only	Before
3/11/2015	Head-On	Injury (Other Visible)	Before
5/23/2017	Rear End	Injury (Complaint of Pain)	After
6/5/2017	Hit Object	Property Damage Only	After
10/31/2017	Broadside	Property Damage Only	After
5/17/2018	Broadside	Injury (Complaint of Pain)	After
7/24/2018	Hit Object	Injury (Severe)	After

Table 42. Berkeley | Comparison Intersection | Crash Data

Date	Type	Severity	Before/After
1/2/2006	Rear End	Injury (Complaint of Pain)	Before
1/6/2006	Rear End	Injury (Complaint of Pain)	Before
2/16/2006	Broadside	Property Damage Only	Before
4/20/2006	Rear End	Injury (Complaint of Pain)	Before
5/16/2006	Sideswipe	Property Damage Only	Before
5/18/2006	Broadside	Property Damage Only	Before
7/22/2006	Sideswipe	Property Damage Only	Before

Date	Type	Severity	Before/After
4/17/2007	Broadside	Property Damage Only	Before
6/4/2007	Broadside	Property Damage Only	Before
7/12/2007	Broadside	Injury (Complaint of Pain)	Before
12/12/2007	Broadside	Injury (Complaint of Pain)	Before
1/28/2008	Broadside	Injury (Complaint of Pain)	Before
4/13/2008	Rear End	Injury (Complaint of Pain)	Before
4/23/2008	Sideswipe	Property Damage Only	Before
7/15/2008	Sideswipe	Property Damage Only	Before
10/25/2008	Not Stated	Injury (Other Visible)	Before
1/15/2009	Broadside	Property Damage Only	Before
4/18/2009	Vehicle/Pedestrian	Injury (Other Visible)	Before
4/26/2009	Other	Property Damage Only	Before
9/13/2009	Head-On	Property Damage Only	Before
1/21/2010	Rear End	Property Damage Only	Before
2/15/2010	Sideswipe	Property Damage Only	Before
2/23/2010	Head-On	Injury (Other Visible)	Before
7/25/2010	Head-On	Injury (Other Visible)	Before
2/14/2012	Sideswipe	Property Damage Only	Before
2/27/2012	Head-On	Injury (Complaint of Pain)	Before
9/6/2012	Broadside	Injury (Complaint of Pain)	Before
10/18/2012	Broadside	Property Damage Only	Before
12/12/2012	Broadside	Injury (Complaint of Pain)	Before
10/30/2013	Broadside	Injury (Other Visible)	Before
6/23/2014	Head-On	Injury (Complaint of Pain)	Before
8/22/2014	Broadside	Injury (Other Visible)	Before
6/17/2015	Other	Injury (Complaint of Pain)	Before
10/9/2016	Sideswipe	Property Damage Only	Before
7/17/2017	Broadside	Property Damage Only	After
5/14/2018	Head-On	Property Damage Only	After
7/9/2018	Broadside	Injury (Complaint of Pain)	After
8/7/2018	Vehicle/Pedestrian	Injury (Complaint of Pain)	After
8/17/2018	Vehicle/Pedestrian	Fatal	After

Salt Lake City crash data was available from January 2010 through August 2018. Table 43 shows crash types at the treated intersection, and Table 44 shows crash types at the comparison intersection. The protected intersection treatment was implemented in October 2015, and both tables indicate whether each crash occurred before or after treatment implementation.

Table 43. Salt Lake City | Treated Intersection | Crash Data

Date	Type	Before/After
2/19/2010	DUI collision with fixed object	Before
2/1/2012	2 vehicle collision	Before
12/8/2012	Intersection related	Before
3/21/2013	Intersection related	Before
6/13/2013	Intersection Related	Before
10/1/2013	Bicycle	Before
10/27/2014	Pedestrian involved, speed related, intersection related	Before
12/14/2014	Drowsy driving, distracted driving, work zone related	Before
1/23/2015	Bicyclist involved, speed related, teenage driver involved	Before
9/9/2015	Work zone related	Before
11/24/2015	Intersection related	After
1/13/2016	Pedestrian involved, intersection related	After
9/1/2016	Bicyclist involved, teenage driver involved	After
1/29/2018	Intersection related	After

Fourteen crashes occurred at Salt Lake City's treated intersection, 10 before and four after the protected intersection was implemented. Three crashes involving a bicyclist or pedestrian occurred in the approximately six years prior to implementation and two crashes involving a bicyclist or pedestrian occurred in the approximately three years following implementation all of which were reported as non-incapacitating. Seven crashes occurred at Salt Lake City's comparison intersection from 2010 to 2018. None of the seven crashes involved a bicyclist or pedestrian.

Table 44. Salt Lake City | Comparison Intersection | Crash Data

Date	Type	Before/After
5/17/2010	Motorcycle involved	Before
8/1/2010	Motorcycle involved, overturn/rollover	Before
8/19/2011	Older driver involved	Before
11/28/2013	Driver left roadway	Before
7/17/2014	Driver left roadway	Before
7/12/2017	None	After
11/19/2017	DUI, collision with fixed object	After

Table 45 is provided to assist the reader but given the small number of crashes and small number of intersections (two), before and after analysis is not appropriate. Differences are anecdotal. The summary table illustrates that the crash data are insufficient to prove or disprove safety benefits from the treated intersections. While additional years of data would help, the sample size of two treated and two comparison intersections is far too small for before-after analysis of crash data, if the goal is development of crash modification factors for protected intersections. Ideally, 100 or more treated intersections would be required for study. Since only 10 such intersections existed in the U.S. at the start of this study, it is not possible to develop such CMFs at this time, but the team hopes that as more are installed, treatment installation dates, motor vehicle, bicycle and pedestrian traffic volumes, crash reports and other records will be kept so that future study can be made of this intersection treatment.

Table 45. Summary of Crash Data

	Berkeley		Salt Lake City	
Annual Crashes	Treated	Comparison	Treated	Comparison
Years Before	11	N/A	5.8	N/A
Years After	1.9	N/A	2.8	N/A
Total All-Severity Crashes Before	19	34	10	5
Total All-Severity Crashes After	5	5	4	2
All-Severity Crashes Before per Year	1.7	3.1	1.7	0.9
All-Severity Crashes After per Year	2.6	2.6	1.4	0.7
Pedestrian Crashes Before	4	1	1	0
Pedestrian Crashes After	0	2*	1	0
Pedestrian Crashes Before per Year	0.4	0.1	0.2	0
Pedestrian Crashes After per Year	0	1.0	0.4	0
Bicyclist Crashes Before	0	0	2	0
Bicyclist Crashes After	0	0	1	0
Bicyclist Crashes Before per Year	0	0	0.3	0
Bicyclist Crashes After per Year	0	0	0.4	0

N/A= Not Applicable; *Includes one pedestrian fatality.

5.7 Conclusions and Recommendations for Future Research

Due to the low numbers of interactions at both treated and comparison sites, only a very dramatic change in behavior between comparison and treated intersections could be measured in this study. No such dramatic difference was consistently observed, and the results are mostly equivocal.

However, some observations can be made from this analysis.

- In general, bicyclist and pedestrian signal compliance was better at comparison sites than at treatment sites in both jurisdictions.
- The percent of left turning motorists yielding to bicyclists was lower at the protected intersections studied than at the comparison. Bicyclists yielded more to turning motorists at the protected intersection.
- The protected intersections studied saw a higher percent of bicyclists making two-stage left turns than at their respective comparison intersections.
- Pedestrians waited in the pedestrian crossing island of the protected intersection roughly half the time in Salt Lake City, but much less often at the Berkeley protected intersection.
- Motor vehicle right-turning speeds were significantly lower at the Salt Lake City protected intersection studied than at the comparison intersections.

These findings show that, for the intersections studied, the protected intersection was associated with some potential safety benefits (more bicyclists making two-stage lefts and lower motorist right turn speed) and some indications that they may be less safe than the comparison intersections (protected intersection had lower bicyclists and pedestrian signal compliance and lower percent of left turning vehicles yielding to bicyclists).

The primary conflict type addressed through the protected intersection designs is right turning vehicles and crossing pedestrians/through-moving bicyclists. The protected intersection design aims to slow the speed of vehicles making those turns, as well as the speed of bicyclists traveling through the intersection. Thus, the significant speed reduction of right turning motorists at the Salt Lake City protected intersection is likely to be associated with improved safety.

The results of the research do not reject the hypothesis that the protected intersections studied improve safety for pedestrians and bicyclists, neither do they prove it. When considering the statistically significant results from this research report along with findings from San Francisco and international studies of protected intersections, we conclude that protected intersections have potential to reduce motorist right turn speeds and as a result have promise as a bicyclist safety countermeasure in the United States.

More research is needed to understand and better quantify the benefits. Specific design details should be studied to help identify the preferred design for protected intersections, especially with respect to the corner curb islands.

Recommendations for further study of protected intersections include:

- Collect additional observations, especially in different contexts.
 - Locations with higher levels of bicycle and pedestrian activity
 - Locations with two-way separated bike lanes at the intersection
 - Locations with one-way streets and two-way bicycle travel
 - Locations with and without median islands
- Understand impacts on setback crossing and drivers' ability to see crossing pedestrians and bicyclists.

CHAPTER 6: Discussion and Future Research Needs

This report documents current research, needs of practitioners, common crash types, and prioritizes research gaps for pedestrian and bicyclist intersections safety countermeasures. This report also contributes additional analysis to the protected intersection literature, but leaves room for much future research. Countermeasures most in need of additional research were identified based on the following priorities:

- Addresses high priority crash types
- Is of interest to interviewees
- Has not already been studied

Pedestrian and bicyclist intersection safety countermeasures that are prioritized for further safety study based on these three criteria are listed below.

- Curb extensions and curb radius reduction
- Four-way stop
- Pedestrian Hybrid Beacon for bicycles
- Protected intersections
- Speed management

These are not the only intersection countermeasures worthy of additional research to determine their safety benefits. A more complete listing is provided in Chapter 4 Table 2. This table demonstrates that there are many countermeasures in need of additional research. Since the protected intersection evaluation and analysis in this report was only able to provide limited evidence of any potential safety benefit for protected intersections, there is still a great need for additional safety research on protected intersections as well.

There are several ongoing research projects that relate to pedestrian or bicyclist safety and whose findings may yield results useful for selecting pedestrian and bicyclist intersection safety countermeasures. Specific ongoing studies include:

- NCHRP 17-84: Pedestrian and Bicycle Safety Performance Functions for the Highway Safety Manual
- NCHRP Synthesis 20-05: Availability and use of Pedestrian Infrastructure Data to Support Active Transportation Planning
- NCHRP 07-25: Guide for Pedestrian and Bicycle Safety at Alternative Intersections and Interchanges (A.I.I.)
- NCHRP 08-102: Bicyclist Facility Preferences and Effects on Increasing Bicycle Trips
- NCHRP 03-133: Traffic Signal Design and Operations Strategies for Non-motorized Users
- Contextual Guidance at Intersections for Protected Bicycle Lanes (National Institute for Transportation and Communities)
- Integrated Implementation of Innovative Intersection Designs (Southeastern Transportation Research, Innovation, Development and Education Center)
- A Safe Systems Approach to Intersection Planning & Design in the United States (Federal Highway Administration)

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APPENDIX A: Detailed Literature Review

The following detailed literature review findings are summarized by location type and specific countermeasure/treatment.

Signalized Intersections

BARNES DANCE

Description and Purpose: A Barnes Dance, also known as a scramble, all-walk phase, or exclusive pedestrian phasing, is a no-motor-vehicle signal phase at intersections allowing pedestrians to cross in any direction.

Summary of Effects: Barnes Dance treatments have been shown to reduce conflicts and crashes involving pedestrians compared to intersections before treatment or control intersections.

According to studies captured in the Synthesis, a Barnes Dance treatment caused a decrease in average crashes involving pedestrians by 44-51 percent compared to 9 percent at untreated locations (Chen et.al., 2013; Chen et.al., 2014; as cited in Thomas et.al. 2016). One of the studies in the Synthesis studied the tradeoffs between four signal- related pedestrian countermeasures in New York City, including Barnes Dance, and found that the Barnes Dance was the second most effective in reducing crashes involving pedestrians. The study recommended Barnes Dance for areas with high pedestrian volumes and moderate vehicular volumes (Chen et.al., 2014; as cited in Thomas et.al. 2016).

In 2015, Zhang and colleagues observed exclusive phasing and found that intersections with traditional concurrent phasing have fewer crashes, but more severe crashes; the study recommended that exclusive pedestrian phasing only be used where high pedestrian compliance would be expected (Zhang et.al. 2015). A 2017 study explored pedestrian compliance and discovered that pedestrians treat exclusive phasing as if it were the more traditional concurrent phasing, minimizing the safety benefits (Ivan et.al. 2017). Another 2017 study found that Barnes dance increased delay for all users (Kothuri et al. 2017)

BICYCLE SIGNALS

Description and Purpose: Intersections that include a bicycle signal and bicycle phasing are a key safety component of streets with bicycle lanes, particularly streets with two-way cycle tracks and high volumes of turning vehicular traffic. Adding a bicycle signal allows a protected phase to be added to the traffic signal cycle for bicycle movement (King 2016).

Summary of Effects: Relatively little research has been done on the safety impacts of bicycle signals. Studies show that signal timing does impact safety, as does having a protected intersection and treatments to move bicyclists safely through intersections. One study found that bicycle signal compliance is about on par with average traditional signal compliance. No research specific to Bicycle Signal Heads was found.

Although specific effects of bicyclists were not calculated, a 2002 study of crash statistics to evaluate the impact of signal timing changes found that intersections where the timing was changed results in a decrease in the number of crashes involving pedestrians (Retting and Chapline 2002; as cited in BIKESAFE 2006). A BIKESAFE study on Bicycle Countermeasures selection found that bicycle signals in Davis, CA, increased bicyclist safety at intersections (BIKESAFE 2006). A 2013 study from Thompson, Paulsen, Monsere, and Figlioizzi found that compliance at bicycle signals was about 90 percent and comparable to traditional signals (Thompson et al. 2013).

BIKESAFE provided no research on Bicycle Signal Heads.

BIKE BOX

Description and Purpose: The bike box provides space for cyclists to position themselves in front of vehicles while stopped at an intersection to improve the predictability of bicyclist stopping position and prevent vehicles from making right turns on red. Also referred to as advanced stop boxes, advanced stop lines, or setbacks, the primary purpose of a bike box is to reduce the likelihood of a right-hook crash at the on-set of the green phase, especially when large vehicles are present. Bike boxes also allow left turning bicyclists to more easily position themselves to make a left turn.



Figure A-43. An uncolored bike box. Photo from Dill, Monsere, and McNeil 2012.

Summary of Effects: Research consistently shows that bike boxes improve bicyclist safety, particularly in terms of bicyclist and motor vehicle compliance (using the bike box correctly and proceeding through the intersection as intended). However, motor vehicle encroachment into bike boxes was identified in multiple studies.

The MassDOT Separated Bike Lane Planning and Design Guide advises that “research has found crash reduction benefits at locations where bicycle crossings are set back from the motorist travel way by a distance of 6 ft. to 16.5 ft. (2 - 5 m)” and that protected intersections reduce conflict to one location, which can be even further reduced by having a signal for turning vehicles (MassDOT 2015).

Dill, Monsere, and McNeil studied bike boxes post-installation to compare motor vehicle encroachment of the crosswalk, yielding, and bicyclist position. They found that bicyclist volumes increased 94 percent post-installation, and number of conflicts decreased by 9 percent (Dill et al. 2012; as cited in BIKESAFE). Hunter analyzed a bike box in Eugene, OR, and found that after installation only 11 percent of bicyclists used the bike box as intended, but when used as intended there were no conflicts (Hunter 2000).

An analysis looking only at rule compliance (i.e., traversing an intersection legally) and facility compliance (i.e., traversing the intersection using a path consistent with its design) determined that intersections with bike boxes yield greater rule and facility compliance than those without. The study also determined that compliance rates increase when two-phase turning facility or advanced green turn signal for bicyclists are added in conjunction with a bike box (Casello et al. 2017).

A study of two intersections in Austin, TX, used a staged approach to determine the effect of bike boxes on bicyclist and motor vehicle behavior: observations were collected under existing conditions, then after bike box markings, and finally after green paint was added to the bike box and bicycle lane. The authors found that 20-26 percent of bicyclists stopped in the bike box after markings were installed, although 90 percent of bicyclists stopped in front of motor vehicles. Once green pavement markings were added, the percent of bicyclists using the bicycle lane to approach the intersection increased from 77 percent to 93 percent, and bicyclists waiting in the bike box increased from 52 percent to 92 percent. A second site saw a significant increase in bicyclists who waited in the bicycle lane or bike box instead of in the motor vehicle queue, 9.1 percent to 15.1 percent. However, motor vehicle encroachment on the bike box was common (Loskorn et.al. 2011).

A study of three sites in Washington, DC (one site included a bicycle box) found that after installation of bicycle facilities, there was a large increase in bicycle volumes. Although the crash results are inconclusive due to short post-installation study times, the researchers postulate that the bike box was still beneficial to both bicyclists and motor vehicles due to the increased separation between modes – even though the bike box was not always used as intended (Parks et.al. 2012).

FLASHING YELLOW ARROW

Description and Purpose: Flashing yellow arrow (FYA) signals are used for protected/permmissive left turns to reduce vehicle delay. They indicate that vehicles may turn left but use caution. They are an alternative to the more common permmissive left green ball signal but are less restrictive than a protected left indicated with a green/red arrow. As an alternative to the green ball, FYA may have benefits to pedestrian crossing safety, but as an alternative to protected-left phasing the FYA are likely to be detrimental.

Summary of Effects: The studies reviewed are not conclusive. No effects of this treatment on crashes involving pedestrians was proven (Van Houten et al. 2014). Washington County in Oregon reprogrammed their signals to ensure that when a pedestrian phase is requested it does not coincide with the flashing yellow arrow. The change was followed by a reduction in crashes (Steyn et al. 2013). Hurwitz and Monsere examined driver behavior with yellow flashing arrow phasing in a simulator and found that 4 to 7 percent of drivers did not focus on pedestrians in the crosswalk (Hurwitz and Monsere 2013). They recommend using protective left phasing instead of yellow flashing arrow when pedestrians are present. There were no studies included in PEDSAFE or BIKESAFE on this treatment.

LEADING BICYCLE INTERVAL (LBI)

Description and Purpose: A leading bicycle interval provides bicyclists with an advanced green signal through an intersection. The intention is to give through-cyclists the chance to get into the line of sight for stopped vehicles and prevent “right hook” conflicts with vehicles turning right.

Summary of Effects: No studies were found specifically on LBIs. A forthcoming study by Sirisha Kothuri will address various signal timing control strategies, including LBIs and split LBIs (Addressing Bicycle-Vehicle Conflicts with Alternate Signal Control Strategies, forthcoming 2017). The Portland Bureau of Transportation performed a literature review and found that leading bicycle intervals provides bicyclist priority over other modes, which increases safety (PBOT 2010).

LEADING PEDESTRIAN INTERVAL (LPI)

Description and Purpose: As described by the Synthesis, “A leading pedestrian interval at a signalized intersection gives pedestrians anywhere from a 3- to 7-s head start before traffic is released by the parallel-path corresponding green signal” (Thomas et al. 2016).

Summary of Effects: Studies on the safety impact of LPIs show that they do have a positive effect on pedestrian safety. CMF estimates for LPIs range from a 5 percent reduction in crashes involving pedestrians to a 59 percent reduction. Overall, LPIs improve safety and reduce conflict for pedestrians and may be even more effective when used in conjunction with other treatments such as No Right Turn on Red signs.

The pedestrian safety Synthesis found that “LPIs, which have been studied predominantly at high pedestrian sites, are another countermeasure that has some crash-based evidence of effectiveness. A CMF of 0.95 (5 percent reduction) for crashes involving pedestrians was estimated for LPIs (FHWA, 2004), but a more recent study estimates reductions of 59 percent in crashes involving pedestrians at 10 treated intersections (significant at the 95 percent confidence level) (Fayish and Gross 2010). Behavioral studies

provide additional support for crash reductions, showing a reduction in conflicts with turning vehicles, particularly left-turning vehicles. RTOR restrictions may also be needed to further reduce conflicts and crashes with right-turning vehicles and improve ability of pedestrians to access the crosswalk when waiting on the curb in some situations such as intersections with high volume of right-turning vehicles (Hubbard et al. 2008)” (Thomas et.al. 2016).

In 1999, King found a 28 percent reduction in crashes involving pedestrians in New York City with LPIs, and a 64 percent reduction when factored for severity (King 1999). Van Houten, Retting, Farmer, and Van Houten found that a three-second LPI in St. Petersburg, FL, reduced conflicts and helped pedestrians to retain their right-of-way when crossing (Van Houten et.al. 2000). A study of Pedsafe II in San Francisco found that LPIs were among the top six of thirteen most effective pedestrian safety treatments, and that the number of vehicles turning right in front of pedestrians was reduced from 6.2 percent to 4 percent (Hua et.al. 2009). New York City DOT evaluated intersections before/after installation of LPIs and found a 14 percent reduction in left-turn pedestrian and bicyclist injuries and a 56 percent reduction in severe or fatal injuries (NYCDOT 2016).

MIXING ZONE TREATMENTS

Description and Purpose: Mixing occurs at intersections where bicyclists approach an intersection in a bicycle facility that requires turning motor vehicles to enter or cross the bicyclist’s travel path. In these designs, there is often no separate bicycle signal or phase. Mixing zones are used in situations where right-of-way is limited, and it is infeasible or cost-preventive to have a protected intersection. In this case, a shared bicycle lane and right-turn lane can be used. Bicycle lanes combined with dedicated vehicle turn lanes become turning zones, the purpose of which is to increase the visibility of the bicyclist and allow vehicular through traffic to pass by in a through lane. In some cases, mixing zone shared lanes are wider than traffic lanes to accommodate bicycles and motor vehicles side-by-side. Different types of mixing zone treatments include mixing zones with yield entry markings, turning zones with flexpost-restricted entry and through bicycle lane, turning zones with unrestricted entry and through bicycle lane, mixing zones with sharrows, and mixing zones with green coloring (Monsere et al. 2015). For the purposes of this report, flexposts are defined as highly reflective flexible post delineators installed in the pavement and are also known as safe-hit delineators or flex stakes.

Summary of Effects: Studies about the safety effect of mixing zones for bicyclists show a range of results; zones with the most success are those that incorporate flexpost-restricted entry and other treatments to supplement the mixing zone. The MassDOT Separated Bike Lane Guide includes evidence that combining mixing zones at intersections with other supportive treatments was associated with a decrease in average annual total crashes (MassDOT 2015).

One study showed that of the different types of mixing zone treatments, those combining a turning zone with flexpost-restricted entry and a through bicycle lane had the highest percentages of observed motor vehicles and bicyclists turning correctly: 86 to 88 percent and 89 to 93 percent, respectively. This type of zone had lower rates of conflict between bicyclists and motor vehicles than other types, including turning zones with unrestricted entry and through bicycle lane, mixed zones with sharrows, and mixed zones with green coloring. This study showed that protecting the bicycle lanes with flexposts on the approaches led to greater compliance than pavement markings alone; however, similar intersections with yield markings to supplement the flexposts showed the highest compliance (93 percent) of vehicles using the lane as intended (Monsere et al. 2015).

Another study examined two locations in Eugene, OR: one with a narrow shared right-turn lane and one with a through bicycle lane and a standard dedicated vehicular right-turn lane. Researchers found that 93

percent of vehicles yielded to bicyclists at the narrow-lane site and only 48 percent yielded at the standard site (Hunter 2000).

A before-and-after study of 17 separated bicycle lanes in 9 cities found that of the 6 sites where analysis included a comparison of intersection versus midblock crashes, the percentage of crashes that occurred at an intersection increased (without accounting for changes in volume). At 10 sites where mixing zones were combined with additional treatments, a decrease in average annual total crashes was observed, and it seems that installing separated bicycle lanes may result in more challenges at intersections (Rothenberg et al. 2016).

FHWA's Separated Bike Lane (SBL) Guide provide resources on mixing zones based on interviews with to 35 U.S. municipalities. Municipalities' decision to employ one-way or two-way separated bicycle lanes depended on context. Municipalities communicated that two-way SBLs are easier to maintain and are good for uninterrupted roadways with few intersections or driveways. However, design challenges for two-way SBLs include intersections, roadways with turning movements, and areas with heavy pedestrian activity (FHWA 2015). FHWA found that bidirectional SBLs may exhibit higher crash rates due to unfamiliarity for drivers yielding to two directions of traffic (Schepers et al. 2010 and Nosal, et al. 2011; cited in FHWA 2015). Areas with large number of intersections, roadways with heavy turning movements and roads with heavy transit use may also complicate operation of SBLs.

NO TURN ON RED SIGNS

Description and Purpose: "No Turn on Red" signs are standard signs used to restricted right turn on red movements. The purpose of this treatment is to eliminate conflicts between turning vehicles and pedestrians or bicyclists crossing during a concurrent walk phase. They are often used in conjunction with Bike Boxes to remind drivers that they are not allowed to turn right on red when a Bike Box is marked.

Summary of Effects: Existing research indicates that "No Turn on Red" signs may be effective at reducing pedestrian-vehicle conflicts. Most studies of this treatment have examined driver behavior – nearly all studies that have examined impacts on crashes have looked at total crashes, and not crashes involving pedestrians specifically. According to studies in the Synthesis, "No Turn on Red" signs have been estimated to reduce all crashes by about 3 percent (CMF = 0.97 (Federal Highway Administration 2018, citing Harkey et al. 2008). Permitting a right turn on red seems to result in an increase in crashes involving pedestrians.

A recent study conducted by Lin, Kourtellis, Wang, and Guo (2016) used Strategic Highway Research Program 2 Naturalistic Driving Study data to examine the effectiveness of four signs aimed at increasing driver yielding to pedestrians at intersections with restricted right turns. The authors found that "No Turn on Red" signs had the highest rate of compliance of all four treatments studied ("Stop here on Red," "No Turn on Red," "Turning Vehicles Yield to Pedestrians," and "Right on Red Arrow after Stop"). The average driver compliance associated with "No Turn on Red" signs in this study was 70 percent.

Dynamic signs and signs that take effect during certain times of day may be more effective at improving driver behavior than static signs, however additional research is needed to confirm this. For example, Pecheux, Bauer, and McLeod (2009) compared the effectiveness of two static signs printed with the messages "No Turn on Red" and "No Turn on Red When Pedestrians in Crosswalk" and a dynamic sign. The dynamic sign was dark during the protected right turn phase, displayed the message "Yield to Pedestrians" during the green phase, and displayed a "No Turn on Red" message during phases when right turn was prohibited and pedestrians had pushed the call button. The active sign was associated with the lowest percentage of No Turn on Red violations.

Research by Preusser, Leaf, DeBartolo, Blomberg, and Levy (1982) indicates that permitting a right turn on red increases crashes involving a motor vehicle making a right turn by 43 percent to 107 percent for pedestrians and by 72 percent to 123 percent for bicyclists (CMF ranging from 1.43 to 2.08). A similar CMF of 1.69 is cited in the Highway Safety Manual, 1st edition, for vehicle-bicycle and crashes involving pedestrians (AASHTO 2010).

PASSIVE DETECTION - BICYCLE

Description and Purpose: Bicycle detection includes technologies such as inductive loop detection systems at or in advance of stop bars that allow signals to passively detect the presence of bicyclists. These are sometimes combined with blue lights that provide feedback to bicyclists indicating that they have been detected. The purpose of this treatment when combined with the feedback light is to reduce risky behaviors conducted by bicyclists, such as signal noncompliance, which may be more likely to occur when bicyclists are unsure whether their presence has been detected by a traffic signal. Passive detection may also be used to extend the all-red phase for cyclist entering a wide intersection just before the change interval so that adequate clearance time is provided.

Summary of Effects: Little is known about the safety benefits of this treatment due to a lack of research on this topic, however, current research suggests that bicyclist comprehension of the pavement markings likely plays a key role in the treatment's effectiveness.

The MUTCD standard 9C-7 bicycle detector marking was developed as part of this treatment in order to indicate to bicyclists where they need to be positioned in order to be detected by the traffic signal. Past research has found that many bicyclists do not understand what the 9C-7 marking means and therefore do not activate it, thereby reducing its potential effectiveness as a treatment (Boot et al., 2012 and Bussey, 2013).

In response to these findings, research conducted by Boudart et al. (2016) sought to determine whether bicyclists understand both the pavement marking, and a blue light feedback device; and whether bicyclists' behavior changes when they are aware that they have been detected by the signal. The authors found that 60 percent of bicyclist intercept survey respondents correctly identified the existing 9C-7 marking and nearly 75 percent understood the meaning of the blue light detector feedback device. The survey results indicated a 30 percent increase of user comprehension when descriptive text was added to the 9C-7 marking. The authors also observed statistically significant increase in bicyclists waiting on the new bicycle detector signal after a descriptive bicycle detector marking was installed at a re-constructed intersection. There was not enough data to determine whether the treatment resulted in a statistically significant reduction in red light running among bicyclists. The results of this study are based on data from only two intersections in Portland, Oregon. Additional studies of this treatment's effectiveness in reducing bicyclist signal noncompliance are needed.

The Portland Bureau of Transportation performed a literature review and found that advanced bicycle detection (using an automated/passive system) reduces bicycle delay but did not associate safety impacts with automated detection (PBOT 2010).

No information on passive detection was found in BIKESAFE.

For more information about passive detection for bicycles, see section on Passive Detection - Pedestrians.

PASSIVE DETECTION - PEDESTRIANS

Description and Purpose: Passive detection, also known as automatic detection, of pedestrians at intersections automatically identifies the presence of a pedestrian waiting to cross and removes the

burden of needing to manually activate a WALK signal. Passive detection has been used to automatically activate the pedestrian signal phase, to monitor and lengthen the amount of crossing time for slower-moving pedestrians, and to increase lighting at midblock crossings. However, difficulty in accurately detecting pedestrian presence and other issues limits the use of this treatment.

Summary of Effects: Studies on the safety impacts of automatic or passive detection of pedestrians show that passive detection may help to reduce pedestrian-vehicle conflicts, reduce the number of late crossings, and reduce the number of pedestrians stranded in the intersection (Hughes et.al. 2001; and Pecheux, Bauer, and McLeod 2009; as cited in PEDSAFE). A study of automatic detection used for enhancing lighting found that motor vehicle yielding and diverting pedestrians increased (Nambisan et.al. 2009).

For more information about passive detection for bicycles, see section on Passive Detection - Bicycles.

PEDESTRIAN COUNTDOWN SIGNALS

Description and Purpose: “Countdown signals are indications designed to begin counting down at the beginning of the clearance interval (flashing “DON’T WALK” symbol) and can be on fixed-time or pushbutton operation. (Sometime pushbuttons are used with fixed time signals to provide locator tones for the blind or to activate Leading Pedestrian Intervals.) They indicate to the pedestrian how much time is left in the crossing phase. According to the MUTCD, countdown pedestrian indications are required for all newly installed traffic signals where pedestrian signals are installed” (Thomas, et.al. 2016). However, not all new or existing signals have pedestrian signals.

Summary of Effects: Studies on the safety impact of Pedestrian Countdown Signals (PCSs) have found mixed results on pedestrian compliance and crash reduction. According to the pedestrian safety Synthesis, “a CMF of 0.75 for pedestrian crashes was developed from one study of the effects of replacing traditional walk/don’t walk pedestrian signals with pedestrian countdown signal heads (Toolbox of Countermeasures and Their Potential Effectiveness for Pedestrian Crashes 2013, citing Markowitz et al. 2006). However, other studies have found more mixed results from widespread installation. Another positive finding: motor vehicle behaviors have not been shown, thus far, to be significantly negatively affected by information from the countdown timers. Pedestrians may or may not comply more often with the walk phase of the signal, and results are somewhat mixed as to whether they tend to be stranded less often in the crosswalk at a signal change. Regarding crashes, as with traditional signals, safety benefits may be affected by the particular phasing and timing strategies used, such as allowing more pedestrian walk time (CMF = 0.49), or providing exclusive phases (0.65) (Toolbox of Countermeasures and Their Potential Effectiveness for Pedestrian Crashes 2013, citing Chen et al. 2012)” (Thomas et.al. 2016). Van Houten, LaPlante, and Gustafson found that installing pedestrian countdown timers reduced pedestrian-vehicle conflicts by 55-70 percent (CMFs of 0.45 to 0.30). (Van Houten, LaPlante, and Gustafson, 2014).

The findings from PEDSAFE suggest that pedestrian countdown signal sites have a mixed impact on the percent of pedestrians who comply with the WALK signal phase, with some studies showing a decrease in compliance and some study sites showing an increase (Zegeer and Huang 2000; Eccles, Tao, and Mangum 2004; Reddy, Datta, McAvoy, Savolainen, Abdel-Aty, Pinapaka 2008; Sharma, Schmitz, Khattak, and Singh 2012; as cited in PEDSAFE). Findings for rates of crashes involving pedestrians after installation of PCSs ranged from a 70 percent reduction (citywide in Detroit, MI) to a 52 percent reduction (when installed at higher-than-average crash rate intersections), to no statistically significant effect (Huitema, Van Houten, and Manal 2014; Markowitz, Sciortino, Fleck, and Lee 2006; Camden, Buliung, Rothman, Macarthur, and Howard 2011; as cited in PEDSAFE). A study also found that PCSs have a positive impact on pedestrians

finishing crossing (Zegeer and Huang 2000; cited in PEDSAFE). Surveyed pedestrians also felt that the PCSs made the crossings “more safe” (Levasseur and Brisbane 2011; cited in PEDSAFE).

Rothman, Cloutier, Macpherson, and Howard (2017) compared the spatial patterns of crashes involving pedestrians pre- and post-installation of PCSs. They found a consistent reduction in crashes (across ages and across locations in Toronto) at the non-PCS locations compared to inconsistent reduction and some increases in crashes at the PCS locations (Rothman, Cloutier, Macpherson, and Howard 2017). Another recent study showed a more consistent reduction in crashes involving pedestrians, particularly for older adults: Kwigizile, Boateng, Oh, and Lariviere (2017) studied the safety effect of PCSs by using crash data in Michigan and found an almost 32 percent reduction in crashes involving pedestrians for all ages and severities, but a particular decrease in crashes involving pedestrians age 65 and older (Kwigizile, Boateng, Oh, and Lariviere 2016).

PROTECTED PHASES

Description and Purpose: Protected phases at intersections provide a way to separate vehicular from pedestrian and bicyclist movements, particularly for left-turns when concurrent phasing would result in a conflict with crossing pedestrians and a left-turning vehicle. Protected phasing can also be used to separate bicycle through movements from right turning vehicles.

Summary of Effects: Relatively few studies on the pedestrian or bicyclist safety impacts of protected phasing have been conducted. Overall, existing studies show that protected-permissive left-turn phasing, protected left-turn phasing, and split phasing had a positive effect on pedestrian safety. One recent study found no significant safety effect.

The Synthesis offered the following on the safety impacts of protected phasing. “Certain types of traffic signal phasing, such as an exclusive phase for pedestrians or restricted left-turn phasing, have been found to reduce both motor vehicle left-turn crashes and crashes involving pedestrians, as compared with traditional concurrent signal phasing (Toolbox of Countermeasures and Their Potential Effectiveness for Pedestrian Crashes 2013, citing Harkey et al. 2008 and Chen et al. 2012).”

Pratt, Songchitruska, and Bonneson studied pedestrian crossings in the path of a signalized left-turn movement. They found that leading protected-permissive left-turn phase, split phasing, and pedestrian recall were effective in reducing conflict rates between vehicles and pedestrians (Pratt, Songchitruska, and Bonneson 2012). Chen, Chen, Ewing, McKnight, Srinivasan, and Roe found that implementing a left turn phase decreased the rate of crashes involving pedestrians by 44.85 percent, compared to a decrease of 11.47 percent at comparison sites (2012).

De Pauw, Daniels, Van Herck, Wets studied signalized intersections with left-turn signals in Flanders, Belgium. They found that left-turn phasing had favorable effects for multiple groups, including bicyclists, and the number of injury crashes decreased significantly which researchers attributed to a decrease in left-turn crashes (De Pauw, Daniels, Van Herck, Wets 2015). A study by Chen, Chen, and Ewing examined the impacts of different kinds of left-turn phasing: permissive, protected-only, and both, and found that changing a signal from permissive to protected/permissive or protected-only did not lead to a significant reduction in crashes (Chen, Chen, and Ewing 2015). Chen, Chen, and Ewing in 2012 found that split phasing reduced crashes involving pedestrian crashes by 39 percent (CMF of 0.61) (Chen, Chen, and Ewing 2012).

SIGNAL TIMING AND DELAY

Description and Purpose: Traffic signal timing can have pedestrian safety impacts. Pedestrian delays encourage crossing behavior against the signal and could have an overall negative impact on perceived

walkability. It is also likely that timing signals to provide a green wave for bicycles may have a positive influence on bicyclist safety by reducing motor vehicle speeds and reducing bicyclist delay, we found no studies of this impact.

Summary of Effects: Studies on minimizing pedestrian delay generally find that by reducing pedestrian delays, safety is improved. Kothuri in 2014 found greater pedestrian compliance with intersections on recall than actuated intersections (Kothuri 2014). In 2017, Kothuri and colleagues tested efficiency-focused pedestrian treatments and found that free operation reduced minor-street pedestrian delays by 60 percent and bicycle delays by 5 percent, and proposed an algorithm that also reduced pedestrian delay from 44.95 seconds to 41.3 seconds (Kothuri et.al. 2017). Kothuri, Reynolds, Monsere, and Koonce in 2013 found that decreased pedestrian delays could be achieved by freeing a coordinated signal during off-peak periods and increasing the permissive length (Kothuri et.al. 2013). Washington County re-programmed its signals with flashing yellow arrows to provide more time to pedestrian phases, and found that it led to a reduction in crashes (Steyn, Quayle, and Boudart 2013).

Signal timing changes have also been found to improve pedestrian and bicyclist safety; in a study in New York that looked at 40 intersections that implemented signal timing changes found that these changes (typically increasing the interval timing on the yellow and all-red intervals) led to an 8 percent reduction in all reportable crashes and a 37 percent reduction in crashes involving pedestrians and bicyclists, a CMF of 0.63 (Retting and Chapline 2002). The Flashing Pedestrian Indicator (FPI), or Flashing Turn Signal Head, is a proposed pedestrian safety treatment, which alternates a flashing right-turn arrow with a flashing pedestrian symbol and indicates to right-turning drivers that there is a pedestrian present in the conflicting crosswalk. A study of its effectiveness found that the FPI resulted in more decisions to yield and more time in making the decision to yield, regardless of pedestrian presence and compared to a standard signal (Boot et.al. 2015).

A study of three sites in Washington, DC (one site with a two-way cycle track on 15th Street NW) observed that on one-way streets, two-way bicycle lanes create signal challenges, particularly for signal progression and signal equipment. After the installation of the two-way cycle track, the number of crashes remained similar after accounting for the increase in bicycle volume (Parks et.al. 2012).

TWO STAGE BICYCLE TURN QUEUE BOXES

Description and Purpose: Two stage turn queue boxes allow bicyclists to make turns more easily and safely at multi-lane signalized intersections. They consist of a painted and marked box adjacent to the crosswalk and the bicycle lane, where bicyclists turning left can first proceed into the intersection and then wait for the next signal to proceed straight again with cross traffic.

Summary of Effects: The NACTO Urban Bikeway Design Guide lists benefits of two stage turn queue boxes:



Figure A-44. 3D concept of a two-stage turn markings based on Portland, OR, and Australian examples. Image from NACTO, 2011.

- “Improves bicyclist ability to safely and comfortably make left turns;
- Provides a formal queuing space for bicyclists making a two-stage turn;
- Reduces turning conflicts between bicyclists and motor vehicles;
- Prevents conflicts arising from bicyclists queuing in bicycle lane or crosswalk;
- Separates turning bicyclists from through bicyclists” (NACTO 2011).

No studies were found in BIKESAFE about this treatment.

Unsignalized Intersections

DRIVEWAY TREATMENTS

Description and Purpose: As vehicles enter and exit the roadway, a potential conflict area exists for pedestrians, bicyclists, and vehicles. Driveways are under-studied elements of the road network despite studies that show up to 19 percent of reported urban traffic crashes involved driveways (Gattis et.al. 2010, 6). There are multiple driveway treatments including raised crossings, colored or other differentiating pavement, removing curb cuts to remove driveways (see also Access Management), reducing the abruptness of surface elevation changes or installing a continuous slope, adding bicycle lanes, and pedestrian standing areas.

Summary of effects: Design guidelines for driveways recommend context-sensitive designs that reduce potential conflict for pedestrians and bicyclists at driveways. Main recommendations include providing pedestrian crossings of at least 2-5 ft. width across driveways with a slope less than 2 percent; minimizing curb cuts in high pedestrian traffic areas; and improving sight distance and visibility through unobstructed sight lines and bicycle lanes (Gattis et.al. 2010; ITE 2010; New Jersey DOT 2008). The project team did not identify any studies of driveway treatments for pedestrians or bicyclists. There is no information on this treatment type in PEDSAFE, BIKESAFE, or the Synthesis.

FOUR-WAY STOP

Description and Purpose: Four-way stop control uses stop signs rather than signals to control traffic at a four-legged intersection. In a four-way stop, all directions of traffic have a stop sign.

Summary of Effects: Compared to intersections with two-way stop control (e.g., an intersection with a major road and a minor road where only traffic on the minor road must stop), all-way stop control is safer for pedestrians.

Lovell and Hauer studied the conversion of intersections from two-way stops to four-way stops and found that four-way stops reduced crashes involving pedestrians by 43 percent (CMF of 0.57) (Lovell and Hauer 1986).

IN-PAVEMENT WARNING LIGHTS

Description and Purpose: In-pavement warning lights (also known as flashing crosswalks) are encased lights placed on both sides of a crosswalk at an uncontrolled intersection. The warning lights may be activated by passive detection of pedestrians waiting in the crossing area or activated by a pushbutton (Thomas et.al. 2016). The purpose of this treatment is to increase the percentage of drivers who yield to pedestrians.

Summary of Effects: This treatment has been the subject of a number of reports; however, the long-term benefits of this treatment remain largely unknown. Thomas (2006) reviewed nine studies on in-pavement warning lights and found that changes in motor vehicle yielding rates and effects on traffic speed varied

by study location - with some locations experiencing improved safety benefits and other locations experiencing no change. In some studies, the increased driver yielding rates did not persist overtime. Research covered in the Synthesis indicates that the need for maintenance and occurrence of lighting malfunctions further limit the benefit of this treatment. Results from Zeeger et al. (2016) indicate that the lights are often only visible to the first car in the queue, suggesting that this treatment may lead to an increase in rear-end crashes. In addition, the pavement lighting may not be visible during daylight hours, limiting the treatment's potential benefits to dark conditions. No studies to date have measured the impact of in-pavement warning lights on the frequency of crashes involving pedestrians (Thomas, 2006). Additional research is needed to determine where this treatment is most appropriate (if at all), and whether the safety improvements associated with this treatment persist long-term.

IN-ROAD SIGNS

Description and Purpose: This treatment involves placing “Yield to Pedestrian” signs in the roadway in the middle of an uncontrolled crosswalk. The purpose of this treatment is to direct drivers to yield to crossing pedestrians (Thomas, et.al. 2016).

Summary of Effects:

According to Zeeger et al. (2016) and the recent pedestrian safety synthesis, driver yielding rates have increased substantially after in-road signs have been installed at crosswalks compared to baseline conditions. Seven of the nine studies reviewed as part of Zeeger et al. (2016) found that the installation of in-road signs was associated with an increase in driver yielding and a decrease in average vehicle speeds. Existing research also indicates that placing signs

closer to the crosswalk is associated with higher yielding rates than placing signs at various distances in advance of the crosswalk. For example, Gedafa et al., 2014 observed a statistically significant increase in yielding rates when in-road signs were placed at the crosswalk as opposed to further away; this study observed yielding rates of approximately 97 percent when “Yield to Pedestrian” signs were placed at the crosswalk.



Figure A-45. In-roadway “Yield to Pedestrian” signs (R1-6 signs) placed on median near crosswalk. Source: <http://www.pedbikeimages.org>, Dan

To date, there have been very few studies on the relationship between in-road signs and crashes. One study reviewed in the Synthesis was unable to draw conclusions about the effect of in-road signs on crashes involving pedestrians. A limitation of this treatment is the maintenance required when signs are damaged and need to be repaired or replaced.

MEDIAN ISLANDS

Description and Purpose: “Medians with pedestrian crossings, and pedestrian median islands—sometimes referred to as center islands, refuge islands, or pedestrian islands—are raised areas intended

to help protect pedestrians who are crossing the road at either intersections or midblock locations. The presence of a median island in the middle of a street or intersection allows pedestrians to focus on one direction of traffic at a time as they cross and gives them a place to wait for an adequate gap between vehicles before finishing the second leg of the crossing. Islands are appropriate for use at both uncontrolled and signalized crosswalk locations” (Thomas 2016). Median islands can also be used by bicyclists.

Summary of Effects: Raised median islands have been found to positively impact pedestrian safety, with CMFs ranging from 0.75 to 0.54 at unsignalized intersections and a consensus that they improve motor vehicle yielding. For bicyclist safety, a CMF of 1.4 from a study of raised islands in the Netherlands (Schepers et.al. 2011) indicates that such islands may reduce instead of improve bicyclist safety. However, the NACTO Urban Bikeway Design Guide does recommend median islands for bicyclist use.

Additionally, the Synthesis notes that “Studies for NCHRP Report 562 found that motorist yielding rates varied widely across sites with median islands and marked crosswalks. Some of the explanatory variables included the number of through lanes and posted speed limit (Fitzpatrick et al. 2006a). Other treatments may be needed to improve yielding at uncontrolled locations or improve pedestrian access and safety at multilane uncontrolled road crossing locations, particularly with higher volumes, higher speeds, or both (see the matrix in Section 3). Treatments recommended by Zegeer et al. (2005) and Fitzpatrick et al. (2006a) include advance stop/yield lines and signs, beacon signals with a red (stop) indication (such as Pedestrian Hybrid Beacons), and signs and flashing beacons, although the latter have tended to have a wider range of effectiveness” (Thomas 2016). In a 2016 study of 1,000 treatment and comparison sites from 14 cities across the U.S., the CMF for pedestrian refuge islands was found to be 0.685 (Zeeger, et.al., 2016).

Stokes, Sarasua, Huynh, Brown, Ogle, Mammadrahimli, Davis, and Chowdury looked specifically at driveway-related crashes in South Carolina, and found that adding a raised median resulted in a Crash Modification Factor of 0.49 (Stokes et.al. 2016).

PEDESTRIAN HYBRID BEACON

Description and Purpose: “Pedestrian hybrid beacons, also known as HAWK beacons (short for High-Intensity Activated Cross Walk Beacon), are traffic control devices developed by Tucson, Arizona, traffic engineer Richard Nassi in the late 1990s as a means of providing safe pedestrian crossings where minor streets intersected with major arterials (Mead et al. 2014; Fitzpatrick and Park 2010a, citing Nassi and Barton 2008). The devices have also been used at midblock crosswalks; there are pedestrian volume warrants outlined by the MUTCD that are significantly lower than for traffic signals (Zegeer et al. 2013, PEDSAFE, Pedestrian Hybrid Beacon (PHB)). The PHB may be located both on the roadside and on mast arms over the major approaches to a crossing location. The head contains two red lenses above a single yellow lens. The lights are normally dark (rest in dark), but when activated, the beacon first displays an interval (a few seconds) of flashing yellow, followed by a steady yellow interval, and then “displays a steady red indication to drivers and a WALK indication to pedestrians, thereby allowing them to cross the major road while traffic is stopped” (FHWA 2014). At the onset of the pedestrian clearance interval (flashing “DON’T WALK” symbol and the countdown timer), the red indications of the PHB changes to an alternating flash pattern to allow drivers to proceed after they have stopped, if the pedestrian has cleared the roadway, thereby contributing to lower vehicle delay than traditional signals. The first PHB was installed in Tucson in 2000. The PHB was considered an experimental treatment until 2009, when it was included for the first time in the MUTCD (FHWA 2009). Pedestrian volume warrants are, however, considerably lower for implementing a PHB than for a traffic signal (Zegeer et al. 2013)” (Thomas et.al. 2016).

Summary of Effects: PHBs have been shown to be effective at improving safety at uncontrolled pedestrian crossings on multiple types of roads. Studies have focused on increased driver yielding as the main safety benefit and multiple studies measuring driver yielding compliance showed results ranging from 89-98 percent depending on the site. PHBs have also been studied for their crash modification factors, and two studies found pedestrian CMFs ranging from 0.453 (55 percent crash reduction) to 0.31 (69 percent crash reduction).

The recent pedestrian safety Synthesis finds “motorist compliance or yielding at crosswalks is improved with PHBs compared with no control and compared with warning type devices. Yielding has generally been consistently high, 90 percent and greater, at study locations with PHBs installed. A CMF estimate of 0.31 (69 percent reduction) for crashes involving pedestrians, and a total crash CMF of 0.71 (29 percent reduction) were also derived from a 2010 FHWA study at 21 sites in Tucson (Fitzpatrick and Park 2010a, 2010b; Fitzpatrick et al. 2011). Fewer pedestrian conflicts have also been observed after PHB installation. In a study conducted by the Texas Transportation Institute, wider crossings were associated with higher yielding rates at sites in Texas, and posted speed did not have an effect on yielding, suggesting that this countermeasure is effective for wider and higher-speed locations. However, results from one study suggest that drivers might use the signal indications—especially when the beacon is in dark mode—to perceive that pedestrians are not present and increase speed. There is also presumably some risk of multiple-threat crash when the flashing red phase is active or if, as some states indicated in survey responses, drivers do not, in fact, stop” (Thomas et.al. 2016).

A 2016 study to develop crash modification factors for four types of uncontrolled pedestrian crossing treatments found that pedestrian hybrid beacons performed best, with a CMF of 0.453 for PHBs and 0.432 for PHBs in conjunction with advance stop/yield signs and markings (Zeeger et.al. 2016). Two studies in Michigan which looked at yielding compliance found that adding an PHB or other device to marked crosswalks achieved the highest compliance (Gates et.al. 2016; Stapleton et.al. 2017). An FHWA report on road user behaviors at PHBs found that driver compliance averaged 96 percent at over 20 sites (even sites with wide crossings and high speeds), and pedestrians activated the PHB 91 percent of the time (FHWA 2016). A 2014 study by Fitzpatrick, Brewer, & Avelar found that across 32 sites, the average yielding rate for PHBs was 89 percent, and noted that cities where the device was highly used has higher driver yielding rates (Fitzpatrick et.al. 2014). BIKESAFE similarly concluded that bicyclists crossing at PHBs may benefit from high yielding, and that crashes involving bicyclists may decrease as well (BIKESAFE).

RAISED CROSSINGS FOR BIKES

Description and Purpose: Raised crossing treatments elevate the pavement where bicyclists travel through an intersection or across a driveway or alley to increase their visibility and to slow the speed of vehicles. This treatment should be used with separated bicycle lane crossings where motor vehicles are required to yield, and are appropriate at minor crossings. Specific crossings include: collector and local street crossings, driveway and alley crossings, channelized right turn lane and roundabout crossings, and intersections with large corner radii (MassDOT 2015).

Summary of Effects: MassDOT found three studies that indicated raised crossing treatments reduced crashes and increased vehicle yielding due to vehicle speed reduction and increased bicyclist visibility (Schepers et al., 2010a; Garder et al., 1998; Schepers et.al., 2010b; as cited in MassDOT 2015). One of these studies, conducted in Sweden, looked at before-and-after effects of installing raised crossings and found that the likely effect of raising the bicycle crossing is a risk reduction of 30 percent compared to a conventional crossing (Garder et.al., 1998). The study also found that motor vehicle speed was reduced by 40 percent and the number of conflicts for pedestrians and motor vehicles also decreased following

the treatment. One study found that raised bicycle crossings increased crashes involving bicyclists, with a CMF of 1.09 (Elvik and Vaa 2004).

RECTANGULAR RAPID FLASH BEACON

Description and Purpose: “The rectangular rapid flash beacon device includes amber LED flashing lights installed to enhance pedestrian crossing warning signs at uncontrolled crosswalks (Zegeer et al. 2013, PEDSAFE, Rectangular Rapid Flash Beacon). The crosswalks should not be controlled by signals, stop signs, or yield control signs. RRFBs are placed on both sides of an uncontrolled crosswalk, below a pedestrian crossing sign, and above an arrow pointing at the crosswalk. The beacons differ from standard flashing beacons by using a rapid flash frequency (approximately 190 times per minute), brighter light intensity, and ability to aim the LED lighting. RRFBs can be automated or pedestrian actuated, and feature an irregular, eye-catching flash pattern to call attention to the presence of pedestrians.” (Thomas et.al. 2016).

Summary of Effects: Rectangular Rapid Flash Beacons have been shown to significantly increase motor vehicle yielding behavior at uncontrolled crosswalks, with motor vehicle yield rates ranging from 34 percent to over 90 percent. Studies have also demonstrated reduced pedestrian-vehicle conflicts, increased stopping distance, and reductions in pedestrians trapped in roadway. This safety benefit extends not just to pedestrians but to bicycles crossing at RRFB locations. “PEDSAFE indicates that RRFBs are good for two-lane streets but may be less suited for use on multilane roads because of the multiple-threat crash potential” (Thomas, et.al. 2016, citing Zegeer et al. 2013).

The recent pedestrian safety synthesis finds “significant increases in motorist yielding behavior at uncontrolled crosswalks up to high rates (90 percent and above) in many locations, including at multilane sites, but yielding rates as low as 34 percent in some cities. Variations in outcomes may relate to time since installation, city of implementation, number of treatments in a city, width of crossing (and possibly number of lanes), traffic speed, and whether the beacon is activated. Higher speed limits (within the range of those studied, 30–45 mph from Texas sites) were associated with slightly higher rates of yielding at 22 sites in three cities; however, all sites also had “School Zone” signs.” (Thomas et.al. 2016). A study exploring the vast differences in yielding rates (the authors identified 19 percent to 98 percent range) analyzed the existing literature to find roadway, traffic, and traffic control device characteristics that are influencing driver yielding. The authors found an average yielding rate with RRFBs of 72 percent and that higher yielding was associated with shorter crossing distances, presence of a median refuge, and a two-leg crossing instead of a four-leg crossing (Fitzpatrick et.al. 2017).

A 2016 study to develop crash modification factors for four types of uncontrolled pedestrian crossing treatments found the CMF for crashes involving pedestrians with RRFBs to be 0.526, although the number of sites where RRFBs were studied was small (Zeeger et.al. 2016). Dougald found that an RRFB installed at a trail crossing had a positive effect on motor vehicle awareness, increased yield rates, and speed reductions (Dougald 2016). Two studies in Michigan which looked at yielding compliance found that adding an RRFB or other device to marked crosswalks achieved the highest compliance (Gates et.al. 2016; Stapleton et.al. 2017). In Texas, different flash patterns for RRFBs were tested and the study found no significant difference in the currently-used 2-5 flash pattern, wig-wag and simultaneous flash pattern, or Blocks pattern; different studies similarly found no difference in yielding if beacons were placed above or below the pedestrian crossing sign or if the signs were rectangular or circular (Fitzpatrick et.al. 2015a; Fitzpatrick et.al. 2016; Fitzpatrick et.al. 2015b). Researchers examining pedestrian behaviors at RRFB sites on a college campus in Virginia found increased perceptions of pedestrian safety and no significant effect between pedestrian crossing behavior after activation of the RRFB or crossing at the RRFB site without activation (Porter et.al. 2016). A study examining multiple safety measures of RRFBs in Florida found

improvement across all measures two weeks after installation: RRFB activation percentages, pedestrian crosswalk usage percentages, pedestrian crosswalk clearance percentages, driver yielding percentages, and driver smooth stopping percentages (Fartash et.al. 2016). A case study in Calgary, Canada found yielding increase from the low-80s before installation of RRFBs to over 95 percent yielding after installation (Mishra 2015).

ROUNDAABOUTS

Description and Purpose: Roundabouts control traffic at intersections without the use of signals or stop signs.

Summary of Effects: Roundabouts have been found to be very effective at reducing vehicular crashes. Studies of their effects on pedestrian and bicyclist safety have found that roundabouts lower speed and may increase yielding for pedestrians, leading to greater pedestrian safety. Some studies have shown that roundabouts are less safe for bicyclists.

The recent pedestrian safety Synthesis found that corridor-wide speed calming measures such as roundabout intersection designs (as well as road diets, speed humps, and tables) have the most evidence of lowering speeds and crashes (Thomas et.al. 2016).

A study of 72 roundabouts in Sweden looking specifically at bicycle and pedestrian safety found that the risk of a crash is about 2.5 times greater for bicyclists in the roundabouts than for bicyclists on a bicycle crossing. These roundabouts saw 67 police reported crashes over the study time period, with no fatalities. 8 of the studied roundabouts accounted for 48 crashes involving bicyclists, and of these roundabouts, 4 had 2 lanes. The study also found that roundabouts with a single lane and central island radius of more than 10 meters had the most reduction in crashes involving bicyclists (Brude and Larsson, 2000). Daniels, Nuyts, and Wets found that converting intersections to roundabouts increased the number of injury-causing crashes involving bicyclists by 27 percent. The study was conducted in Flanders-Belgium and produced CMFs ranging from 1.77 for fatal or serious injuries in urban areas to 1.01 for serious and minor injuries in rural areas (Daniels, Nuyts, and Wets 2008).

Another study in Minnesota focused on yielding at two roundabouts: a 1-lane roundabout and a 2-lane roundabout. Researchers found drivers entering the roundabout were more likely to yield to pedestrians than drivers exiting the roundabout (Hourdos, 2012). A study considering use of roundabouts by people with visual impairments studied 3 roundabouts in North Carolina and Colorado found that crossing treatments improved delay and risk (Schroeder et.al., 2011). The researchers recommend that roundabouts should have narrow lanes near crosswalks, detectable warnings, and wide splitter islands. More research was needed around the risk of raised crosswalks to those with visual impairments in roundabouts.

A study looking at the placement of midblock crosswalks between two-lane roundabouts found that placing the crosswalks 15, 20, and 30m from the exit of the roundabout was an ideal balance of pedestrian safety benefits with traffic performance and emissions (Fernandes 2015).

The NCHRP Report 672 explores the planning, design, construction, maintenance, and operation of roundabouts. The report includes the following guidance for improving roundabout safety for bicyclists and pedestrians: setting the crossing location back from the entrance line; including landscaping strips to separate vehicular and pedestrian traffic and guide pedestrians to crossings; implement speed control to reduce crash severity; use two-lane entrances/exits for major roadways and one-lane entries/exits on minor roadways; design refuge areas to be a minimum 6' to accommodate bicyclists, wheelchairs, and strollers; include accessibility features such as detectable warnings, ramps that connect sidewalks and

crosswalks, splitter islands, and signalized pedestrian crossings; terminate bicycle lanes or shoulders in advance of the roundabout to merge bicyclists, and reserve bicycle ramps for complex or multi-lane roundabouts (Rodegerdts et.al., 2010).

Both Signalized and Unsignalized

ACCESS MANAGEMENT

Description and Purpose: Access management treatments alter how and where roadways are accessed in order to mitigate conflicts that can occur at intersections and driveways. Access management treatments include reducing the number of driveways, exclusive turn lanes, pedestrian refuges, and restricting movements at intersections and driveways.

See individual treatments for more detail: No Turn on Red (NOTR) signs, Driveways, and Median Islands.

ADVANCE STOP/YIELD LINES

Description and Purpose: Advanced yield or stop lines are a pavement marking placed 20-50 ft. in advance of a crosswalk, intended to increase the distance between where drivers have stopped or yielded and the crosswalk. To alert drivers where to stop, these markings can be supplemented with “Stop Here for Pedestrians” or Yield Here to Pedestrians” signs (R1-5 or R1-5a).

Summary of Effects: Studies on the impact of advanced yield or stop lines on pedestrian safety show that conflicts are reduced by increasing the stopping distance for vehicles and improving pedestrian visibility, particularly in multi-threat situations. Overall, studies found that motor vehicle yielding does not greatly increase with the addition of advance stop/yield lines or signs. One study found a CMF of 0.75 for advanced yield or stop markings and signs (Zeeger et.al. 2016).

The recent pedestrian safety synthesis and PEDSAFE find that conflicts are reduced with advance stop bars or yield markings and signs. “Although the increases in percentages of motorists who yield or stop tends to be small, the distance at which motorists stop or yield has increased, which improves sight lines between motorists in other lanes and pedestrians and helps reduce the multiple-threat crash type (Malenfant and Van Houten 1990; Van Houten 1998; Van Houten and Malenfant 1992; Van Houten, Malenfant, and McCusker 2001; Van Houten et al. 2002)” (Thomas et.al. 2016).

A 2016 study on mostly urban, multi-lane intersections and midblock locations in the U.S. developed Crash Modification Factors (CMFs) for four different treatments including advance yield or stop markings and signs. Advanced yield or stop markings and signs were found to have a CMF of 0.75, or a 25 percent reduction in risk of crashes involving pedestrians (Zeeger et.al. 2016).

COLORLED PAVEMENT MARKINGS THROUGH INTERSECTIONS

Description and Purpose: Colored pavement markings through intersections are intended to provide bicyclists with a clear, highly visible pathway through an intersection. They also help to alert motor vehicles to the presence of bicycle through-traffic. The pavement can be colored with paint, Durable Liquid Pavement Markings (DLPM), thermoplastic, or colored asphalt.

Summary of effects: Studies on the safety of colored bicycle lane treatments at intersections show that colored bicycle lanes appear to create a safer cycling environment. The colored bicycle lanes have the effect of increasing both bicyclist and motor vehicle awareness and improving motor vehicle yielding. Bicyclists are more likely to use the colored bicycle lanes to navigate through an intersection than prior to

the application of colored pavement treatments. Colored bicycle lanes were found to have a non-significant positive effect on crash rates.

Hunter, Harkey, Stewart, and Birk found that after blue bicycle lane treatment installation in Portland, more motor vehicles yielded to bicyclists, more bicyclists complied with the marked path, and motor vehicle-cyclists conflicts decreased (Hunter et.al. 2000; cited in BIKESAFE). Hunter, Srinivasan, and Martell in 2008 had similar findings with green pavement used in an intersection in Florida; 95.8 percent of motor vehicles yielded to bicyclists compared to 86.7 percent in the before period (Hunter et.al. 2008; cited in BIKESAFE).

For treatments not involving colored pavement, see Mixing Zones (Monsere et.al., 2015; Hunter 2000).

CROSSWALK MARKING

Description and Purpose: A crosswalk marking helps identify a preferred crossing location at uncontrolled locations, or helps reinforce legal crosswalks at intersections through high-visibility markings. High visibility styles of crosswalk markings, such as continental or ladder-style, are typically preferred over traditional parallel line markings.

Summary of Effects: Studies referenced in both the recent pedestrian safety synthesis and PEDSAFE found that there are safety benefits associated with high-visibility crosswalk markings (Zegeer et al. 2013, Thomas et.al. 2016). These resources cite studies that developed crash modification factors for high-visibility crosswalk markings of 0.52 (Chen et al., 2012) and 0.63 (Feldman, Manzi and Mitman, 2010) when applied at general crossing locations and school crossing locations, respectively. These markings, when compared with traditional treatments, were also found to be more conspicuous and visible to drivers ((McGrane and Mitman 2013), especially relevant for their selection at uncontrolled locations.

Selecting the appropriate location to mark (or not mark) a crosswalk has been the focus of numerous studies (Carter et al. 2006). The most notable of these was an assessment of crash risk based on the characteristics of marked and unmarked crosswalk locations, which found that crash risk increases at uncontrolled marked crosswalks as vehicle speeds, volumes, and number of lanes increases (Zegeer, et. al 2002). Agencies are advised to plan carefully for locations where they anticipate marking crosswalks, using supplemental treatments such as raised median islands, advance yield lines, and beacon systems to support marked crosswalks at such locations.

CURB EXTENSION

Description and Purpose: Curb extensions (also known as “bulb-outs,” or “neck downs”), decrease the width of a roadway through the physical extension of a curb line or sidewalk in order to reduce vehicle speed, reduce crossing distance for pedestrians, and improve visibility between pedestrians and oncoming motor vehicles (PEDSAFE).

As described in the recent pedestrian safety Synthesis, “curb extensions are installations at intersections or midblock locations that extend the sidewalk or curb line into the street or parking lane, thus reducing the street width and crossing time. They should not extend into bicycle travel lanes or shoulders (Zegeer et al. 2013, PEDSAFE, Curb Extensions). Curb extensions have the potential to enhance pedestrian safety in several ways: by making pedestrians and motorists more visible to each other; when used at intersections, by keeping motor vehicles from parking too close to the intersection and blocking sight lines; by reducing crossing distance (which may also allow less signal time to be devoted to the pedestrian phase at signalized locations); and by narrowing radii at intersections, which may slow turning traffic. Curb extensions also tend to allow for better placement of curb ramps and prevent ramps from being

blocked by vehicles that park at the corner. They are useful in conjunction with parking restriction policies, such as daylighting” (Thomas et.al. 2016).

Summary of effects: Crash effects or behavioral effects of curb extensions for pedestrians have not been widely studied, although existing studies do indicate that curb extensions increase motor vehicle yielding, which has a clear safety benefit for pedestrians.

The recent pedestrian Synthesis noted that “although curb extensions are expected to have safety benefits based on exposure and design principles, behavioral or crash effects of curb extensions have not yet been widely studied. Curb extensions may be effective at improving motorist yielding as part of a larger package of treatments or designs (including crosswalks with advance stop/ yield lines and median islands). As mentioned, curb extensions are expected to slow turning speeds by narrowing the effective turn radius, and to improve sight lines between pedestrians and motorists (but this depends on implementation - for example tall shrubberies should not be used as part of landscaped areas). They may also help to lower overall crash severity based on a New York City study that attempted to measure crash effects at six locations (King 1999)” (Thomas et.al. 2016).

A 2015 study examined three safety countermeasures at zebra crossings to identify the most effective: curb extensions, parking restrictions, and advanced yield markings. This study used an advanced driving simulator with no-treatment and treatment scenarios on urban two-lane roads. By analyzing speed data combined with the conditions, effects of safety measures were revealed. The curb extension had the greatest effect on driver’s speed behavior when approaching the zebra crossing (confirmed by a driver questionnaire about countermeasure effectiveness, in which over 80 percent of drivers perceived curb extensions to be effective). Curb extensions also had the lowest number of instances of non-yielding (Bella and Silvestri 2015).

CURB RADIUS REDUCTION

Description and Purpose: Curb radius reductions force sharper turns and thus reduce turning speeds for vehicles; they also create larger waiting areas for crossing pedestrians. Actual curb radius is the radius of the curb, compared to the effective radius which is the path of a turning vehicle and which may be influenced by on-street parking, bicycle lanes, and advance stop lines. The recommended actual curb radius of 5-10 feet is recommended with an effective curb radius of 15-20 feet on urban streets with high pedestrian volumes.

Important considerations for curb radius reduction are as follows (Zegeer, et al. 2013, PEDSAFE, Curb Radius Reduction):

- Parking or bicycle lanes at an intersection can increase the effective radius.
- If curb radius is small in an area with high traffic volumes, vehicles may drive over the curb.
- Consider emergency vehicle access.

Summary of effects: The recent pedestrian safety Synthesis found that “no research studies were identified that have evaluated the safety or behavioral effects for pedestrians of reducing corner radii, which as mentioned, are intended to slow motorist turning speeds, and increase chances for motorists to perceive and yield to crossing pedestrians. Slower speeds are also expected to reduce the severity of crashes that do occur. There is an equation to estimate a CMF for all crashes based on the skew angle, (which can also lead to wide radii at some corners). The equation can be used to estimate effects compared to an intersection without any skew (all right angles). (See FHWA’s Speed Management Toolkit by Thomas, et al., 2015 or the CMF Clearinghouse)” (Thomas et.al. 2016).

In a study measuring the safety of various strategies to reduce crashes between right-turning vehicles and bicycles at signalized intersections in Oregon, a driving simulator was used to investigate causal factors of right-hook crashes. In the second experiment, treatments including signage, pavement markings, curb radii, and protected intersection designs were evaluated using 22 scenarios with 28 participants who turned right 596 times. All of the treatments had some positive effect on measured driver performance, but the magnitudes of the differences were unable to be mapped (Hurwitz et.al. 2015).

No research on curb radius reduction was identified in BIKESAFE or PEDSAFE.

GATEWAY TREATMENT

Description and Purpose: The Gateway Treatment is a new approach to the standard placement of one in-street Local Law Yield to Pedestrians in Crosswalk sign (MUTCD sign type R1-6) installed in the centerline of a roadway or on a median. This new approach involves placing three in-street signs between same direction travel lanes and placing additional signs on the right and left side of the road at each approach. These signs are placed in conjunction with a new or pre-existing crosswalk. The purpose of this treatment is to increase the percentage of drivers yielding to pedestrians on multilane roads.

Summary of Effects: Existing research suggests that the Gateway Treatment increases safety among pedestrian-vehicle interactions at crosswalks through increases in driver yielding. The relationship between the Gateway Treatment and frequency of crashes involving pedestrians has not yet been documented.

After studying the installation of the Gateway Treatment at three multilane sites in Michigan, Bennett, Manal, and Van Houten (2014) found that the Gateway Treatment is much more effective than the traditional approach of installing one in-street sign, and that the yielding levels achieved by this new approach were comparable to those of more expensive treatments like the Pedestrian Hybrid Beacon (also known as HAWK), or the Rectangular Rapid Flashing Beacon. Driver yielding rates at the treatment sites increased from less than 35 percent during baseline conditions to approximately 80 percent after installation of the full Gateway Treatment. When used in conjunction with HAWKs, the Gateway Treatment was associated with driver yielding rates as high as 95 percent (Bennett, Manal, and Van Houten, 2014).

In 2015 Bennett and Van Houten (2015) found that the use of the R1-6 signs and their initial setup of the signs placed in the street is critical to the success of the Gateway Treatment – treatments that used blank signs or placed signs on the curb were associated with lower yielding rates than the original Gateway Treatment.

It is possible that this treatment may also reduce vehicle speed by giving drivers the perception of a narrower roadway, however the validity of this theory has not been studied. Two noteworthy disadvantages of the Gateway Treatment include the signs not being able to remain in place in the winter in locations that receive heavy snowfall, and that signs will need to be replaced if struck by vehicles multiple times.

INTERCHANGE TREATMENTS

Description and Purpose: Interchange locations can be especially problematic for bicyclists and pedestrians. While nonmotorized users are often not exiting or entering freeways themselves, entrance and exit ramps often intersect with bicycling and walking routes, creating opportunities for conflicts between bicyclists, pedestrians, and motor vehicle traffic. For this reason, a number of treatments discussed elsewhere in this memorandum can be used to address safety at interchange locations. In particular, these include:

- Raised Median Islands
- Crosswalk Markings
- Mixing Zone Treatments

Using these treatments in combination requires a careful assessment of the particular type of interchange location. The Institute of Transportation Engineers (ITE) developed the Recommended Design Guidelines to Accommodate Pedestrians and Bicycles at Interchanges (Mitman and Ridgway, 2014) to provide agencies with tools for making these decisions.

Summary of Effects: Effectiveness of these particular treatments will vary based on which one is selected, and in what combination they are used. Individual sections written up in this memorandum include specific measures of effectiveness for these treatments.

LIGHTING

Description and Purpose: Illumination at crosswalks and along the roadway can help increase visibility for pedestrians and bicyclists, particularly at approaches to crosswalks.

Summary of Effects: Studies on the safety impacts of illumination show that increasing or adding lighting to crosswalks, road segments, and intersections improves pedestrian and bicyclist safety by reducing crashes, increasing compliance, and improving visibility. Overall, the studies examining lighting impacts on pedestrian safety primarily found changes in driver yielding, while studies of lighting and bicyclist safety found links between lighting and crashes. CMFs for crashes involving pedestrians range from 0.19-0.58 (Elvik and Vaa 2004; Ye et.al. 2008). Only one recent study found that lighting was associated with more crashes involving bicyclists and pedestrians (Niaki et.al. 2016). The authors speculate that this might be because lighting is installed at high crash locations.

The recent pedestrian safety synthesis finds that “pedestrian safety and motorist and driver behaviors such as pedestrians’ use of crosswalks and driver yielding are improved by the use of enhanced illumination at crossing locations. Lighting is estimated to reduce injury crashes at intersections by 27 percent (CMF 0.73), and reduce all severity of crashes at intersections by 21 percent (CMF of 0.79). For segments, lighting improvements are estimated to reduce injury crashes by 23 percent (CMF 0.77) and all crashes by 20 percent (CMF 0.80) (Toolbox of Countermeasures and Their Potential Effectiveness for Pedestrian Crashes 2013, citing Harkey et al. 2008)” (Thomas et al. 2016).

The Handbook of Road Safety Measures found that providing intersection illumination resulted in CMFs ranging from 0.58 (nighttime) to 0.19 for crashes involving pedestrians (Elvik and Vaa 2004). Ye, Pendyala, Washington, Konduri, and Oh found a CMF of 0.56 for adding intersection lighting to rural two-lane intersections (Ye et al. 2008). A recent Minnesota Department of Transportation literature review found safety improvements from lighting at roundabouts for total crashes, but did not look specifically at crashes involving bicyclists and pedestrians.

Many studies finding potential safety improvements from lighting examine the behavior of drivers and pedestrians but do not examine crashes (PEDSAFE). However, a study by Kim et al in North Carolina (2007) found a “lack of streetlights at night was associated with a 111 percent increase in the probability of a fatal outcome, and a less than 100 percent increase in the probability of an incapacitating injury.” Relatedly, a Dutch study of crash data by Wanvik (2009) found “that road lighting was associated with a 60 percent decrease in bicyclist injury collisions.” (BIKESAFE). Wang et al (2017) report similar conclusions from a study in Tampa Bay, Florida, that found reductions in fatal and severe crashes involving pedestrians and bicyclists at intersections, as well as reductions in head-on and angled crashes for motor vehicles, due to lighting.

NUMBER OF THROUGH LANES AND TURN LANES

Description and Purpose: The number of lanes on a roadway determine how far pedestrians must cross at an intersection and how many conflict points might exist for turning traffic and bicyclists or pedestrians. Efforts have been made to reduce the number and width of lanes through ‘road diets’ that not only reduce the number of lanes but provide space to implement additional pedestrian and bicyclist safety treatments such as adding bicycle lanes, median crossing islands, and reducing speed.

Summary of Effects: Studies on the number of lanes show that an increased number of lanes has a negative impact on pedestrian safety. Overall, reducing the number of lanes, such as through implementation of ‘road diet’ treatments, result in fewer crashes involving pedestrians. One study did note an increase in crashes involving pedestrians and bicyclists after a ‘road diet’ treatment but did not account for possible increases in numbers of pedestrians and bicyclists present. The impact of having right-turn lanes on pedestrian safety has not been widely studied; one study found that conventional right-turn lanes have substantially more crashes involving pedestrians than intersections with channelized right-turn lanes or shared through/turn lanes. For a deeper look at bicyclist treatments for right-turn lanes see the section on Mixing Zones.

The recent pedestrian safety Synthesis found that reducing the number of lanes through a ‘road diet’ results in fewer crashes involving pedestrians between and at intersections. “A study of road diets at 460 sites in New York City found that road diets have contributed to a trend of fewer crashes involving pedestrians between intersections, and decreases in total and injury crashes at segments and intersections abutting the road diet segments. The range of expected effects on total crashes is from 19 percent to 47 percent reduction. The expected effects are in the lower range in large urban areas, and in the higher range for main roads passing through smaller towns and suburban areas. An average 29 percent reduction (CMF of 0.71) for total crashes across all sites was estimated (Toolbox of Countermeasures and Their Potential Effectiveness for Pedestrian Crashes 2013, citing Harkey et al. 2008). Road diets have also been found to reduce motor vehicle operating speeds, which are also associated with expected reductions in injury and fatal crashes” (Thomas et.al. 2016).

The Federal Highway Administration, as cited in PEDSAFE, lists benefits of four-to-three lane road diets:

- “Decreasing vehicle travel lanes for pedestrians to cross, therefore reducing the multiple-threat crash” (FHWA ‘Road Diet’). Further evidence of the benefits of reducing roads with four or more lanes can be found in a 2001 study by Zegeer, et al., which found that two- and three-lane roads were associated with reduced pedestrian crash risk when compared to roads with four or more lanes (Zeeger, et.al. 2002).”
- “Providing room for a pedestrian crossing island” (FHWA ‘Road Diet’). Pedestrian islands have been demonstrated to increase pedestrian safety on roadways.”
- “Improving speed limit compliance and decreasing crash severity when crashes do occur” (FHWA ‘Road Diet’). Reducing vehicle speeds decreases vehicle stopping distance as well as decreases the likelihood of a pedestrian-vehicle crash resulting in a fatality. A 1987 U.K. Department of Transportation study found that the risk of pedestrian death is 5 percent at impacts of 20 mi/h, 45 percent at impacts of 30 mi/h, and 85 percent at impacts of 40 mi/h (U.K. DOT 1987), indicating that decreasing average vehicle speed decreases injury severity and the probability of pedestrian fatality in the event of a collision.”
- “Creating a buffer between pedestrians and traffic through the addition of bicycle lines and on-street parking (FHWA ‘Road Diet’). Such buffers increase safety by leading to lower vehicle speeds

as well as an increase the perception of safety for pedestrians traveling on adjacent sidewalks, leading to greater pedestrian mobility.”

An analysis done by Chen, Chen, Ewing, McKnight, Srinivasan, and Roe looked at the average rate of crashes involving pedestrians on roadway segments that had received a road diet project, and found that it increased by 3.48 percent at intersection treatment sites compared to a 17.86 percent decrease at comparison sites. Although not mentioned, this may not be adjusted for increased pedestrian volumes post-installation (Chen et.al. 2012). Safety performance function (SPF) models for high-frequency and high-severity types of crashes involving pedestrians at intersections were developed by Thomas, Lan, Sanders, Frackelton, Gardner, and Hintze. They found that larger number of lanes (as well as larger number of intersection legs and other factors) were associated with more crashes involving pedestrians (2017). The same study aimed to develop SPF models for types of intersection crashes bicyclists: total crashes involving bicyclists at intersections, crashes in which the bicyclist and driver were traveling in opposite directions, and crashes in which the bicyclist and driver were traveling perpendicularly to one another. Variables such as larger numbers of lanes and two-way center turn lanes were predictors of crashes involving bicyclists (Thomas et.al. 2017). Chen and Zhou studied crash profiles in the city of Seattle and found that: densities of 4-way intersections and more-than-five-way intersections and areas with an averagely higher posted speed limit are positively correlated with the frequency and risk of crashes involving pedestrians (2016). Shawky, Garib, and Al-Herthei examined factors such as road and site characteristics of crashes for crossing pedestrians in Abu Dhabi (AD) UAE. Factors that that have a significant impact on the crash-injury severity for pedestrians are the speed limit and road width in terms of number of lanes (Shawky et.al. 2014). Potts, Bauer, Torbic, and Ringert (2013) studied differences in rates of crashes involving pedestrians between channelized right-turn lanes, conventional right-turn lanes, and shared through- and right-turn lanes using data from almost 400 intersections in Toronto. They found significantly more (70-80 percent) crashes involving pedestrians for conventional right-turn lanes compared to the other two right-turn lane designs (Potts, Bauer, Torbic, and Ringert 2013).

For bicycles, road diets have been found to increase incidence of crashes involving bicyclists by 5.9 percent at intersection road diet sites, versus a 25.6 percent decrease at comparison intersections. However, bicycle volumes before and after implementation of a road diet were not recorded so results do not take into account the possible increase in volume of bicyclists using the facilities (Chen et.al. 2012; as cited in BIKESAFE).

A 2015 pilot study examined crash rates for non-motorized mode users and found that among other factors, providing dedicated right turn lanes can significantly reduce the number of crashes involving pedestrians and bicyclists (Burbridge 2015). PEDSAFE does not include any studies on improvements for right-turn lane design.

PEDESTRIAN CROSSWALK YELLOW WARNING SIGN

Description and Purpose: A pedestrian crosswalk yellow warning sign is a sign used to alert drivers and bicyclists of a pedestrian crossing. It is typically used in locations where road users are not expecting a pedestrian crossing (Synthesis). The purpose of this treatment is to improve pedestrian safety by increasing the number of drivers yielding to pedestrians.

Summary of Effects: While a number of studies have been conducted on this treatment, its effect on pedestrian safety remains unclear (Synthesis). The impacts of pedestrian crosswalk warning signs vary considerably across existing studies and are dependent on whether the standard yellow sign is implemented by itself or in combination with other treatments or enhancements. The inclusion of signs with additional countermeasures such as design improvements or lighting may result in increased driver

yielding (Synthesis). In addition, signs which are enhanced with flashing beacons or lights have been shown to be more effective when activated manually or automatically by pedestrians than those that blink continuously (Fitzpatrick et al., 2006). According to the Synthesis, a few studies have observed a reduction in pedestrian-vehicle conflicts after installing a pedestrian crosswalk yellow warning sign but a robust CMF has not yet been derived.

A recent study conducted by Yuan and Dulaski (2017) compared driver yielding rates to crosswalks using the pedestrian crosswalk yellow warning sign during different times of day. The authors found that there was a large decrease in driver yielding rates from daytime (77 percent) to nighttime (33 percent), as well as an increase in driver approaching speeds in nighttime conditions compared to daytime conditions. These findings suggest that the safety improvements associated with this treatment are limited to daytime conditions, further indicating that this treatment may be more effective when enhanced with lighting or other treatments that are not as limited by the reduction in visibility associated with nighttime conditions.

PARKING RESTRICTIONS/DAYLIGHTING

Description and Purpose: “Vehicles that are parked too close to pedestrian crossings obscure the visibility of both drivers and pedestrians. This increases the risk of dart-dash-type pedestrian collisions at both intersections and midblock crosswalks. Eliminating parking from the roadway near a crossing can improve pedestrian and driver sight lines and make it easier for pedestrians to cross. It is important to install physical barriers to negate the possibility of illegal parking. The removal of parking spaces can also free up the roadway for other uses, such as curb bulb outs/extensions, landscaping, or bicycle parking (Zegeer et al. 2013, PEDSAFE, Parking Restrictions)” (Thomas et.al. 2016)

Summary of Effects: Restricting parking near intersections has been shown to reduce crashes. By restricting parking near intersections, also known as daylighting, the visibility of both pedestrians and motor vehicles is increased. There are only a handful of existing studies of the safety effects of parking restrictions, but the overall results show that they have a positive effect on driving behavior: studies showed that when parking restrictions were present, pedestrians were more visible to motor vehicles and there was less aggressive driver behavior (Bella & Silvestri 2015; Obeid et.al. 2017). As found in the recent Synthesis, parking restrictions had a CMF of 0.7 (Toolbox of Countermeasures and Their Potential Effectiveness for Pedestrian Crashes 2013, citing Gan et al. 2005; Thomas et.al. 2016).

The recent pedestrian safety synthesis recommends that “Typically, jurisdictions should restrict vehicles from parking within at least 20 ft of an intersection. In its Urban Street Design Guide, NACTO (n.d.) suggests 20–25 ft restricted areas” (Thomas et.al. 2016).

PROTECTED INTERSECTIONS

Description and Purpose: Protected intersections are a Dutch concept that involves a reconfiguration of intersection geometry to better protect bicyclists. It can be used with separated bicycle lanes and other pavement markings approaching the intersection, but has a few key intersection features: introduction of physical separation such as islands or bollards to reduce the available turning radius for cars and re-aligning bicycle lanes and paths away from travel lanes to better position through-bicyclists to avoid conflict with right-turning cars (Gilpin et.al. 2015). These re-alignments are sometimes referred to as ‘bend-outs.’ The treatment creates a recessed crossing for bicyclists, which provides space for motor vehicles to turn right, have greater ability to see and react to approaching bicyclists, and yield before encountering bicycle conflicts. “The bend-out design positions bicyclists downstream on the side street away from the intersection, allowing vehicles to complete turning movements before interacting with bicyclists” (Gilpin et.al. 2015).

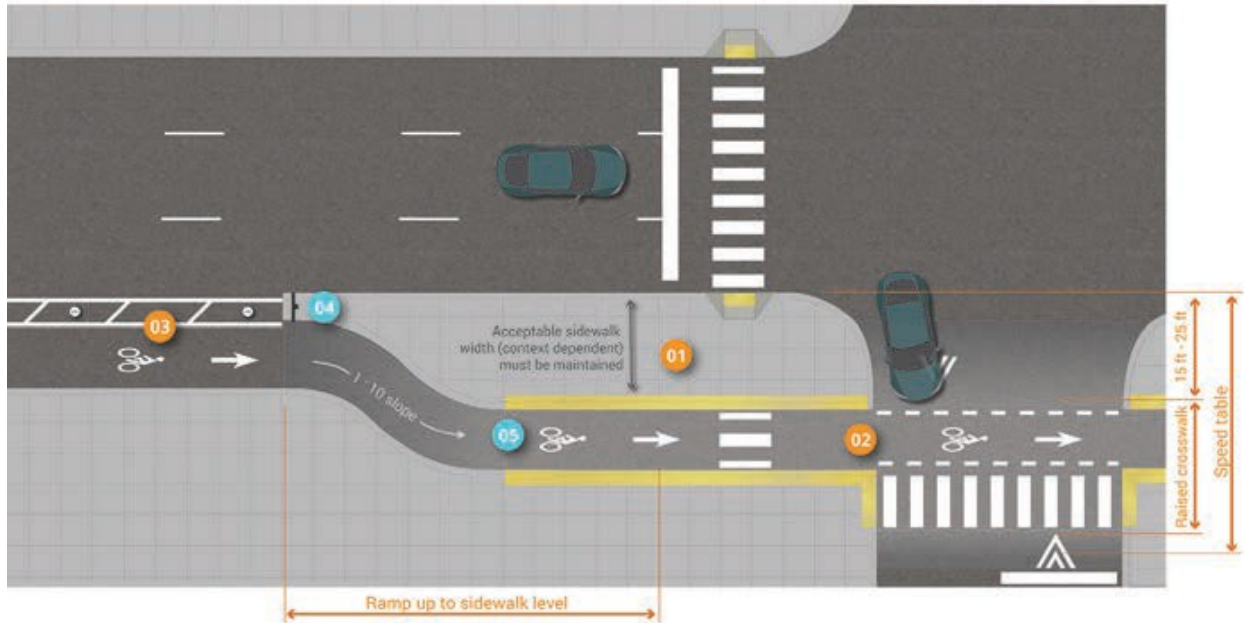


Figure A-46. 'Bend Out' design recommendation for separated bike lanes at intersections. Source: FHWA 2015.

Summary of Effects: No studies were found focused on Dutch intersections, protected intersections, or bend-outs specifically. However, these types of treatments are new in the US and studies are currently underway to assess their effectiveness.

The MASSDOT Separated Bike Lane Planning and Design Guide 2015 notes that “protected intersections are preferable to mixing zones” and “protected intersections reduce opportunities for conflicts involving bicyclists and motor vehicles to one location - where turning vehicles cross the SBL.” “Bicycle and pedestrian crossings set back from the intersection create space for turning motorists to yield to bicyclists and pedestrians. Research has found crash reduction benefits at locations where bicycle crossings are set back from the motorist travel way by a distance of 6 ft. to 16.5 ft. (2 - 5 m)” (Schepers et.al. 2011; Brüde and Larsson 1992; Schnüll and Lange 1992; Welleman and Dijkstra 1988; as cited in MASSDOT 2015). A study in 2015 examining mixing zones and intersections with turning zones and a through bicycle lane (TBL) found that “protected TBLs were more effective than mixing zones with pavement markings through conflicts at increasing driver and cyclist design compliance” (Monsere 2015).

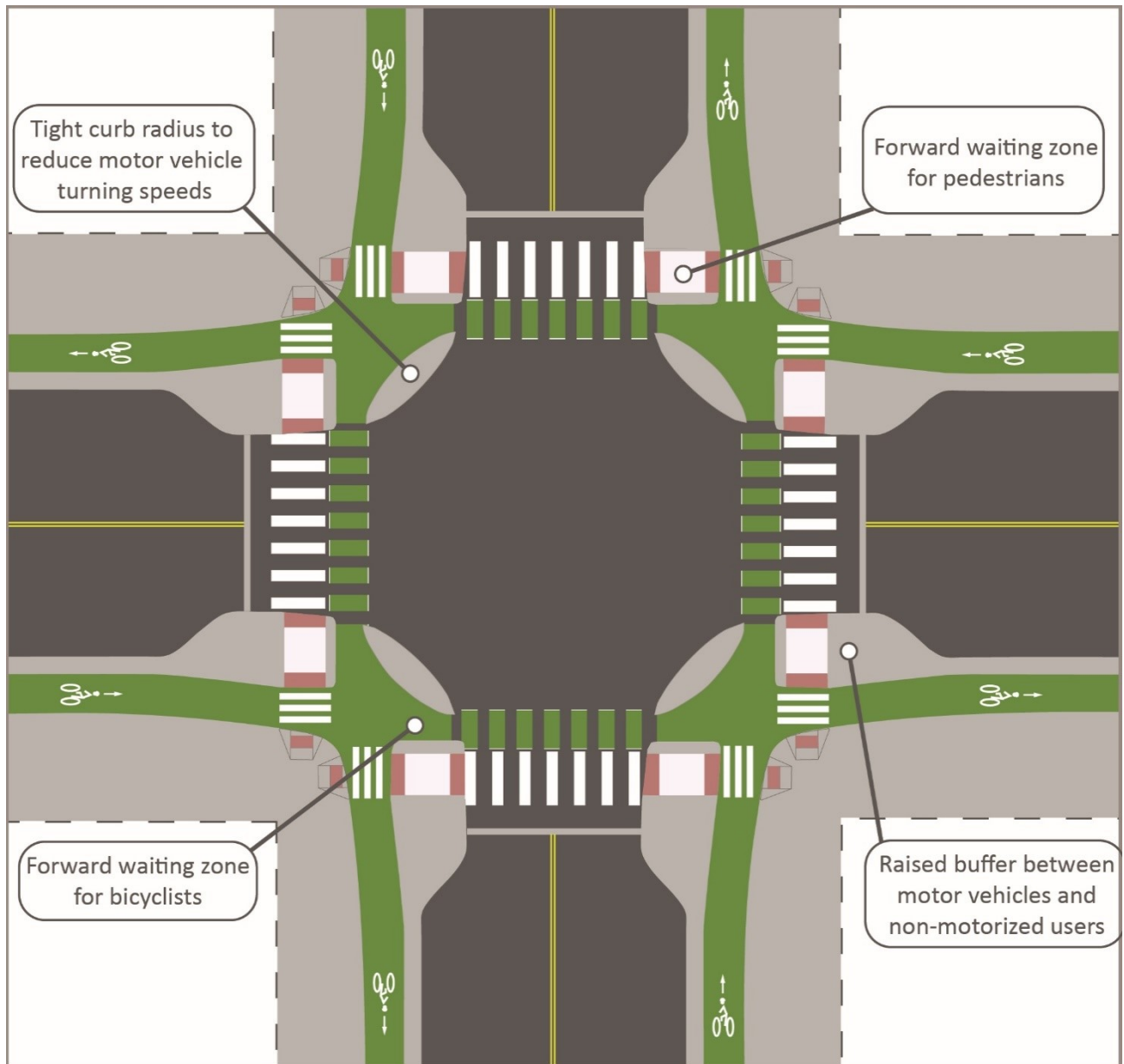


Figure A-47. The protected intersection features a package of treatments that work together to separate conflict points.

RAISED CROSSINGS

Description and Purpose: As noted in the Synthesis and according to PEDSAFE, “raised pedestrian crossings are a type of vertical traffic control measure that can reduce vehicle speeds, reduce the need for curb ramps (though truncated domes should still be included), and enhance the pedestrian crossing environment. A raised pedestrian crossing is similar to a raised intersection, but it is only the width of a crosswalk, usually 10 to 15 ft. Construction involves providing ramps on each vehicle approach, which elevates the entire crosswalk to the level of the sidewalk. The crosswalks can be built with a variety of materials, including asphalt, concrete, stamped concrete, or brick pavers (Zegeer et al 2013, PEDSAFE, Speed Tables, Raised Pedestrian Crossings countermeasures).” Raised pedestrian crosswalks or speed

tables can be applied at midblock locations, low speed streets, collector roads, shopping centers, and campuses. Raised crossing can also be used for bicyclist crossings.

Summary of Effects: While raised crossing treatments should result in lower speeds and better pedestrian visibility, studies have not measured the effect on motor vehicle yielding and crashes involving pedestrians. Speeding reduction is better documented and speed tables have been shown to cause speed reductions in five studies cited by FHWA (Engineering Speed Management Countermeasures: A Desktop Reference of Potential Effectiveness in Reducing Speed 2014).

The City of Boulder found that while multiple crossing treatments, including raised crossings, result in a significant increase in driver compliance and yielding at crosswalks, some may lead to higher frequencies of crashes involving motor vehicles and crashes involving pedestrians at high-volume locations (City of Boulder, 2011). The Handbook of Road Safety Measures found that installing raised pedestrian crosswalks resulted in a CMF of 0.55 for crashes involving pedestrians (Elvik and Vaa 2004).

SPEED MANAGEMENT

Description and Purpose: Speed management, is beneficial to pedestrians and bicyclists to slow motor vehicle speeds at intersections where pedestrians and bicyclists cross or otherwise interact with motor vehicle traffic. Traffic calming is one approach to speed management. “The objective of traffic calming is usually to reduce traffic speed or volume, reduce conflicts between local traffic and through traffic, make it easier for pedestrians to cross the road, and reduce traffic noise. Traffic calming can be applied both in residential areas and on roads that have commercial roadside development” (PEDSAFE).

“Measures that are part of traffic calming include:

- Narrowing driving lanes, often by widening sidewalks
- Installing chokers or curb bulbs
- Using cobblestone in short sections of the road
- Providing raised crosswalks or speed humps
- Installing transverse rumble strips, usually at the start of the treated roadway segment
- Providing parking bays” (PEDSAFE).

Other approaches to speed management include changing signal progression along a corridor to slow vehicle speeds or “slow zones” as applied by the city of New York which are an application of traffic calming measures to an entire residential area rather than just individual roadways.

Summary of Effects: Traffic calming includes the application of many individual treatments that have been studied in detail and can be found throughout this report. Specific effects of corridor-wide traffic calming on pedestrians have not been formally studied. However, combinations of treatments that make up traffic calming have been found to lower vehicle speeds and increase yielding, which both have a positive impact on pedestrian safety and crash rates.

The pedestrian safety Synthesis found no formal studies that assess corridor-wide traffic calming on pedestrians; “However, studies on area-wide traffic-calming measures of effectiveness are the closest approximation to corridor-wide speed calming available. Area-wide traffic calming, using a combination of multiple types of countermeasures, has resulted in an estimated percent reduction in fatalities and injury collisions for all road users (Elvik and Vaa 2004). Treatments commonly applied in traffic-calming schemes, such as street closures, cul-de-sacs, diverters, traffic circles, shared street design, chicanes, flares/chokers, speed humps, speed limit signs and speed zones, enforcement programs, walkways, parking controls, and other signage, have also yielded reductions in vehicle speeds. Results of the meta-

analysis indicated that these types of schemes reduced the number of total injury collisions for all road users by about 11 percent across all urban road types according to the CMF Clearinghouse review of this study (CMF ID 588 Details, n.d.). Effects were greatest on residential/local streets (18 percent; CMF ID 587 Details, n.d.) and lower on main roads (6 percent; CMF ID 586 Details, n.d.)” (Thomas et.al. 2016).

Various measures of traffic calming, such as raised crosswalks, have been studied for their impact on pedestrian safety and are covered in detail in PEDSAFE (see Appendix A). BIKESAFE also covers the research on traffic calming’s impact on bicyclists. The studies covered in BIKESAFE conclude that the speed is a factor in bicyclist safety; severity of crash increased with an increase in vehicle speeds and injuries decrease with a decrease in vehicle speeds (Kim et.al. 2007; Grundy et.al. 2009). Bicyclist injury is reduced by half at intersections with speeds of 30km/h or less (Harris et.al. 2013).

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APPENDIX B: Coding Procedure

Post Encroachment Time (PET)

The team defined Post Encroachment Time (PET) for this project as “the time difference between when one road user passes out of an area of possible collision and when the conflicting road user arrives at the same area” (Puscar et al. 2018). In this study, a road user could be a pedestrian, bicyclist, or motor vehicle. **The team defined “events” as when road users paths of travel intersected each other with PET of five seconds or less for the following two road users types:** bicycle-motorist, pedestrian-motorist, bicycle-pedestrian, and bicycle-bicycle. The team did not analyze motorist-motorist or pedestrian-pedestrian events. A PET of five seconds or less was based on similar studies (Table B-1).

Table B-1. Table of Conflict Definitions from the literature

Definition	PET Threshold or Description	Modes	Reference
Conflict	≤ 5 sec	Bicyclist-motorist	Zagenehpour et al 2015
Dangerous conflict	≤ 1.5 sec		
Conflict	≤ 3 sec	Bicyclist-motorist	Kassim et al 2014
Event	≤ 5 sec	Bicyclist-motorist	Hurwitz et al, 2015
Undescribed event	3-5 sec		
Undescribed event	2.5-3 sec		
Low risk	1.6-2.0		
Moderate risk	1.0-1.5 sec		
High risk	≤ 0.9 sec		
Conflict	≤ 1 second	Pedestrian-motorist	Ni and Wang 2015 (China)
Conflict	< 4 seconds	Bicyclist-motorist	Puscar et al 2018
Conflict	series of events that could lead to a collision	Bicyclist-motorist	Monsere et al 2015
Precautionary conflict	low-risk interaction where a minor change in direction or speed was needed to avoid a conflict		
Minor Conflict	precautionary braking or change of direction, or both		
Substantial conflict	emergency braking or change of direction, or both		
Major conflict	near collision with emergency braking or change of direction, or both		

Coders used Adobe Prelude to locate the time stamps of each event in intersection footage; coders identified the specific frame in which a road user entered or exited the potential conflict area. These time stamps were used to determine PET. Any obvious interaction, such as slowing or yielding, regardless of whether it seemed to fit strictly within the five-second PET interval, was coded.

Coding Procedure

Prior to beginning the coding, a team of three people performed an inter-rater reliability test to ensure the video would be processed with similar judgement. The coding procedure enables coders to reduce intersection footage to three types of data: motor vehicle turning movements, pedestrian and bicyclist travel, and events with or without interactions. Video footage was gathered by four cameras for each intersection, and each camera looked at one leg of the intersection. The coder was responsible for analyzing certain observations based on the visibility of the angle of the camera (Figure B-1, B-2):

- Pedestrians traveling away or toward the camera
- Bicyclists traveling toward the camera and going straight, taking a right turn, or beginning a two-stage left
- Bicyclists taking a left turn from the motor vehicle left turn lane on the leg in question
- Bicyclists traveling away from the camera (heading the wrong way)



Source: Map data ©2019 Google Earth

Figure B-1 Coded Ped/Bike Maneuvers from Single Camera



Source: Map data ©2019 Google Earth

Figure B-2 Coded MV Maneuvers from Single Camera

For each observed road user, the coder followed a series of steps, listed below, based on the type of road user and if an event or interaction occurred. Adobe Prelude allowed the coder to apply a time stamp and a series of descriptors to a road user or event. The time stamps of pedestrians and bicyclists, if not involved in an event, were based on the time the user first enters the crosswalk or passes the curb.

The coding procedure varies some between the two protected intersection locations. The team coded whether the pedestrian pressed the crossing button for Berkeley (when visible), but this step was omitted for Sale Lake City as the camera angle prevented sight of some of the buttons. The crossing signal changed automatically in Salt Lake City, so pressing the button is not necessary for the pedestrian to cross in compliance.

Any out of the ordinary actions were narrated in separate notes. If movement was unclear in the camera angle, coders could refer to footage from the camera mounted to the right of the initially viewed camera to get a position of the bicycle when entering intersection, etc. The coder could also mark unclear events for a second person to review.

Coding Procedure for Pedestrians and Bicyclists

1. **Type of road user**
 - a. Pedestrian; Bike
2. **Platoon:** The number of multiple pedestrians or bicyclists travelling together through the intersection
3. **Approach:** The position a pedestrian or bicyclist enters the intersection relative to the camera. During the final processing, approach was connected to cardinal directions.
 - a. Near; Far
4. **Signal:** The action of a pedestrian or bicyclist when arriving at intersection regarding traffic signal for their direction of travel
 - a. Arrive & exit green; Arrive & exit yellow; Arrive green, exit red; Arrive yellow, exit red; Arrive red, wait for green; Right on red; Ran light; Jump green
5. **Wait position:** Where the user waits if arriving at intersection on red light and waits for green
 - a. Curb; Protected (in dedicated bike lane or island before motor vehicle lane); Beyond
6. **Initial position:** Where on the street a pedestrian travels through intersection or bicyclist approaches intersection
 - a. Bike lane; Crosswalk; Motor vehicle through lane; Motor vehicle left lane; Sidewalk (exclusively used for bicyclists)
7. For pedestrians only:
 - a. **Pedestrian signal:** The crosswalk signal display when pedestrian steps into the intersection
 - i. Walkman; Stop Hand; Flashing
 - b. **Pedestrian crossing compliance:** Whether pedestrian uses crosswalk travelling through the intersection
 - i. Uses crosswalk; Veers out
 - c. **Pedestrian button:** Whether pedestrian presses the crosswalk button
 - i. Berkeley pedestrians only
 - ii. Use Button; No Button
 - d. **Cross options:** The maneuver of the pedestrian in relation to the median island
 - i. Salt Lake City pedestrians only – 300 South has a median island
 - ii. Two-stage, Half cross, Half near, Half far
 1. 2-stage = User crosses half of intersection, stops to wait at median island, then continues across second half
 2. Half cross = User crosses half of intersection before turning to travel along median sidewalk
 3. Half near = User travels along median sidewalk before turning to cross near half of intersection
 4. Half far = User travels along median sidewalk before turning to cross far half of intersection
8. For bikes only:

- a. **Bike maneuver:** Action bicyclist takes in the intersection
 - i. Thru; Right; Regular left; Two-stage left; Two-stage through; U-turn
 - 1. Regular left = same left turn a motor vehicle would take
 - 2. Two-stage left = left turn done crossing two legs of intersection
 - 3. Two-stage through = bicyclist entering the intersection facing traffic, crossing their initial leg at the crosswalk, and continuing with their initial direction traveling with traffic
- b. **Bike direction:** The direction a bicyclist is travelling relative to traffic
 - i. With traffic; Against

Coding Procedure for Turning Motor Vehicles

The team also coded motor vehicle turning movements for all studied intersections. Only certain motor vehicles turns were coded from camera angles: motor vehicles turning left from the left leg or turning right from the far leg (Figure B-3). Additional notes, such as if the motor vehicle encroached on the crosswalk or drove over an extended curb, were added if needed.

- 1. **Motor vehicle type:** Category of the motor vehicle involved in event
 - a. Passenger; Truck/Bus; Motorcycle
 - i. Passenger = car, pickup truck, SUV, passenger van
 - ii. Truck/Bus = tractor trailer/semi, delivery/work vehicle, moving van
 - iii. Motorcycle = also moped or motorized scooter
- 2. **Motor vehicle approach:** The intersection leg relative to the camera that the motor vehicle is initially travelling on
 - a. Near; Far; Left; Right
- 3. **Motor vehicle maneuver:** Action motor vehicle takes in intersection
 - a. Left; Right; Through; U-turn

Coding Procedure for Events

Coders performed the procedures listed above to describe the trajectories of both road users in the event. Coders then followed these steps to describe the event. If the second road user was another pedestrian or bike, this information was written in the notes and reduced to the regular format later.

- 1. **Event type:** The intersection users involved in the event
 - a. Pedestrian-Motor vehicle; Bike-Motor vehicle; Bike-Bike; Bike- Pedestrian
- 2. **First party:** user that arrived at the potential conflict area first
 - a. Pedestrian; Bike; Motor vehicle
- 3. **Second party:** user that arrived at the potential conflict area second
 - a. Pedestrian; Bike; Motor vehicle
- 4. **Interaction:** The behavior of one or both parties during the event
 - a. 1 cross 2; 1 Gradual slow/stop/trajectory change; 1 Sudden slow/stop/trajectory change; 2 Gradual slow/stop/trajectory change; 2 slow/stop/trajectory change; Both slow/stop/change trajectories; Collision
- 5. **Yielder:** The party which yields to the other, if necessary
 - a. No Need; User 1 Yielded (U1 Yield); User 2 Yielded (U2 Yield); Near Miss

Each leg's data was compiled together to provide a comprehensive overview of movements and events in the intersection.

Types of Interactions

For the majority of the time, interactions corresponded to a certain type of yielder. Coders agreed upon and used these patterns shown in Table B-2. This table encompasses the majority but not all interactions:

Table B-2. Common Interactions and Corresponding Yielders

Interaction: "1 Cross 2" applies to an interaction in which a road user crosses in front of a person or vehicle within approximately five seconds and there was no observable change in speed or trajectory. Change in speed could be rapid slowing, stopping, or pause in momentum than normal for the turn. Yielder: No obvious yield required or observed; or there could be a Near Miss.
Interaction: "Gradual slow/stop/trajectory change" if User 1 or User 2 observably slowed, paused, and/or came to a near or full stop to allow the other party to clear the area OR changed their path of travel. Yielder: User 1 or 2 intentionally yielded; or there could be a Near Miss.
Interaction: "Sudden slow/stop/trajectory change" if User 1 or User 2 slowed or stopped suddenly, swerved, or in the case of the pedestrian jumped (any direction) or ran out of the way. Yielder: This would most likely be coded as a "Near Miss"

Coding Roster

Coders analyzed the same legs of the treated and comparison intersections to reduce bias.

Table B-3. Salt Lake City Coders

Wednesday, May 23, 2018	7:00AM to 9:00AM	Noon to 1:00PM	4:00PM to 6:00PM
North - Camera B	M. Vann	M. Vann	M. Vann
South - Camera D	K. Heuser	K. Heuser	K. Heuser
East - Camera C	K. Heuser	K. Heuser	K. Heuser
West - Camera A	S. Blank	S. Blank	S. Blank
Thursday, May 24, 2018	7:00AM to 9:00AM	Noon to 1:00PM	4:00PM to 6:00PM
North - Camera B	M. Vann	M. Vann	M. Vann
South - Camera D	K. Heuser	M. Vann	K. Heuser
East - Camera C	K. Heuser	K. Heuser	K. Heuser
West - Camera A	S. Blank	S. Blank	S. Blank
Saturday, May 26, 2018	7:00AM to 9:00AM	Noon to 2:00PM	4:00PM to 6:00PM
North - Camera B	N/A	M. Vann	N/A
South - Camera D	N/A	M. Vann	N/A
East - Camera C	K. Heuser	K. Heuser	K. Heuser
West - Camera A	S. Blank	S. Blank	S. Blank

Where N/A = leg not coded

Table B-4. Berkeley Coders

Wednesday, May 23, 2018	7:00AM to 9:00AM	Noon to 1:00PM	1:00PM to 4:00PM	4:00PM to 6:00PM
North - Camera B	N/A	N/A	N/A	N/A
South - Camera D	S. Blank	M. Vann	M. Vann	S. Blank
East - Camera C	K. Heuser	M. Vann	K. Heuser	M. Vann
West - Camera A	N/A	N/A	N/A	N/A
Thursday, May 24, 2018	7:00AM to 9:00AM	Noon to 1:00PM	1:00PM to 4:00PM	4:00PM to 6:00PM
North - Camera B	N/A	N/A	N/A	N/A
South - Camera D	N/A	K. Heuser	K. Heuser	K. Heuser
East - Camera C	N/A	S. Blank	M. Vann	M. Vann
West - Camera A	N/A	N/A	N/A	N/A
Saturday, May 26, 2018	7:00AM to 9:00AM	Noon to 2:00PM	1:00PM to 4:00PM	4:00PM to 6:00PM
North - Camera B	N/A	N/A	N/A	N/A
South - Camera D	S. Blank	M. Vann	K. Heuser	N/A
East - Camera C	S. Blank	M. Vann	K. Heuser	N/A
West - Camera A	N/A	N/A	N/A	N/A

Where N/A = leg not coded

Appendix B References

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APPENDIX C: Signal Timing Details

The signal timing details below were observed from the video collected and do not always match the signal timing plans provided by the cities. (FWD is Flashing Don't Walk.)

The Alameda and Hopkins (Treated Intersection)

- Countdown timers? Yes

AM and Midday (Weekday and Saturday)

- Cycle length: 65 seconds
- The Alameda
 - o Green time: 34 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 27 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 20 seconds
- Hopkins
 - o Green time: 21 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 40 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 14 seconds

PM (Weekday)

- Cycle length: 70 seconds
- The Alameda
 - o Green time: 38 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 28 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 20 seconds
- Hopkins
 - o Green time: 21 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 45 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 14 seconds

PM (Saturday)

- Cycle length: 65 seconds
- The Alameda
 - o Green time: 34 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 27 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 20 seconds
- Hopkins
 - o Green time: 21 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 40 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 14 seconds

The Alameda and Marin (Comparison Intersection)

- Countdown timers? Yes

AM and Midday (Weekday and Saturday)

- Cycle length: 65 seconds
- The Alameda
 - o Green time: 25 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 36 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 18 seconds
- Marin
 - o Green time: 30 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 31 seconds
 - o Pedestrian Walk: 14 seconds
 - o Pedestrian FDW: 16 seconds

PM (Weekday)

- Cycle length: 70 seconds
- The Alameda
 - o Green time: 25 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 41 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 18 seconds
- Marin
 - o Green time: 35
 - o Yellow Clearance: 4
 - o Red Clearance: 31
 - o Pedestrian Walk: 14 seconds
 - o Pedestrian FDW: 16 seconds

PM (Saturday)

- Cycle length: 65 seconds
- The Alameda
 - o Green time: 25 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 36 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 18 seconds
- Marin
 - o Green time: 30 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 31 seconds
 - o Pedestrian Walk: 14 seconds
 - o Pedestrian FDW: 16 seconds

200 W and 300 S (Treated Intersection)

- Countdown timers? Yes

6:30 AM – 9:30 AM (Weekdays)

- Cycle Length: 60 seconds (30 second offset)
- 200 W
 - o Green time: 26 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 30 seconds
 - o Pedestrian Walk: 5 seconds
 - o Pedestrian FDW: 20 seconds
- 300 S
 - o Green time: 24 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 32 seconds
 - o Pedestrian Walk: 5 seconds
 - o Pedestrian Clearance: 18 seconds

9:30 AM – 3:30 PM (Weekdays) and All Day (Saturday)

- Cycle Length: 72 seconds (0 second offset)
- 200 W
 - o Green time: 31 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 37 seconds
 - o Pedestrian Walk: 5 seconds
 - o Pedestrian FDW: 20 seconds
- 300 S
 - o Green time: 31 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 37 seconds
 - o Pedestrian Walk: 5 seconds
 - o Pedestrian Clearance: 18 seconds

3:30 PM – 6:30 PM (Weekdays)

- Cycle Length: 60 seconds (38 second offset)
- 200 W
 - o Green time: 26 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 30 seconds
 - o Pedestrian Walk: 5 seconds
 - o Pedestrian FDW: 20 seconds
- 300 S
 - o Green time: 24 seconds
 - o Yellow Clearance: 4 seconds
 - o Red Clearance: 32 seconds
 - o Pedestrian Walk: 5 seconds
 - o Pedestrian Clearance: 18 seconds

200 E and 300 S (Comparison Intersection)

- Countdown timers? Yes

6:30 AM – 9:30 AM (Weekdays)

- Cycle Length: 60 seconds (12 second offset)
- 200 E
 - o Green time: 25.7 seconds
 - o Yellow Clearance: 3.4 seconds
 - o Red Clearance: 30.9 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 17 seconds
- 300 S
 - o Green time: 22.8 seconds
 - o Yellow Clearance: 3 seconds
 - o Red Clearance: 34.2 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian Clearance: 12 seconds

9:30 AM – 3:30 PM (Weekdays) and All Day (Saturday)

- Cycle Length: 72 seconds (30 second offset)
- 200 E
 - o Green time: 28.7 seconds
 - o Yellow Clearance: 3.4 seconds
 - o Red Clearance: 39.9 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 17 seconds
- 300 S
 - o Green time: 31.8 seconds
 - o Yellow Clearance: 3 seconds
 - o Red Clearance: 37.2 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian Clearance: 12 seconds

3:30 PM – 6:30 PM (Weekdays)

- Cycle Length: 60 seconds (43 second offset)
- 200 E
 - o Green time: 25.7 seconds
 - o Yellow Clearance: 3.4 seconds
 - o Red Clearance: 30.9 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian FDW: 17 seconds
- 300 S
 - o Green time: 22.8 seconds
 - o Yellow Clearance: 3 seconds
 - o Red Clearance: 34.2 seconds
 - o Pedestrian Walk: 7 seconds
 - o Pedestrian Clearance: 12 seconds

